Unified analysis of discontinuous Galerkin methods for elliptic problems: detailed proofs

Supplement to the Lean formalization of D. N. Arnold, F. Brezzi, B. Cockburn and L. D. Marini (2002)

Abstract

We give precise statements and detailed proofs of the principal quantitative results of Arnold–Brezzi–Cockburn–Marini for all nine methods in their comparison table: the four fully stable and adjoint-consistent methods, ordinary NIPG, the pure-penalty and NIPG superpenalty variants, Baumann–Oden, and original Bassi–Rebay. The different regularity hypotheses, energy norms, kernel conditions, and projected-error conclusions are retained explicitly. The domain is a bounded convex polygon and meshes are conforming, uniformly shape-regular triangulations, with no quasiuniformity requirement. The proofs include the analytic regularity, traces, polynomial and conforming approximation, and conservative vector interpolation needed to close the arguments. The Lean formalization in this folder (`Showcase.lean` and the library `LeanCode`) proves the results stated in `Showcase.lean`. Its proof route is described in the blueprint (`blueprint/Blueprint.pdf`); in places it differs from the arguments given here, for example in the proof of the H^2 regularity on convex polygons, which the Lean proof obtains without the Fourier transform.

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1 The statements

This document covers the principal quantitative results in Sections 3–5 and all nine methods compared in Table 6.1 of [1]. We first specify the four stable, consistent, adjoint-consistent methods of Section 5.1. We then state the distinct superpenalty, NIPG, and weak-stability conclusions, followed by a source-coverage table. The label *Bassi* in the first four-method theorem means the stabilized method of Bassi et al. [13]; *BR* means the original Bassi–Rebay method [10]. Likewise Brezzi [22] and the pure-penalty Brezzi [23] method are different. Reference numbers in these method names are those of the original article.

1.1 Domain, spaces, meshes, and traces

Let Ω be a nonempty bounded open convex polygon in $\mathbb{R}^2$. We remove redundant collinear vertices, so that its interior angles lie strictly between 0 and π. All functions and vector spaces are real unless a Fourier transform is used in a proof.

For an open set D, $H^m(D)$ consists of $L^2(D)$ equivalence classes whose distributional partial derivatives through order m belong to $L^2(D)$. More explicitly, $g = \partial_i v$ means $\int_D g\phi = -\int_D v\,\partial_i\phi$ for every $\phi \in C_c^\infty(D)$, and higher derivatives are defined iteratively. We use the ordered-derivative convention

$$|v|^2_{H^m(D)} = \sum_{i_1,\dots,i_m\in\{1,2\}} \|\partial_{i_1}\cdots\partial_{i_m} v\|^2_{L^2(D)}, \qquad \|v\|^2_{H^m(D)} = \sum_{j=0}^{m} |v|^2_{H^j(D)}.$$

For $m = 0$, the empty derivative is v. In particular, the second-derivative seminorm uses the full Frobenius norm of the Hessian. This convention is equivalent, with constants depending only on m, to the usual multiindex convention. Set

$$H^1_0(D) = \overline{C_c^\infty(D)}^{\,H^1(D)}.$$

All boundary and edge values below are Sobolev traces. Thus definitions do not depend on arbitrary representatives of L^2 classes.

A conforming triangulation $\mathcal{T}_h$ is a finite collection of closed nondegenerate triangles whose union is $\overline{\Omega}$, whose interiors are contained in Ω and disjoint, and whose pairwise intersections are empty, a common vertex, or an entire common edge. Each edge belongs either to exactly two triangles, or to exactly one triangle and $\partial\Omega$. Write $\mathcal{E}_h^\circ, \mathcal{E}_h^\partial$ for these two types and $\mathcal{E}_h = \mathcal{E}_h^\circ \cup \mathcal{E}_h^\partial$. Let $h_K = \operatorname{diam} K$, let ρ_K be its inradius, let h_e be the length of e, and let $h = \max_K h_K$. The shape bound is

$$\max_{K\in\mathcal{T}_h} h_K/\rho_K \le \kappa.$$

There is no lower bound on h_K/h: quasiuniformity is not assumed.

Fix an integer $p \ge 1$. Let $\mathbb{P}_p(K)$ denote the restrictions of bivariate polynomials of total degree at most p. The discrete spaces and the comparison space are

$$V_h = \{v \in L^2(\Omega) : v|_K \in \mathbb{P}_p(K) \text{ a.e. for every } K\}, \quad \Sigma_h = V_h^2, \quad W_h = V_h + \big(H^2(\Omega)\cap H^1_0(\Omega)\big).$$

Both discrete spaces are finite dimensional. The symbols $\nabla_h, D_h^2, \operatorname{div}_h$ mean differentiation separately on each triangle.

On an interior edge shared by K^+ and K^-, with their outward unit normals n^+, n^-, define

$$\{v\} = \tfrac12(v^+ + v^-), \quad [\![v]\!] = v^+n^+ + v^-n^-, \qquad \{q\} = \tfrac12(q^+ + q^-), \quad [\![q]\!] = q^+\cdot n^+ + q^-\cdot n^-.$$

Here v is scalar and q is vector valued. On a boundary edge, set $[\![v]\!] = vn$ and $\{q\} = q$. We do not use a scalar average or a vector jump on boundary edges. All these definitions are unchanged if the labels $+$ and $-$ are exchanged.

For scalar or vector functions, $(a,b)_\Omega$ denotes the integral of their product or Euclidean inner product. On a face, write $\langle a,b\rangle_e$ for the corresponding edge integral with respect to length measure.

1.2 Lifting operators and the four bilinear forms

For $\phi \in L^2(e;\mathbb{R}^2)$, define $r_e\phi \in \Sigma_h$ by

$$(r_e\phi,\tau)_\Omega = -\langle \phi, \{\tau\}\rangle_e \qquad (\tau \in \Sigma_h). \tag{1.1}$$

For an interior edge and $q \in L^2(e)$, define $l_e q \in \Sigma_h$ by

$$(l_e q,\tau)_\Omega = -\langle q, [\![\tau]\!]\rangle_e \qquad (\tau \in \Sigma_h). \tag{1.2}$$

These definitions have unique solutions by finite-dimensional Riesz representation; the proof, support properties, and uniform estimates are given below. Choose a constant vector $\beta_e \in \mathbb{R}^2$ on each interior edge. For $v \in W_h$, set

$$R(v) = \sum_{e\in\mathcal{E}_h} r_e([\![v]\!]), \qquad L_\beta(v) = \sum_{e\in\mathcal{E}_h^\circ} l_e(\beta_e\cdot[\![v]\!]), \qquad A_\beta(v) = R(v) + L_\beta(v).$$

For positive edge parameters η_e, define

$$A_j(w,v) = \sum_{e\in\mathcal{E}_h} \frac{\eta_e}{h_e}\langle [\![w]\!], [\![v]\!]\rangle_e, \tag{1.3}$$

$$A_r(w,v) = \sum_{e\in\mathcal{E}_h} \eta_e (r_e([\![w]\!]), r_e([\![v]\!]))_\Omega. \tag{1.4}$$

Finally set

$$g(w,v) = (\nabla_h w, \nabla_h v)_\Omega, \quad C(w,v) = \sum_{e\in\mathcal{E}_h}\langle \{\nabla_h w\}, [\![v]\!]\rangle_e, \quad D_\beta(w,v) = \sum_{e\in\mathcal{E}_h^\circ}\langle [\![\nabla_h w]\!], \beta_e\cdot[\![v]\!]\rangle_e.$$

Each expression is defined for $w, v \in W_h$. The four methods in this document are *defined* by the following forms:

$$B_{\mathrm{IP}}(w,v) = g(w,v) - C(w,v) - C(v,w) + A_j(w,v), \tag{1.5}$$

$$B_{\mathrm{Bassi}}(w,v) = g(w,v) - C(w,v) - C(v,w) + A_r(w,v), \tag{1.6}$$

$$B_{\mathrm{Brezzi}}(w,v) = g(w,v) - C(w,v) - C(v,w) + (R(w),R(v))_\Omega + A_r(w,v), \tag{1.7}$$

$$\begin{aligned} B_{\mathrm{LDG}}(w,v) = {} & g(w,v) - C(w,v) - C(v,w) - D_\beta(w,v) - D_\beta(v,w) \\ & + (A_\beta(w), A_\beta(v))_\Omega + A_j(w,v). \end{aligned} \tag{1.8}$$

The negative signs in front of D_β in (1.8) follow from the numerical flux $\widehat{u} = \{u\} - \beta\cdot[\![u]\!]$ and the minus sign in (1.2). They correct the positive signs printed in [1, Eq. (3.27)]. The direct derivation below also agrees with the completed-square formula in its Table 4.1. This is a correction to that displayed expansion, not a change to the method defined by the paper's fluxes.

1.3 The exact and discrete problems and their norms

For $f \in L^2(\Omega)$, the exact solution is the unique $u \in H_0^1(\Omega)$ satisfying

$$(\nabla u, \nabla v)_\Omega = (f,v)_\Omega \qquad (v \in H_0^1(\Omega)). \tag{1.9}$$

Existence, uniqueness, and $u \in H^2(\Omega)$ are proved below for the stated convex polygon. Higher regularity is *not* inferred from $f \in L^2$: the high-order error estimates additionally assume $u \in H^{p+1}(\Omega)$.

For any one of the four forms, the discrete solution is defined by

$$u_h \in V_h, \qquad B_h(u_h, v_h) = (f, v_h)_\Omega \quad (v_h \in V_h). \tag{1.10}$$

Define the nonnegative quantities

$$G_h(v)^2 = \sum_K \|\nabla v\|^2_{L^2(K)}, \quad H_h(v)^2 = \sum_K h_K^2 \|D^2 v\|^2_{L^2(K)}, \quad J_h(v)^2 = \sum_e h_e^{-1} \|[\![v]\!]\|^2_{L^2(e)}$$

and the norm

$$N_h(v) = \left(G_h(v)^2 + H_h(v)^2 + J_h(v)^2\right)^{1/2}, \qquad v \in W_h. \tag{1.11}$$

That this is a norm is proved below. It is uniformly equivalent to the paper's lifting-based norm (4.2). Using (1.11) makes the public error statement more elementary; it does not weaken it.

Theorem 1.1 (Uniform stability and optimal errors)**.** *Fix $p \geq 1$ and a shape bound $\kappa > 0$. There is a positive number $\eta_*(p, \kappa)$, independent of the domain and mesh, with the following property.*

Fix a bounded open convex polygon Ω, numbers $0 < \eta_- \leq \eta_+ < \infty$, and $b \geq 0$. Choose one of the four methods, subject to

method	*additional requirement*
IP	$\eta_- > \eta_*(p, \kappa)$
Bassi	$\eta_- > 3$
Brezzi	*none*
LDG	*none.*

There exist constants $C_b, C_s, C_E, C_0 > 0$, depending only on $\Omega, p, \kappa, \eta_-, \eta_+, b$ and the chosen method, such that the following statements hold for every *conforming mesh with $h_K/\rho_K \leq \kappa$,* every *edge-parameter family satisfying $\eta_- \leq \eta_e \leq \eta_+$, and, for LDG,* every *family of constant edge vectors with $|\beta_e| \leq b$.*

(i) *The form B_h is symmetric and bilinear on W_h. It satisfies*

$$|B_h(w, v)| \leq C_b N_h(w) N_h(v) \quad (w, v \in W_h), \qquad B_h(v_h, v_h) \geq C_s N_h(v_h)^2 \quad (v_h \in V_h).$$

(ii) *For every $f \in L^2(\Omega)$, problem (1.10) has exactly one solution u_h.*

(iii) *If the weak solution u of (1.9) belongs to $H^{p+1}(\Omega)$, then*

$$N_h(u - u_h) \leq C_E h^p |u|_{H^{p+1}(\Omega)}, \qquad \|u - u_h\|_{L^2(\Omega)} \leq C_0 h^{p+1} |u|_{H^{p+1}(\Omega)}. \tag{1.12}$$

In particular, the constants precede the quantifiers over the mesh, coefficients, right-hand side, and exact solution. They cannot silently depend on any of those variable data.

The proof gives an explicit sufficient choice $\eta_* = 6c_+(p, \kappa)$, where c_+ is the uniform polynomial lifting upper-bound constant constructed below. No optimality of this threshold is claimed. For the Bassi method, the number 3 comes from the three edges of a triangle; it is tied to the normalization (1.4).

1.4 Flux correspondence and consistency statements

Write $\mathcal{P}_e(w) = \eta_e h_e^{-1}[\![w]\!]$ and $\mathcal{Q}_e(w) = \eta_e\{r_e([\![w]\!])\}$. For a discrete pair $(w,\sigma) \in V_h \times \Sigma_h$, define the following single edge values of the vector numerical flux:

method	$\widehat{\sigma}$
IP	$\{\nabla_h w\} - \mathcal{P}_e(w)$
Bassi	$\{\nabla_h w\} + \mathcal{Q}_e(w)$
Brezzi	$\{\sigma\} + \mathcal{Q}_e(w)$
LDG	$\{\sigma\} + \beta_e[\![\sigma]\!] - \mathcal{P}_e(w)$.

The $\beta_e[\![\sigma]\!]$ term is omitted on boundary edges. The scalar numerical flux is $\widehat{w} = \{w\}$ on interior edges for IP, Bassi, and Brezzi, and $\widehat{w} = \{w\} - \beta_e \cdot [\![w]\!]$ for LDG. In every case $\widehat{w} = 0$ on $\partial\Omega$.

Theorem 1.2 (Flux and primal formulations)**.** *For each of the four methods, a pair $(u_h, \sigma_h) \in V_h \times \Sigma_h$ satisfies*

$$\begin{aligned} (\sigma_h, \tau)_K &= -(u_h, \operatorname{div}\tau)_K + \langle \widehat{u}_h, \tau \cdot n_K\rangle_{\partial K} && (\tau \in \mathbb{P}_p(K)^2), \\ (\sigma_h, \nabla v)_K &= (f, v)_K + \langle \widehat{\sigma}_h \cdot n_K, v\rangle_{\partial K} && (v \in \mathbb{P}_p(K)) \end{aligned}$$

on every triangle if and only if u_h satisfies (1.10) and

$$\sigma_h = \nabla_h u_h + R(u_h) \quad \textit{for IP, Bassi, and Brezzi,} \qquad \sigma_h = \nabla_h u_h + A_\beta(u_h) \quad \textit{for LDG.}$$

For every collection of triangles, summation of the second equation with test function 1 gives balance of $\int f$ against the outgoing numerical flux on the boundary of that collection; all internal face contributions cancel.

Theorem 1.3 (Consistency, adjoint consistency, and orthogonality)**.** *For each of the four forms, if $u, \psi \in H^2(\Omega) \cap H_0^1(\Omega)$ and $-\Delta u = f$, $-\Delta\psi = g$ in $L^2(\Omega)$, then*

$$B_h(u, v) = (f, v)_\Omega, \qquad B_h(v, \psi) = (v, g)_\Omega \quad (v \in W_h).$$

Consequently, a discrete solution satisfies $B_h(u - u_h, v_h) = 0$ for every $v_h \in V_h$.

These are statements about the explicitly defined extended forms. For a nondiscrete u, its gradient need not belong to Σ_h. In particular, testing a flux equation against Σ_h identifies only an L^2 projection; it does not justify identifying a discrete auxiliary variable with ∇u. The proofs below keep that distinction explicit.

1.5 The remaining principal statements of the source

The four-method theorem is one part of the paper's conclusions. To specify the other entries of its Table 6.1, retain all definitions above and introduce the following forms on the same comparison space:

$$B_N(w, v) = g(w, v) - C(w, v) + C(v, w) + A_j(w, v), \tag{1.13}$$

$$B_{P,j}(w, v) = g(w, v) + A_j(w, v), \qquad B_{P,r}(w, v) = g(w, v) + A_r(w, v), \tag{1.14}$$

$$B_{\mathrm{BO}}(w, v) = g(w, v) - C(w, v) + C(v, w), \tag{1.15}$$

$$B_{\mathrm{BR}}(w, v) = g(w, v) - C(w, v) - C(v, w) + (R(w), R(v))_\Omega. \tag{1.16}$$

These are ordinary NIPG, the two pure-penalty methods (Babuška–Zlámal [7] and Brezzi et al. [23]), Baumann–Oden, and original Bassi–Rebay, respectively. Thus all nine methods in the original paper now have explicit definitions in this section.

For a superpenalty, use bounded positive coefficients $\eta_- \le \eta_e \le \eta_+$ but replace the penalty by

$$A_j^{\mathrm{sup}}(w,v) = \sum_e \eta_e h_e^{-(2p+1)} \langle [\![w]\!], [\![v]\!] \rangle_e, \tag{1.17}$$

$$A_r^{\mathrm{sup}}(w,v) = \sum_e \eta_e h_e^{-2p} (r_e([\![w]\!]), r_e([\![v]\!]))_\Omega. \tag{1.18}$$

The associated enhanced norm is

$$\mathcal{N}_{a,h}(v)^2 = G_h(v)^2 + H_h(v)^2 + A_a^{\mathrm{sup}}(v,v), \qquad a \in \{j, r\}.$$

This is a different norm from N_h; the stronger jump weight must not be suppressed in the statement. The three superpenalized methods are $g + A_j^{\mathrm{sup}}$, $g + A_r^{\mathrm{sup}}$, and $g - C + C^{\mathrm{op}} + A_j^{\mathrm{sup}}$, where $C^{\mathrm{op}}(w,v) = C(v,w)$.

Theorem 1.4 (NIPG and the superpenalized methods)**.** *Fix the domain, degree $p \ge 1$, shape bound, and bounds $0 < \eta_- \le \eta_+ < \infty$ before choosing a mesh, edge coefficients, or forcing. The three superpenalized discrete problems and the ordinary NIPG discrete problem have unique solutions for every $f \in L^2(\Omega)$. If the exact solution belongs to $H^{p+1}(\Omega)$, then ordinary NIPG satisfies*

$$N_h(u - u_h) \le C_E h^p |u|_{H^{p+1}(\Omega)}, \qquad \|u - u_h\|_{L^2(\Omega)} \le C_L h^p |u|_{H^{p+1}(\Omega)}.$$

For each of the three superpenalized methods,

$$\mathcal{N}_{a,h}(u - u_h) \le C_E h^p \|u\|_{H^{p+1}(\Omega)}, \qquad \|u - u_h\|_{L^2(\Omega)} \le C_L h^{p+1} \|u\|_{H^{p+1}(\Omega)}.$$

In the superpenalized NIPG case the full norm of u can be replaced by its highest-order seminorm. The constants are uniform over the fixed mesh and coefficient class. There is no quasiuniformity assumption. The positive lower bound on the coefficients suffices; no additional large-penalty threshold is needed for these three forms.

For ordinary NIPG the stated h^p rate is a guaranteed upper bound, not a universal lower bound on the error or a claim that faster convergence is impossible in individual cases. Theorems 5.8 and 5.10 prove Theorem 1.4, including its uniform constants.

Theorem 1.5 (Baumann–Oden)**.** *For fixed $p \ge 2$, fixed Ω, and fixed shape bound κ, every admissible mesh and every $f \in L^2(\Omega)$ give a unique solution of*

$$B_{\mathrm{BO}}(u_h, v_h) = (f, v_h)_\Omega \qquad (v_h \in V_h).$$

If $u \in H^{p+1}(\Omega)$, it satisfies, with uniform constants,

$$N_h(u - u_h) \le C_E h^p |u|_{H^{p+1}(\Omega)}, \qquad \|u - u_h\|_{L^2(\Omega)} \le C_L h^p |u|_{H^{p+1}(\Omega)}.$$

For $p = 1$, a general solvability assertion is false: there is a one-triangle mesh and a constant forcing for which no discrete solution exists.

Theorem 6.4 and Proposition 6.5 prove these assertions. In particular the energy estimate controls the full jump term, not just the broken gradient; the quadratic testing argument supplies that additional step explicitly.

For original Bassi–Rebay define

$$\mathcal{D}_h v_h = \nabla_h v_h + R(v_h), \qquad Z_h = \{v_h \in V_h : \mathcal{D}_h v_h = 0\}.$$

Let P_{p-1} be the elementwise L^2 orthogonal projection onto polynomials of degree at most $p-1$. Thus on each element $(P_{p-1}v)|_K$ is the unique such polynomial whose difference from v is orthogonal to every polynomial of degree at most $p-1$ on that element.

Theorem 1.6 (Original Bassi–Rebay)**.** *For $p \geq 1$, the kernel is exactly*

$$Z_h = \{v_h \in V_h : P_{p-1}(v_h) = 0 \text{ on every element and } \{v_h\} = 0 \text{ on every interior edge}\}.$$

The discrete equation $B_{\mathrm{BR}}(u_h, v_h) = (f, v_h)_\Omega$ for all $v_h \in V_h$ is solvable if and only if f is orthogonal to Z_h. When it is solvable, its solution set is one affine coset $u_h + Z_h$; the flux $\sigma_h = \mathcal{D}_h u_h$ and the projection $P_{p-1}u_h$ are unique.

In particular, solvability holds whenever f is an elementwise polynomial of degree at most $p-1$. For such data, if the exact solution belongs to $H^{p+1}(\Omega)$, every discrete solution satisfies

$$\|\nabla u - \sigma_h\|_{L^2(\Omega)} \leq C_\sigma h^p |u|_{H^{p+1}(\Omega)}, \qquad \|P_{p-1}(u - u_h)\|_{L^2(\Omega)} \leq C_0 h^{p+1} |u|_{H^{p+1}(\Omega)}.$$

The constants depend only on the fixed domain, degree, and shape bound. They are chosen before the mesh and the admissible data.

Theorems 6.11, 6.12, and 6.13 prove this statement. The projected estimate is the assertion of the source's (5.28). Replacing its left side by the full L^2 error is false: the explicit kernel example after Theorem 6.12 demonstrates this even for zero forcing.

1.6 Coverage of the original paper

Source result	Exact content retained	Proof here
Section 3.2, (3.8)–(3.11)	Flux elimination and primal equivalence	Theorem 4.4
Section 3.3, (3.13)–(3.16)	Consistency, orthogonality, and sufficient conditions for adjoint consistency	Theorems 4.6, 4.7
Sections 4.1–4.3	Uniform boundedness, penalty stability, and approximation	Sections 3, 4, 5
Section 5.1; first five method rows of Table 6.1	Four optimal methods and ordinary NIPG	Theorems 1.1, 5.10
Section 5.2, (5.9)–(5.12)	Two pure superpenalties and superpenalized NIPG	Theorem 5.8
Section 5.3.1, (5.20) and following discussion	Baumann–Oden for $p \geq 2$; failure of a general $p = 1$ statement	Theorem 6.4, Proposition 6.5
Section 5.3.2, (5.23)–(5.28)	BR kernel, compatibility, flux error, and projected scalar error	Theorems 6.11–6.13

The historical survey, qualitative solver comparisons, suggestions about extensions, and open problems in the original paper are not theorems being certified. Nor is a general converse involving full vector-flux single-valuedness asserted: tangential changes are invisible to the scheme, as explained in Remark 4.8. The source's quantitative main results for all nine methods are the theorems above, with their different hypotheses and norms preserved.

1.7 Foundations and proof strategy

We use standard measure theory and functional analysis: completeness of L^2, the Cauchy–Schwarz inequality, Fubini's theorem, dominated convergence, the Riesz representation theorem, smooth cutoffs and mollifiers, the change-of-variables formula, the Plancherel theorem on $\mathbb{R}^d$ and Fourier series on the circle. We do not assume a package of polygonal Sobolev or finite-element theory. Bounded-domain weak derivatives, density, traces, Poisson regularity, interpolation, liftings, and all error estimates are proved in this document. The Fourier normalizations, weak-derivative multiplier theorem, and interval sine/cosine bases are derived in Sections 7.2–7.3. No elliptic-regularity, trace, Bramble–Hilbert, Morley, or Raviart–Thomas theorem is left as an additional assumed premise of the main results.

The proof has three useful separations. First, local polynomial approximation is constructed by derivative moments and therefore does not require nodal evaluation of a Sobolev function. Second, discrete solvability uses injectivity of a map between finite-dimensional spaces of the same dimension. Third, the energy and duality arguments use only quantitative inequalities already proved for the concrete forms.

2 Analytic foundations and polygonal Poisson regularity

This section proves the analytic fact needed in the duality argument, including existence: if $G \subset \mathbb{R}^2$ is a bounded open convex polygon and $g \in L^2(G)$, then the weak zero-Dirichlet solution satisfies

$$-\Delta\psi = g, \qquad \psi \in H^2(G) \cap H_0^1(G), \qquad \|\psi\|_{H^2(G)} \le C_G \|g\|_{L^2(G)}. \tag{2.1}$$

The proof below does not assume elliptic boundary regularity, a trace theorem, or density in $H^2(G) \cap H_0^1(G)$. In particular it does not use a second-derivative estimate on smooth functions to assert surjectivity of the Dirichlet Laplacian. It first constructs the weak solution, and then proves regularity directly at the vertices, edges, and interior.

2.1 Definitions and elementary approximation

All integrals are Lebesgue integrals. Functions in L^2 are identified when they agree almost everywhere. For an open set $U \subset \mathbb{R}^d$, a multi-index $\alpha \in \mathbb{N}_0^d$, and $u, v \in L^1_{\mathrm{loc}}(U)$, the assertion $D^\alpha u = v$ in the weak sense means

$$\int_U u D^\alpha \varphi = (-1)^{|\alpha|} \int_U v\varphi \quad (\varphi \in C_c^\infty(U)).$$

Weak derivatives, when they exist, are unique: a locally integrable function which integrates to zero against every smooth compactly supported test function is zero almost everywhere. This follows by convolution with a smooth approximate identity on each relatively compact open subset and then by local L^1 convergence of that convolution. Set

$$H^m(U) = \{u \in L^2(U) : D^\alpha u \in L^2(U),\ |\alpha| \le m\}, \qquad |u|^2_{H^j(U)} = \sum_{i_1,\dots,i_j=1}^{d} \|\partial_{i_1} \cdots \partial_{i_j} u\|^2_{L^2(U)},$$

where $|u|_{H^0(U)} = \|u\|_{L^2(U)}$ and $\|u\|^2_{H^m(U)} = \sum_{j=0}^m |u|^2_{H^j(U)}$. Thus our norm counts ordered derivative indices; it is equivalent to the usual multi-index norm, since each derivative of order j occurs between one and $j!$ times. Weak mixed derivatives commute because the derivatives of smooth test functions commute. Here m is a nonnegative integer; $H^0 = L^2$. The notation $H_0^1(U)$ means the closure of $C_c^\infty(U)$ in $H^1(U)$, rather than an initially assumed pointwise boundary condition. We write $|D^2 u|^2 = \sum_{i,j=1}^d |\partial_i \partial_j u|^2$ for the Frobenius norm, which counts a mixed second derivative twice when $d = 2$. The subscript "loc" means membership on every open subset whose closure is compact in the indicated open set. We use $A \Subset B$ to mean that $\overline{A}$ is compact and contained in B.

Completeness of L^2, Cauchy–Schwarz, Fubini, and Riesz are standard. The weak Sobolev spaces have the concrete closed-graph Hilbert model of Lemma 7.3. The Fourier transform below is the angular-frequency transform of Lemma 7.4; its weak-derivative rule in both directions is proved in Lemma 7.5. Completeness and Parseval for the interval sine and cosine systems are proved in Lemma 7.6, including the Haar-to-length normalization. These are forward references to proofs, not extra assumptions: those proofs depend only on standard facts, cutoffs, and weak integral identities, not on the polygon regularity theorem.

Lemma 2.1 (Sobolev completeness and elementary operations)**.** *$H^m(U)$ is a Hilbert space. Restriction to an open subset preserves weak derivatives. Multiplication by a smooth function whose derivatives through order m are bounded on U maps $H^m(U)$ continuously to itself, with the usual Leibniz formula. Affine invertible changes of variables preserve H^m locally and obey the usual chain rule for weak derivatives. The zero extension of a member of $H_0^1(U)$ belongs to $H^1(\mathbb{R}^d)$, and its first weak derivatives are the zero extensions of the corresponding derivatives on U.*

Proof. If (u_k) is Cauchy in H^m, each $D^\alpha u_k$ has a limit v_α in L^2. Passing to the limit in the defining identity against a fixed test function shows $D^\alpha v_0 = v_\alpha$. This proves completeness; the sum of the L^2 inner products gives the Hilbert structure. Restriction follows because a compactly supported test function in the smaller set is also a test function in the larger set after zero extension. For multiplication, insert the product of the multiplier and a test function in the definition of a first weak derivative, expand that test function's derivative, and then induct on m. The resulting finite sum gives the asserted norm bound. For an affine change of variables, change variables in the integral against a test function; its classical chain rule and the constant Jacobian give the weak chain rule, again by induction. Finally, choose $u_k \in C_c^\infty(U)$ tending to u in $H^1(U)$. Their zero extensions are Cauchy in $H^1(\mathbb{R}^d)$ with exactly the same norms. Their L^2 limit is the zero extension of u, and their derivative limits are the zero extensions of its derivatives. Completeness gives the last assertion. □

The approximation inputs below are supplied by Lemma 7.2, which derives affine L^2 continuity from the density of smooth compactly supported functions in L^2. For a nonnegative smooth kernel ρ supported in the unit ball with integral one and $\rho_\varepsilon(x) = \varepsilon^{-d}\rho(x/\varepsilon)$, the proved Jensen–Fubini estimate is

$$\|\rho_\varepsilon * f - f\|_{L^2}^2 \le \int \rho_\varepsilon(y)\|f(\cdot - y) - f\|_{L^2}^2 \, dy \longrightarrow 0. \tag{2.2}$$

The same proof with the first power gives local L^1 approximation after cutting off. Support control and density of $C_c^\infty(U)$ in $L^2(U)$ are also proved there. These arguments do not assume density of compactly supported smooth functions in the full $H^m(U)$, which would be false.

Lemma 2.2 (One-dimensional fundamental theorem and local coordinate changes)**.** *If a locally integrable function on an interval has weak derivative $f \in L^1_{\mathrm{loc}}$, it has a locally absolutely continuous representative whose classical derivative is f almost everywhere. If $\Phi : U \to V$ is a smooth diffeomorphism of open sets, $u \in H^m_{\mathrm{loc}}(V)$, and m is a finite integer, then $u \circ \Phi \in H^m_{\mathrm{loc}}(U)$ with the classical chain rules interpreted as weak derivative identities.*

Proof. On a relatively compact interval, an indefinite integral of f is absolutely continuous and has weak derivative f, by Fubini. Subtract it from the given function. A distribution with zero derivative on an interval is constant: every compactly supported smooth test function of integral zero is the derivative of a compactly supported smooth function, so it is annihilated; subtracting a fixed test function of integral one then shows that the distribution acts by a constant times the integral. Uniqueness of distributions represented by locally integrable functions proves the assertion, and overlapping intervals give the locally absolutely continuous representative.

For the second assertion, fix a compact subset $K \Subset U$ and choose an open neighborhood of $\Phi(K)$ with compact closure in V. On a slightly smaller such neighborhood, mollification of u converges to u with all derivatives through order m in L^2: extend the derivatives locally by zero, use (2.2), and observe that convolution commutes with weak derivatives wherever the kernel stays inside V. The mollified functions are smooth. Their classical chain rules are finite sums of derivatives of the mollified functions composed with Φ, multiplied by smooth

functions formed from derivatives of Φ. On a neighborhood of K these multiplier functions and the Jacobian of Φ^{-1} are bounded. Change of variables therefore makes each sum converge in L^2 on that neighborhood. Passing to the limit against compactly supported tests proves the weak chain rules there. Since K was arbitrary, the assertion is local on U. □

Lemma 2.3 (Smooth density on bounded convex sets)**.** *Let U be a bounded nonempty open convex subset of $\mathbb{R}^d$ and let $m \geq 0$ be an integer. Restrictions to U of functions smooth on a neighborhood of $\overline{U}$ are dense in $H^m(U)$.*

Proof. Choose $x_0 \in U$ and $r > 0$ with $B_r(x_0) \subset U$. For $0 < \lambda < 1$ put $D_\lambda x = x_0 + \lambda(x - x_0)$ and $U_\lambda = D_\lambda^{-1}(U)$. Convexity implies

$$D_\lambda x + B_{(1-\lambda)r}(0) \subset U \quad (x \in \overline{U}). \tag{2.3}$$

To verify this also for $x \in \partial U$, fix $|b| < (1-\lambda)r$ and choose $x_k \in U$ tending to x. The points $x_0 + b/(1-\lambda) + \lambda(x - x_k)/(1-\lambda)$ belong to $B_r(x_0)$ for large k. Their convex combinations with x_k, with respective coefficients $1-\lambda$ and λ, equal $D_\lambda x + b$ and belong to U. This proves (2.3). In particular a fixed neighborhood of $\overline{U}$ lies in U_λ; for example its radius may be $(1-\lambda)r/(2\lambda)$.

For $u \in H^m(U)$ define $u_\lambda = u \circ D_\lambda$ on U_λ. Lemma 2.1 gives $D^\alpha u_\lambda = \lambda^{|\alpha|}(D^\alpha u) \circ D_\lambda$. Extend each $D^\alpha u$ separately by zero as an L^2 function on $\mathbb{R}^d$; no assertion about derivatives of these zero extensions is needed. Continuity of dilation and translation in L^2 gives

$$\|\lambda^{|\alpha|}(D^\alpha u) \circ D_\lambda - D^\alpha u\|_{L^2(U)} \longrightarrow 0, \qquad \lambda \uparrow 1.$$

At a fixed λ, extend u_λ by zero outside U_λ and convolve it with ρ_ε, taking ε smaller than the distance from $\overline{U}$ to the complement of U_λ. The convolution is smooth on a neighborhood of $\overline{U}$ and, there, its weak derivatives are the convolutions of the corresponding derivatives of u_λ. This follows by the weak derivative identity and Fubini, because the convolution kernel never meets ∂U_λ there. Formula (2.2), applied to the zero extension of each derivative, proves convergence to u_λ in $H^m(U)$ as $\varepsilon \downarrow 0$. Choose $\lambda_k \uparrow 1$, and then choose ε_k so that this last error is smaller than $1/k$. The resulting smooth sequence tends to u in $H^m(U)$. □

2.2 Weak existence and the interior Fourier estimate

Lemma 2.4 (Poincaré inequality and weak Poisson solution)**.** *If a bounded open set $U \subset \mathbb{R}^2$ is contained in a rectangle $(a,b) \times (c,d)$ and $L = b - a$, then*

$$\|v\|_{L^2(U)} \leq L\|\nabla v\|_{L^2(U)} \qquad (v \in H_0^1(U)). \tag{2.4}$$

For every $g \in L^2(U)$ there is exactly one $\psi \in H_0^1(U)$ such that

$$\int_U \nabla\psi \cdot \nabla v = \int_U gv \qquad (v \in H_0^1(U)), \tag{2.5}$$

and

$$\|\nabla\psi\|_{L^2(U)} \leq L\|g\|_{L^2(U)}, \qquad \|\psi\|_{L^2(U)} \leq L^2\|g\|_{L^2(U)}. \tag{2.6}$$

Proof. For $v \in C_c^\infty(U)$, extend it by zero to the rectangle. The fundamental theorem of calculus and Cauchy–Schwarz give $|v(x,y)|^2 \leq L\int_a^b |\partial_x v(s,y)|^2\, ds$. Integration in (x,y) proves (2.4), first for these smooth functions and then for $H_0^1(U)$ by its definition. The gradient inner product is consequently a complete Hilbert inner product on $H_0^1(U)$, because its norm is equivalent to the H^1 norm. The functional $v \mapsto \int_U gv$ has norm at most $L\|g\|_{L^2}$ for this inner product. Riesz gives (2.5). Testing it with ψ gives (2.6). The difference of two solutions, tested against itself, has zero gradient and therefore zero L^2 norm by (2.4). □

Lemma 2.5 (Full-plane second derivatives)**.** *If $V \in H^1(\mathbb{R}^2)$ and $-\Delta V = F \in L^2(\mathbb{R}^2)$ in the weak distributional sense, then $V \in H^2(\mathbb{R}^2)$ and*

$$\|D^2 V\|_{L^2(\mathbb{R}^2)} = \|F\|_{L^2(\mathbb{R}^2)}. \tag{2.7}$$

Proof. The distributional equation gives $|\xi|^2 \widehat{V} = \widehat{F}$. Each $\xi_i \xi_j \widehat{V}$ belongs to L^2, since $|\xi_i \xi_j| \le |\xi|^2$. The converse derivative rule for the Fourier transform identifies its inverse transform with $-\partial_i \partial_j V$. Plancherel and $\sum_{i,j} \xi_i^2 \xi_j^2 = |\xi|^4$ give (2.7). □

Lemma 2.6 (Reflection at a straight Dirichlet edge)**.** *Write $P = \{(x, y) : y > 0\}$. If $v \in H_0^1(P)$, $F \in L^2(P)$, and $\int_P \nabla v \cdot \nabla \eta = \int_P F\eta$ for every $\eta \in H_0^1(P)$, then $v \in H^2(P)$ and*

$$\|D^2 v\|_{L^2(P)} \le \|F\|_{L^2(P)}. \tag{2.8}$$

Proof. Define the odd reflections $V(x, y) = v(x, y)$ for $y > 0$ and $V(x, y) = -v(x, -y)$ for $y < 0$, and define F_o in the same way. Approximating v in $H^1(P)$ by $C_c^\infty(P)$ and reflecting those functions proves $V \in H^1(\mathbb{R}^2)$: their odd reflections are smooth across the line because each is zero in a neighborhood of it, and all squared H^1 differences double under reflection.

Given $\varphi \in C_c^\infty(\mathbb{R}^2)$, put $\eta(x, y) = \varphi(x, y) - \varphi(x, -y)$ on P. This function belongs to $H_0^1(P)$. Here is a direct verification without a trace theorem. It is smooth, compactly supported in the closed half-plane, and satisfies $|\eta(x, y)| \le Cy$ near $y = 0$. Choose a smooth q with $q = 0$ on $(-\infty, 1]$, $q = 1$ on $[2, \infty)$, and bounded derivative. Then $q(y/\varepsilon)\eta \in C_c^\infty(P)$ and converges in $H^1(P)$ to η. The only additional derivative term is $\varepsilon^{-1} q'(y/\varepsilon)\eta$, whose squared integral is $O(\varepsilon)$; all other errors have bounded integrands supported in a strip of width 2ε and bounded horizontal length. Changing variables $y \mapsto -y$ in the lower half-plane now gives

$$\int_{\mathbb{R}^2} \nabla V \cdot \nabla \varphi = \int_P \nabla v \cdot \nabla \eta = \int_P F\eta = \int_{\mathbb{R}^2} F_o \varphi.$$

Apply Lemma 2.5. The squared norms on both sides of (2.7) double under reflection, which proves (2.8) (indeed, equality). □

2.3 A direct estimate at a convex corner

The following strip calculation supplies the boundary regularity at a vertex. It is useful to specify its energy space without appealing to boundary traces. Fix $0 < \omega < \pi$, put

$$I = (0, \omega), \quad Q = \mathbb{R} \times I, \quad \nu_n = \frac{n\pi}{\omega}, \quad e_n(\theta) = \sqrt{\frac{2}{\omega}} \sin(\nu_n \theta) \quad (n \ge 1),$$

and let X be the closure of $C_c^\infty(Q)$ in the norm

$$\|w\|_X^2 = \int_Q (|w_t|^2 + |w_\theta|^2).$$

For $h \in C_c^\infty(I)$, integration by parts shows that the cosine coefficient of h' with index $n \ge 1$ is ν_n times the sine coefficient of h, and its constant cosine coefficient is zero. Parseval therefore proves

$$\|h\|_{L^2(I)}^2 \le \nu_1^{-2} \|h'\|_{L^2(I)}^2. \tag{2.9}$$

Integration in t shows that the X norm and $H^1(Q)$ norm are equivalent on $C_c^\infty(Q)$. Thus X is precisely that closure in $H^1(Q)$, is complete, and embeds continuously in $L^2(Q)$.

Lemma 2.7 (Spectral description of the strip energy space)**.** *For $w \in X$, let $w_n(t) = \int_I w(t,\theta)e_n(\theta)\,d\theta$. Then*

$$\|w\|_X^2 = \sum_{n=1}^{\infty} \int_{\mathbb{R}} (|w_n'|^2 + \nu_n^2 |w_n|^2)\,dt. \tag{2.10}$$

Conversely, any sequence $w_n \in H^1(\mathbb{R})$ for which the sum on the right is finite defines, by its sine series, a unique member of X.

Proof. For a smooth compactly supported w, Fubini and the sine Parseval identity give the equality for the t derivative. Integration by parts in θ, followed by cosine Parseval as above, gives the equality for the θ derivative. Approximation gives (2.10) for $w \in X$.

For the converse, first take a finite sum with coefficients in $C_c^\infty(\mathbb{R})$. Each angular factor e_n is a limit in $H^1(I)$ of compactly supported smooth functions: multiply it by $q(\theta/\varepsilon)q((\omega-\theta)/\varepsilon)$, with q as in Lemma 2.6. Since e_n vanishes linearly at both endpoints, the squared norm of each cutoff-derivative error is $O(\varepsilon)$, and the other errors tend to zero by integration over the shrinking endpoint intervals. Thus the finite sum belongs to X. Any $H^1(\mathbb{R})$ coefficient is a limit of compactly supported smooth coefficients: first multiply by a smooth cutoff equal to one on $[-R,R]$, with derivative bounded by C/R, and let $R \to \infty$; then convolve with a mollifier. The product rule, integrability of the function and its derivative, and (2.2) prove convergence. Finally truncate the sine series. The assumed summability and the already proved identity make these truncations Cauchy in X. Their L^2 limit is the claimed sine series, and completeness supplies its membership in X. □

Lemma 2.8 (A shifted strip operator)**.** *For every $f \in L^2(Q)$ there exists a function $z \in X$ with $z_{tt}, z_{t\theta}, z_{\theta\theta} \in L^2(Q)$ satisfying*

$$-z_{tt} - 2z_t - z - z_{\theta\theta} = f \quad \text{in distributions on } Q. \tag{2.11}$$

It satisfies

$$\|z\|_{L^2}^2 + \|z_t\|_{L^2}^2 + \|z_\theta\|_{L^2}^2 + \|z_{tt}\|_{L^2}^2 + 2\|z_{t\theta}\|_{L^2}^2 + \|z_{\theta\theta}\|_{L^2}^2 \le C_\omega \|f\|_{L^2}^2. \tag{2.12}$$

If, in addition, $f(t,\theta) = 0$ for almost every $t > T$, then $\widetilde{w}(t,\theta) = e^t z(t,\theta)$ belongs to X and solves

$$-\widetilde{w}_{tt} - \widetilde{w}_{\theta\theta} = e^t f. \tag{2.13}$$

Proof. Let f_n be the sine coefficients of f, and take the Fourier transform in t, with frequency ξ. Define

$$\widehat{z}_n(\xi) = \frac{\widehat{f_n}(\xi)}{\xi^2 + \nu_n^2 - 1 - 2i\xi}. \tag{2.14}$$

For real f, the numerator has Hermitian symmetry and the denominator $d_n(\xi) = \xi^2 + \nu_n^2 - 1 - 2i\xi$ satisfies $d_n(-\xi) = \overline{d_n(\xi)}$. Thus $\widehat{z}_n$ has Hermitian symmetry and each inverse transform z_n is real; equivalently one may take real parts throughout. Since $\nu_1 > 1$, the number $c_\omega = 1 - \nu_1^{-2}$ is positive and

$$|\xi^2 + \nu_n^2 - 1 - 2i\xi| \ge \xi^2 + \nu_n^2 - 1 \ge c_\omega(\xi^2 + \nu_n^2). \tag{2.15}$$

Plancherel and sine Parseval give

$$\sum_n \int_{\mathbb{R}} (\xi^2 + \nu_n^2)^2 |\widehat{z}_n(\xi)|^2\,d\xi \le c_\omega^{-2} \|f\|_{L^2(Q)}^2. \tag{2.16}$$

Because $\nu_n \ge \nu_1 > 1$, this controls the zero- and first-order norms as well. Lemma 2.7 constructs $z \in X$. The partial sine sums have second weak derivatives which are Cauchy in L^2:

use the sine system for z_{tt} and $z_{\theta\theta}$ and the orthonormal cosine system with positive indices for $z_{t\theta}$. Passing to the limit against a compactly supported test function proves that these limits are the weak derivatives of z. Their squared norms are, respectively, the sums with multipliers ξ^4, ν_n^4, and $\xi^2\nu_n^2$. This proves (2.12). Formula (2.14), first in each mode and then in the convergent series, proves (2.11).

It remains to justify multiplication by e^t; this step cannot be inferred from $z \in H^2(Q)$ alone. Suppose $f = 0$ on $t > T$. Each $z_n \in H^2(\mathbb{R})$ satisfies there

$$-z_n'' - 2z_n' + (\nu_n^2 - 1)z_n = 0.$$

Solving this constant-coefficient ordinary differential equation gives

$$z_n(t) = z_n(T)e^{-(\nu_n+1)(t-T)} \qquad (t \geq T). \tag{2.17}$$

For completeness, an H^2_{loc} solution has an absolutely continuous first derivative; factoring the differential operator into $(\partial_t + 1 - \nu_n)(\partial_t + 1 + \nu_n)$ and integrating the two first-order equations gives a linear combination of $e^{(\nu_n-1)t}$ and $e^{-(\nu_n+1)t}$. The former is excluded by $z_n \in L^2(T,\infty)$; the continuous representative supplies the value at T.

The boundary values in (2.17) have the summability needed below. For $a \in H^1(\mathbb{R})$ and $\nu > 0$,

$$\nu|a(T)|^2 \leq \int_{\mathbb{R}} (|a'|^2 + \nu^2|a|^2)\,dt. \tag{2.18}$$

To see this, the absolutely continuous representative obeys $|a(T)|^2 = -\int_T^\infty (|a|^2)'$; its limit at infinity is zero. That limit follows because $(|a|^2)' = 2\,\mathrm{Re}(a'\overline{a})$ is integrable and $|a|^2$ is integrable. Now use $2\nu|a'||a| \leq |a'|^2 + \nu^2|a|^2$. Consequently

$$\sum_n \nu_n|z_n(T)|^2 \leq \sum_n \int_{\mathbb{R}} (|z_n'|^2 + \nu_n^2|z_n|^2) < \infty.$$

On $t \leq T$, multiplication by e^t and its derivative is bounded, so the sum of the squared t- and angular-derivative norms of $e^t z_n$ is finite there. On $t > T$, (2.17) gives

$$e^t z_n(t) = e^T z_n(T)e^{-\nu_n(t-T)}, \qquad \int_T^\infty \left(|(e^t z_n)'|^2 + \nu_n^2|e^t z_n|^2\right) dt = e^{2T}\nu_n|z_n(T)|^2.$$

Each coefficient is locally H^2 across T, and these estimates show that it is globally H^1. Summation and Lemma 2.7 prove $\widetilde{w} \in X$. Multiplying (2.11) by e^t in distributions proves (2.13). Its right side belongs to $L^2(Q)$ because f is supported in $t \leq T$. Thus the distributional equation also holds tested against every member of X, by density and the angular Poincaré inequality. □

Lemma 2.9 (Convex-sector regularity)**.** *Let*

$$S_\omega = \{(r\cos\theta, r\sin\theta) : r > 0,\ 0 < \theta < \omega\}, \qquad 0 < \omega < \pi.$$

Suppose $v \in H_0^1(S_\omega)$ vanishes for $r > R$, $F \in L^2(S_\omega)$ vanishes for $r > R$, and $-\Delta v = F$ weakly. Then $v \in H^2(S_\omega)$ and

$$\|D^2 v\|_{L^2(S_\omega)} \leq C_\omega \|F\|_{L^2(S_\omega)}. \tag{2.19}$$

Proof. Use $t = \log r$ and write

$$w(t,\theta) = v(e^t\cos\theta, e^t\sin\theta), \qquad f(t,\theta) = e^t F(e^t\cos\theta, e^t\sin\theta).$$

The change of variables gives

$$\int_{S_\omega} |\nabla v|^2 = \int_Q (|w_t|^2 + |w_\theta|^2), \qquad \|f\|_{L^2(Q)} = \|F\|_{L^2(S_\omega)}. \tag{2.20}$$

These statements, including $w \in X$, do not require any unknown weighted estimate on v. To verify them, take $v_k \in C_c^\infty(S_\omega)$ tending to v in H^1. Their transforms w_k belong to $C_c^\infty(Q)$, and the classical polar-coordinate calculation gives equality of the gradient differences. The angular Poincaré inequality makes (w_k) Cauchy in X. On compact substrips, the coordinate Jacobian and its inverse are bounded, so its limit is the displayed transform of v. The equality follows by passage to the limit. The equality for f is simply the area formula $dx = e^{2t}\,dt\,d\theta$.

Transforming a compactly supported smooth test function gives

$$-w_{tt} - w_{\theta\theta} = e^t f \quad \text{weakly on } Q. \tag{2.21}$$

Indeed the polar gradient inner product becomes the strip gradient inner product, and the right-hand integral becomes $\int_Q e^{2t} F(e^t, \theta)\eta = \int_Q e^t f \eta$. The right side is $L^2(Q)$, since $f = 0$ for $t > \log R$; thus (2.21) extends to tests in X.

Construct z from f by Lemma 2.8, and let $\widetilde{w} = e^t z \in X$. This constructed function and the original $w \in X$ solve exactly the same equation. Test their difference equation with $w - \widetilde{w}$. Its X norm is zero, and (2.9) then makes its L^2 norm zero. Therefore

$$w = e^t z. \tag{2.22}$$

Notice that membership of the originally unknown $e^{-t}w$ in L^2 was neither assumed nor used: it has been proved by this construction and uniqueness argument.

We finish by calculating the Cartesian second derivatives. On an annulus strictly inside the sector, the coordinate change is a smooth diffeomorphism with bounded derivatives on compact subsets. The weak chain rule there follows from Lemma 2.2; hence the usual polar Hessian formula is valid. In the orthonormal radial and angular frame its entries are

$$v_{rr} = r^{-2}(w_{tt} - w_t), \qquad (D^2 v)_{r\theta} = r^{-2}(w_{t\theta} - w_\theta), \qquad (D^2 v)_{\theta\theta} = r^{-2}(w_{\theta\theta} + w_t).$$

Substitute (2.22), use $dx = e^{2t}\,dt\,d\theta$, and use invariance of the Frobenius norm under an orthogonal change of frame. This gives

$$\|D^2 v\|^2_{L^2(S_\omega)} = \|z_{tt} + z_t\|^2_{L^2(Q)} + 2\|z_{t\theta}\|^2_{L^2(Q)} + \|z_{\theta\theta} + z_t + z\|^2_{L^2(Q)}. \tag{2.23}$$

More formally, first use this identity on compact annular subsectors and then exhaust S_ω. Lemma 2.8 bounds the result by $C_\omega \|f\|^2_{L^2(Q)}$. These locally defined second derivatives are therefore in $L^2(S_\omega)$ globally and are the weak derivatives on the sector: every compactly supported test function is contained in a compact annular subsector. The vertex lies on the boundary, so no test function in the sector meets it and no distributional term there is being discarded. Since v was already in H^1, this proves $v \in H^2$ and (2.19). □

2.4 Localization and the polygon theorem

Theorem 2.10 (Dirichlet regularity on a convex polygon)**.** *Let $G \subset \mathbb{R}^2$ be an arbitrary bounded nonempty open convex polygon. For each $g \in L^2(G)$ there is a unique $\psi \in H^2(G) \cap H_0^1(G)$ with $-\Delta\psi = g$ in distributions. It is the weak solution of (2.5), and it satisfies (2.1), with a constant depending only on the fixed polygon.*

Proof. Lemma 2.4 already supplies a unique weak solution $\psi \in H_0^1(G)$ and its H^1 bound. We prove its additional regularity. Remove any redundant collinear vertices from the polygon's description. At every remaining vertex, convexity gives an interior angle $\omega \in (0, \pi)$. A sufficiently small ball centered at the vertex meets G in the intersection of that ball with a rotated and translated copy of S_ω. At a nonvertex boundary point, a small ball meets G in a half-ball. At an interior point it is contained in G.

Choose finitely many such open balls covering $\overline{G}$, with slightly smaller concentric balls still covering it, and choose a smooth partition of unity χ_j on a neighborhood of $\overline{G}$ subordinate to the larger balls. One explicit construction is to take nonnegative smooth bump functions supported in the larger balls and positive on their smaller balls, add them, and divide each bump by their positive sum on a neighborhood of $\overline{G}$. Multiply by a further smooth cutoff equal to one near $\overline{G}$ and compactly supported in that neighborhood to obtain smooth functions defined everywhere. Then

$$\sum_j \chi_j = 1 \quad \text{near } \overline{G}, \qquad \operatorname{supp} \chi_j \Subset B_j. \tag{2.24}$$

All these functions and their first two derivatives have fixed bounds.

Fix j and put $v_j = \chi_j \psi$ on G. Regard v_j as a function on the associated model domain M_j, namely the full plane, half-plane, or sector, by setting it zero outside the ball's intersection with G. Then $v_j \in H_0^1(M_j)$. In fact, if $\psi_k \in C_c^\infty(G)$ tends to ψ in $H^1(G)$, the same localization and zero extension of $\chi_j \psi_k$ belongs to $C_c^\infty(M_j)$: its support stays away from both the model-domain boundary and the boundary of B_j. The product estimate in Lemma 2.1 gives convergence in $H^1(M_j)$. This also justifies the zero extension without presupposing any trace property.

On G, the distributional product rule gives

$$-\Delta(\chi_j \psi) = F_j := \chi_j g - 2\nabla \chi_j \cdot \nabla \psi - (\Delta \chi_j)\psi. \tag{2.25}$$

Define F_j to be zero on the remainder of M_j. It is in $L^2(M_j)$, is supported in B_j, and

$$\|F_j\|_{L^2(M_j)} \le C_j \big(\|g\|_{L^2(G)} + \|\psi\|_{H^1(G)} \big). \tag{2.26}$$

The equation $-\Delta v_j = F_j$ holds in all of M_j. To check this without a possible interface term at the ball, take a test function on M_j and multiply it by a smooth cutoff equal to one on the supports of χ_j and its first two derivatives and supported in B_j. All integrals involving v_j and F_j are unchanged. The new test function, restricted to G, is an admissible interior test function when the original test has compact support in M_j; the model-domain description is valid throughout B_j. Thus (2.25) proves the assertion. Since $v_j \in H_0^1(M_j)$ and $F_j \in L^2(M_j)$, the identity extends by the defining H^1 closure to all tests in $H_0^1(M_j)$ whenever that formulation is used.

For the full-plane case apply Lemma 2.5; for the half-plane case apply Lemma 2.6; for the vertex case apply Lemma 2.9, after the rigid coordinate change. At a vertex both localized functions vanish outside some finite radius, as required. These three lemmas give

$$v_j \in H^2(M_j), \qquad \|D^2 v_j\|_{L^2(M_j)} \le C_j \|F_j\|_{L^2(M_j)}.$$

By (2.24), $\psi = \sum_j v_j$ on G. The sum is finite, so restriction and completeness of the weak derivative identities imply $\psi \in H^2(G)$ and

$$\|\psi\|_{H^2(G)} \le C_G \big(\|g\|_{L^2(G)} + \|\psi\|_{H^1(G)} \big) \le C_G \|g\|_{L^2(G)},$$

where the last step is (2.6). Any distributional solution in $H^2(G) \cap H_0^1(G)$ satisfies (2.5): first test with $C_c^\infty(G)$ and then use its density in $H_0^1(G)$. Uniqueness therefore follows from Lemma 2.4. □

Remark 2.11 (What convexity supplies)**.** The only geometric input in the regularity proof is the strict inequality $\nu_1 = \pi/\omega > 1$ at each genuine polygonal vertex. It makes the symbol bound (2.15) positive. The argument allows arbitrary convex polygons and does not require a rectangular domain or angles bounded away from π uniformly across different polygons. Its constant may depend on the fixed polygon, exactly as in (2.1).

3 Meshes, traces, and constructive polynomial approximation

This section fixes all geometric and approximation constants used below. No lower bound on the ratio of the sizes of two nonadjacent elements is assumed. Throughout, $\Omega \subset \mathbb{R}^2$ is the bounded convex polygon in the statement of the problem.

3.1 Geometry and scaling

A triangulation $\mathcal{T}_h$ is a finite collection of nondegenerate closed triangles whose union is $\overline{\Omega}$, whose interiors are disjoint, and whose pairwise intersections are either empty, a common vertex, or a complete common edge. Boundary edges lie on $\partial\Omega$. We integrate over triangle interiors without changing their notation. The sets of interior and boundary edges are $\mathcal{E}_h^\circ$ and $\mathcal{E}_h^\partial$; put $\mathcal{E}_h = \mathcal{E}_h^\circ \cup \mathcal{E}_h^\partial$. Each interior edge belongs to exactly two triangles and each boundary edge to exactly one. Write

$$h_K = \operatorname{diam} K, \qquad h_e = |e|, \qquad h = \max_{K\in\mathcal{T}_h} h_K, \qquad \frac{h_K}{\rho_K} \le \kappa,$$

where ρ_K is the inradius and the same finite κ applies to every mesh in the family. This last inequality is our precise shape-regularity assumption. Constants depending on κ are independent of the number and sizes of the triangles.

Let s_K denote the semiperimeter and $a_{K,e}$ the altitude over e. The elementary area identities give

$$|K| = \rho_K s_K = \tfrac{1}{2} h_e a_{K,e}, \qquad s_K \ge h_K, \quad a_{K,e} \le h_K, \quad h_e \le h_K.$$

Consequently

$$\frac{h_K^2}{\kappa} \le |K| \le \frac{h_K^2}{2}, \qquad \frac{2h_K}{\kappa} \le h_e \le h_K, \qquad \frac{2h_K}{\kappa} \le a_{K,e} \le h_K. \tag{3.1}$$

For the last lower bound use $s_K \ge h_e$; for the edge lower bound use $a_{K,e} \le h_K$ and $s_K \ge h_K$. In particular, triangles sharing an edge have comparable diameters, with ratio at most $\kappa/2$. This is a local consequence of shape regularity, not a quasiuniformity assumption.

Use the fixed reference triangle $\widehat{K} = \{(\xi_1, \xi_2) : \xi_1 > 0, \xi_2 > 0, \xi_1 + \xi_2 < 1\}$. An affine vertex map $F_K(\xi) = b_K + B_K\xi$ maps its closure onto K. The two columns of B_K have length at most h_K, so

$$\|B_K\| \le \sqrt{2} h_K, \qquad |\det B_K| = 2|K|, \qquad \|B_K^{-1}\| \le \frac{\kappa}{\sqrt{2} h_K}.$$

Here and below matrix norms are Euclidean operator norms. Indeed, the small singular value of B_K is its determinant magnitude divided by its large singular value, and (3.1) supplies the lower bound. For w on K write $\widehat{w} = w \circ F_K$. Direct substitution in the integrals gives

$$\|w\|_{L^2(K)}^2 = |\det B_K| \|\widehat{w}\|_{L^2(\widehat{K})}^2, \qquad \|w\|_{L^2(e)}^2 = \frac{h_e}{|\widehat{e}|} \|\widehat{w}\|_{L^2(\widehat{e})}^2. \tag{3.2}$$

The second formula uses the constant speed of the affine map on an edge. The chain rule gives $\nabla\widehat{w} = B_K^T(\nabla w) \circ F_K$ and $D^2\widehat{w} = B_K^T(D^2 w \circ F_K) B_K$. The corresponding identity for the

order-j derivative tensor has one factor B_K^T for each derivative. In particular its Euclidean norm is bounded above by $\|B_K\|^j$ times the original tensor norm, and the converse uses $\|B_K^{-1}\|^j$. These identities first follow for smooth functions and then for Sobolev functions by smooth approximation and change of variables.

For clarity, all Sobolev seminorms in the proof use the ordered Cartesian derivatives:

$$|w|_{H^j(K)}^2 = \sum_{i_1,\dots,i_j=1}^{2} \|\partial_{i_1}\cdots\partial_{i_j} w\|_{L^2(K)}^2 \quad (j \ge 1), \qquad |w|_{H^0(K)} = \|w\|_{L^2(K)}. \tag{3.3}$$

Thus $|w|_{H^2(K)} = \|D^2 w\|_{L^2(K)}$ uses the Frobenius norm: the mixed derivative occurs twice. In multiindex notation the coefficient of $\|D^\alpha w\|^2$, $|\alpha| = j$, is $j!/\alpha!$; hence this convention is equivalent to the usual multiindex convention with constants depending only on j. Set $\|w\|_{H^s(K)}^2 = \sum_{j=0}^{s} |w|_{H^j(K)}^2$.

3.2 A direct Poincare inequality and a trace construction

We use the smooth approximation on convex sets proved in Lemma 2.3: if D is a bounded open convex set and $w \in H^m(D)$, there are functions smooth on neighborhoods of $\overline{D}$ converging to w in $H^m(D)$. Thus none of the following assertions requires pointwise values of a Sobolev function.

Lemma 3.1 (Mean-zero Poincare inequality)**.** *For a bounded open convex set $D \subset \mathbb{R}^2$, let $d = \operatorname{diam} D$ and $w_D = |D|^{-1}\int_D w$. Then*

$$\|w - w_D\|_{L^2(D)} \le \sqrt{2}\, d\, \|\nabla w\|_{L^2(D)} \qquad (w \in H^1(D)).$$

Proof. For a smooth w, the fundamental theorem of calculus on the segment from y to x, convexity, and Cauchy–Schwarz imply

$$|w(x) - w(y)|^2 \le d^2 \int_0^1 |\nabla w((1-t)y + tx)|^2\, dt.$$

For $0 \le t \le 1/2$, hold x fixed and change variables $z = (1-t)y + tx$ in the y integral. The Jacobian inverse is $(1-t)^{-2} \le 4$ and the transformed region is contained in D. For $1/2 \le t \le 1$ hold y fixed and change x instead; the Jacobian inverse is $t^{-2} \le 4$. Integrating in x, y, t therefore gives

$$\int_D \int_D |w(x) - w(y)|^2\, dx\, dy \le 4d^2 |D| \|\nabla w\|_{L^2(D)}^2.$$

Expansion of the square on the left gives exactly $2|D|\|w - w_D\|_{L^2(D)}^2$. Smooth approximation passes both the average and the two norms to the limit. □

Lemma 3.2 (Triangle traces)**.** *Restriction of smooth functions to an edge $e \subset \partial K$ extends uniquely to a bounded linear map $\operatorname{tr}_{K,e} : H^1(K) \longrightarrow L^2(e)$. It obeys*

$$\|\operatorname{tr}_{K,e} w\|_{L^2(e)}^2 \le \frac{\kappa}{2}\big(3h_K^{-1}\|w\|_{L^2(K)}^2 + h_K \|\nabla w\|_{L^2(K)}^2\big). \tag{3.4}$$

The same estimate summed over the components holds for vector fields.

Proof. Let a be the vertex opposite e. The vector $x - a$ is tangential to the other two edges, and $(x-a)\cdot n_K = a_{K,e}$ on e. For a smooth real-valued w, integration of $\operatorname{div}((x-a)w^2) = 2w^2 + 2w(x-a)\cdot\nabla w$ gives

$$a_{K,e}\int_e w^2 = 2\int_K w^2 + 2\int_K w(x-a)\cdot\nabla w \le 3\|w\|_{L^2(K)}^2 + h_K^2 \|\nabla w\|_{L^2(K)}^2.$$

The last step uses $|x-a| \le h_K$ and $2ab \le a^2+b^2$. For completeness, the integration identity used here is an elementary triangle identity: on $\widehat{K}$, integration of $\partial_1 F_1$ over $0<\xi_1<1-\xi_2$ gives the values on the sloping and vertical edges, and integration of $\partial_2 F_2$ over $0<\xi_2<1-\xi_1$ gives the values on the sloping and horizontal edges. Adding yields the outward normal flux, with the sloping-edge arc length factor canceling its normal's $1/\sqrt{2}$ factor. Affine change of variables gives the identity on K. Now use $a_{K,e} \ge 2h_K/\kappa$. If smooth $w_n \to w$ in $H^1(K)$, the estimate for $w_n - w_l$ shows that the restrictions form a Cauchy sequence in $L^2(e)$. Its limit defines the trace, is independent of the sequence by the same estimate, and satisfies (3.4). Uniqueness follows from density. □

Traces below are always these L^2 traces; there is no evaluation at vertices. If $w \in H^1(\Omega)$, its two traces on an interior edge coincide. Indeed, approximate w in $H^1(\Omega)$ by functions smooth near $\overline{\Omega}$, restrict to the two triangles, and use Lemma 3.2; each approximating function has equal traces. If $w \in H^2(\Omega)$, apply the same argument also to the two components of ∇w. If $w \in H^1_0(\Omega)$, approximate it by $C^\infty_c(\Omega)$ in $H^1(\Omega)$, as in the definition of H^1_0. Each approximation is zero on every boundary edge, so the trace of w is zero there. In particular,

$$[\![w]\!] = 0 \text{ on all edges for } w \in H^2(\Omega)\cap H^1_0(\Omega), \qquad [\![\nabla w]\!] = 0 \text{ on interior edges for } w \in H^2(\Omega), \tag{3.5}$$

with the jumps defined next.

3.3 Edge algebra and integration by parts

On an interior edge choose names K^+ and K^- for its triangles and let n^+, n^- be their outward unit normals. Thus $n^- = -n^+$. For scalar v and vector q, with one-sided traces denoted by superscripts, define

$$\{v\} = \tfrac{1}{2}(v^+ + v^-), \quad [\![v]\!] = v^+ n^+ + v^- n^-, \qquad \{q\} = \tfrac{1}{2}(q^+ + q^-), \quad [\![q]\!] = q^+ \cdot n^+ + q^- \cdot n^-.$$

These definitions are unchanged by exchanging the names + and −. On a boundary edge with exterior normal n, define $[\![v]\!] = vn$ and $\{q\} = q$. Scalar averages and vector jumps are needed only on interior edges; the second sum below is explicitly restricted to those edges. The notation ∇_h, div_h, and Δ_h means that the derivative is taken separately in each triangle.

Lemma 3.3 (Green identities with Sobolev traces)**.** *If $w \in H^1(K)$ and $q \in H^1(K)^2$, then*

$$\int_K \nabla w \cdot q + \int_K w \operatorname{div} q = \sum_{e\subset\partial K} \int_e (\operatorname{tr}_{K,e} w)(\operatorname{tr}_{K,e} q)\cdot n_K. \tag{3.6}$$

In particular, for $z \in H^2(K)$ and $w \in H^1(K)$,

$$\int_K \nabla z \cdot \nabla w = -\int_K (\Delta z) w + \sum_{e\subset\partial K} \int_e w\, \nabla z \cdot n_K.$$

Proof. For smooth functions apply the elementary triangle divergence identity proved in Lemma 3.2 to wq. Approximate w and the components of q in $H^1(K)$ by smooth functions. All volume integrals converge by Cauchy–Schwarz and the boundary integrals converge by the trace estimate and Cauchy–Schwarz on each edge. The specialization $q = \nabla z$ is permitted because $z \in H^2(K)$. □

Lemma 3.4 (Mesh Green identity)**.** *For piecewise H^1 scalar w and piecewise H^1 vector q,*

$$\sum_K \int_K (\nabla w \cdot q + w \operatorname{div} q) = \sum_{e\in\mathcal{E}_h} \int_e \{q\}\cdot[\![w]\!] + \sum_{e\in\mathcal{E}_h^\circ} \int_e [\![q]\!]\{w\}. \tag{3.7}$$

Proof. Sum (3.6). For an interior edge, set $n = n^+$; expansion gives the pointwise identity

$$w^+q^+ \cdot n^+ + w^-q^- \cdot n^- = \tfrac{1}{2}(q^+ + q^-) \cdot (w^+ - w^-)n + \tfrac{1}{2}(w^+ + w^-)(q^+ - q^-) \cdot n.$$

The single boundary contribution is $wq \cdot n = \{q\} \cdot [\![w]\!]$. All products are integrable because both factors are in $L^2(e)$. □

3.4 Polynomial inverse estimates and edge extensions

Write $\mathbb{P}_p(K)$ for polynomials of total degree at most the fixed integer $p \geq 1$, and set

$$V_h = \{v \in L^2(\Omega) : v|_K \in \mathbb{P}_p(K) \text{ for all } K\}, \qquad \Sigma_h = V_h^2.$$

We record explicitly the finite-dimensional argument behind the polynomial estimates. On $\widehat{K}$, express a polynomial in the monomial basis. Its squared $L^2(\widehat{K})$ norm is a positive definite quadratic form in its coefficients: if it is zero, continuity makes the polynomial zero in the open triangle, and successive one-variable polynomial identities make every coefficient zero. The Euclidean unit sphere of coefficients is compact, so this quadratic form has a positive minimum there. Every derivative norm and every edge L^2 norm is a continuous quadratic form on the same finite space, hence has a finite maximum on that sphere. Dividing the maxima by the positive minimum proves, for each fixed degree p,

$$\|\nabla \widehat{q}\|_{L^2(\widehat{K})} \leq C_p \|\widehat{q}\|_{L^2(\widehat{K})}, \qquad \|\widehat{q}\|_{L^2(\widehat{e})} \leq C_p \|\widehat{q}\|_{L^2(\widehat{K})}. \tag{3.8}$$

Take the maximum over the three reference edges. This argument is a proof of the needed norm comparisons, rather than an appeal to an unstated finite element estimate.

Lemma 3.5 (Polynomial inverse estimates)**.** *There is $C_{\mathrm{inv}} = C_{\mathrm{inv}}(p, \kappa)$ such that, for $q \in \mathbb{P}_p(K)$ and each edge $e \subset \partial K$,*

$$\|\nabla q\|_{L^2(K)} \leq C_{\mathrm{inv}} h_K^{-1} \|q\|_{L^2(K)}, \qquad \|q\|_{L^2(e)} \leq C_{\mathrm{inv}} h_K^{-1/2} \|q\|_{L^2(K)}. \tag{3.9}$$

There is $I = I(p, \kappa)$ for which

$$\sum_K h_K^2 \|D^2 v_h\|_{L^2(K)}^2 \leq I^2 \sum_K \|\nabla v_h\|_{L^2(K)}^2 \qquad (v_h \in V_h). \tag{3.10}$$

For $p = 1$ one may take $I = 0$.

Proof. The chain rule, (3.2), and (3.8) give

$$\|\nabla q\|_K^2 \leq |\det B_K| \|B_K^{-1}\|^2 C_p^2 \|\widehat{q}\|_{\widehat{K}}^2 \leq C(p, \kappa) h_K^{-2} \|q\|_K^2.$$

Similarly the squared edge estimate has scale factor $h_e/(|\widehat{e}|\,|\det B_K|) \leq C(\kappa) h_K^{-1}$. Apply the derivative estimate separately to $\partial_1 v_h$ and $\partial_2 v_h$, which have degree at most $p - 1$, and sum. Their gradient norms sum to the full Frobenius Hessian norm. Multiplication by h_K^2 and summation give (3.10). The Hessian of every affine polynomial is zero. □

Lemma 3.6 (An edge polynomial has a controlled extension)**.** *For $q \in \mathbb{P}_p(e)$ there is $Q \in \mathbb{P}_p(K)$ such that $Q|_e = q$ and*

$$\|Q\|_{L^2(K)} \leq C_{\mathrm{ext}}(p, \kappa) h_e^{1/2} \|q\|_{L^2(e)}. \tag{3.11}$$

This holds component by component for vector polynomials as well.

Proof. Choose the reference vertex ordering so that e is the image of the edge $\widehat{e} = \{(t,0) : 0 \le t \le 1\}$. If $\widehat{q}(t) = \sum_{j=0}^{p} c_j t^j$, define $\widehat{Q}(\xi_1, \xi_2) = \widehat{q}(\xi_1)$. This is a polynomial of total degree at most p with the required edge values, and direct integration even gives

$$\|\widehat{Q}\|^2_{L^2(\widehat{K})} = \int_0^1 (1-t)|\widehat{q}(t)|^2\,dt \le \|\widehat{q}\|^2_{L^2(\widehat{e})}.$$

Set $Q = \widehat{Q} \circ F_K^{-1}$. Since $|\det B_K| = h_e a_{K,e}$, scaling gives $\|Q\|_K^2 \le a_{K,e}\|q\|_e^2$. Finally $a_{K,e} \le h_K \le (\kappa/2)h_e$, proving the claim with $C_{\mathrm{ext}} = \sqrt{\kappa/2}$, in fact independent of p for this particular extension. No constraint is imposed on the other two edge traces, nor is one needed for the lifting argument. □

3.5 The broken norm and the trace bounds it controls

Let $H^2(\mathcal{T}_h)$ be the space of $L^2(\Omega)$ functions whose restriction to each K belongs to $H^2(K)$. For v in this space define

$$G_h(v)^2 = \sum_K \|\nabla v\|^2_{L^2(K)}, \qquad H_h(v)^2 = \sum_K h_K^2 \|D^2 v\|^2_{L^2(K)},$$
$$J_h(v)^2 = \sum_{e \in \mathcal{E}_h} h_e^{-1} \|[\![v]\!]\|^2_{L^2(e)}, \qquad E_h(v)^2 = G_h(v)^2 + J_h(v)^2,$$
$$N_h(v)^2 = G_h(v)^2 + H_h(v)^2 + J_h(v)^2.$$

The edge norm of a vector jump is its Euclidean L^2 norm. Boundary edges are included in J_h; this is essential for definiteness.

Lemma 3.7 (Definiteness)**.** *N_h is a norm on $H^2(\mathcal{T}_h)$, and hence on*

$$W_h := V_h + \big(H^2(\Omega) \cap H_0^1(\Omega)\big).$$

Also E_h is a norm on $H^2(\mathcal{T}_h)$.

Proof. Each expression is the norm of a vector of linear maps into a finite direct sum of L^2 spaces, so homogeneity and the triangle inequality hold. If $E_h(v) = 0$, then $\nabla(v|_K) = 0$ and Lemma 3.1 implies $v|_K = c_K$ almost everywhere. The zero interior jump forces the constants on neighboring triangles to agree. The triangle adjacency graph is connected: choose a point strictly inside the first triangle and outside the finitely many supporting lines of mesh edges. Choose a point strictly inside the second triangle and outside the finitely many lines joining the first point to mesh vertices. The connecting segment meets no mesh vertex and is contained in no edge line. Convexity puts the segment in Ω; its crossings then give a finite chain of triangles sharing whole edges. There is a boundary edge because the nonempty polygon boundary is the union of boundary edges. The zero boundary jump forces the adjacent constant to vanish, and connectedness forces all c_K to vanish. Thus E_h, and therefore N_h, is definite. □

Lemma 3.8 (Gradient traces and the discrete norm)**.** *There is $T = T(\kappa)$ such that*

$$F_h(v)^2 := \sum_K \sum_{e \subset \partial K} h_e \|\nabla(v|_K)\|^2_{L^2(e)} \le T^2\big(G_h(v)^2 + H_h(v)^2\big) \qquad (v \in H^2(\mathcal{T}_h)). \tag{3.12}$$

In particular

$$\sum_{e \in \mathcal{E}_h} h_e \|\{\nabla_h v\}\|^2_{L^2(e)} \le F_h(v)^2, \qquad \sum_{e \in \mathcal{E}_h^\circ} h_e \|[\![\nabla_h v]\!]\|^2_{L^2(e)} \le 2F_h(v)^2. \tag{3.13}$$

On V_h the enhanced norm obeys

$$E_h(v_h) \le N_h(v_h) \le \sqrt{1 + I^2}\, E_h(v_h). \tag{3.14}$$

Proof. Apply (3.4) to each component of ∇v, multiply by $h_e \le h_K$, and add over the three edges of K. This gives

$$\sum_{e\subset\partial K} h_e\|\nabla v\|_e^2 \le \frac{3\kappa}{2}\big(3\|\nabla v\|_K^2 + h_K^2\|D^2 v\|_K^2\big),$$

so $T^2 = 9\kappa/2$ is one permissible choice. The first bound in (3.13) follows on an interior edge from $|(a+b)/2|^2 \le (|a|^2+|b|^2)/2$ and on a boundary edge by its one-sided definition. The second uses $|a\cdot n^+ + b\cdot n^-|^2 \le 2(|a|^2+|b|^2)$. Finally (3.14) follows from (3.10) and the definitions. □

3.6 A polynomial constructed by derivative averages

The construction here avoids point values and makes the $p=1$, H^2 case particularly transparent. For an integer $m \ge 1$ and $u \in H^{m+1}(K)$, define $\Pi_K^m u \in \mathbb{P}_m(K)$ by

$$\int_K D^\alpha(u - \Pi_K^m u) = 0 \qquad \text{for every multiindex } \alpha \text{ with } |\alpha| \le m. \tag{3.15}$$

Here $D^\alpha = \partial_1^{\alpha_1}\partial_2^{\alpha_2}$ and $\alpha! = \alpha_1!\alpha_2!$. To see existence constructively, fix any $x_0 \in K$, start with $P = 0$, and for $k = m, m-1, \dots, 0$ replace P by

$$P + \sum_{|\alpha|=k} \frac{(x-x_0)^\alpha}{\alpha!}\frac{1}{|K|}\int_K D^\alpha(u-P).$$

At degree k, the new summand's αth derivative is the specified constant for $|\alpha| = k$; all derivatives of higher order vanish. Thus this step enforces the degree-k moment equations without disturbing equations enforced at earlier steps. After the last step all equations hold. For uniqueness, a polynomial in the kernel has constant order-m derivatives with zero means, so its degree is at most $m-1$; descending in degree shows that it is zero. This also shows that the construction is independent of x_0, is linear in u, and reproduces every polynomial of degree at most m.

Lemma 3.9 (Local polynomial approximation)**.** *For $0 \le j \le m+1$,*

$$|u - \Pi_K^m u|_{H^j(K)} \le (\sqrt{2}h_K)^{m+1-j}|u|_{H^{m+1}(K)}. \tag{3.16}$$

In particular, with $m = p$, the local L^2, H^1 seminorm, and H^2 seminorm errors have orders h_K^{p+1}, h_K^p, and h_K^{p-1} respectively. For full Sobolev norms and $0 \le s \le 2$,

$$\|u - \Pi_K^p u\|_{H^s(K)} \le C(p, s, \operatorname{diam}\Omega) h_K^{p+1-s}|u|_{H^{p+1}(K)}. \tag{3.17}$$

No shape-regularity constant is needed in (3.16) itself.

Proof. Put $z = u - \Pi_K^m u$. Every derivative $D^\alpha z$ of order at most m has mean zero by (3.15). Apply Lemma 3.1 to each ordered derivative of order $j \le m$ and add its square over all ordered indices. The convention (3.3) gives exactly

$$|z|_{H^j(K)}^2 \le 2h_K^2|z|_{H^{j+1}(K)}^2.$$

At order $m+1$ the polynomial differentiates to zero, so $|z|_{H^{m+1}(K)} = |u|_{H^{m+1}(K)}$. Iterating proves (3.16), including its endpoint $j = m+1$. The full-norm statement follows by summing the squares for $0 \le j \le s$ and using $h_K \le \operatorname{diam}\Omega$; for example the squared constant is bounded by $\sum_{j=0}^s 2^{p+1-j}(\operatorname{diam}\Omega)^{2(s-j)}$. When $p = 1, s = 2$, the highest term is the exact identity $\|D^2(u - \Pi_K^1 u)\|_K = \|D^2 u\|_K$, so this case requires no additional regularity or interpretation. □

For $1 \le m \le p$ define the discontinuous polynomial $\Pi_h^m u \in V_h$ by $(\Pi_h^m u)|_K = \Pi_K^m(u|_K)$. The values on neighboring triangles are not matched.

Lemma 3.10 (Approximation in the enhanced DG norm)**.** *Let $1 \le m \le p$ and $u \in H^{m+1}(\Omega) \cap H_0^1(\Omega)$. Then*

$$N_h(u - \Pi_h^m u)^2 \le C_{\mathrm{app}}(m,\kappa)^2 \sum_K h_K^{2m} |u|^2_{H^{m+1}(K)} \le C_{\mathrm{app}}(m,\kappa)^2 h^{2m} |u|^2_{H^{m+1}(\Omega)}. \tag{3.18}$$

In particular $m = p$ gives the primal order h^p, whereas $m = 1$ gives the order-h dual approximation for every fixed $p \ge 1$:

$$\inf_{z_h \in V_h} N_h(z - z_h) \le C_{\mathrm{app}}(1,\kappa) h \, \|D^2 z\|_{L^2(\Omega)} \quad (z \in H^2(\Omega) \cap H_0^1(\Omega)). \tag{3.19}$$

Proof. Write $\eta = u - \Pi_h^m u$. Formula (3.16), with $j = 1, 2$, gives

$$\|\nabla \eta\|_K^2 + h_K^2 \|D^2 \eta\|_K^2 \le \big(2^m + 2^{m-1}\big) h_K^{2m} |u|^2_{H^{m+1}(K)}.$$

This formula also holds for $m = 1$, using the endpoint $j = 2$. On an interior edge, $|[\![\eta]\!]|^2 = |\eta^+ - \eta^-|^2 \le 2(|\eta^+|^2 + |\eta^-|^2)$; on a boundary edge $|[\![\eta]\!]|^2 = |\eta|^2$. Hence

$$J_h(\eta)^2 \le 2 \sum_K \sum_{e \subset \partial K} h_e^{-1} \|\eta|_K\|_e^2.$$

Use $h_e^{-1} \le \kappa/(2h_K)$ and the trace estimate to bound the right-hand side by

$$C(\kappa) \sum_K \big(h_K^{-2} \|\eta\|_K^2 + \|\nabla \eta\|_K^2\big) \le C(m,\kappa) \sum_K h_K^{2m} |u|^2_{H^{m+1}(K)},$$

where the last step uses (3.16) with $j = 0, 1$. Adding proves the first estimate, and $h_K \le h$ proves the second. Taking $\Pi_h^1 z$ in the infimum proves (3.19). All edge estimates were made with the size of an incident triangle, so arbitrary grading away from neighboring elements is allowed. □

The last proof actually applies to any piecewise H^{m+1} function when its error is formed element by element. The global membership and zero-boundary hypotheses stated above ensure, in addition, that the errors used later belong to W_h and that the exact solution has vanishing scalar jumps as in (3.5).

4 Flux elimination and the four stable methods

All vector spaces and bilinear forms in this section are real. The mesh is a finite conforming triangulation of the convex polygon Ω, with $h_K/\rho_K \le \kappa$ as specified above, and $p \ge 1$ is fixed. Write

$$X = H^2(\mathcal{T}_h), \qquad W_h = V_h + (H^2(\Omega) \cap H_0^1(\Omega)), \qquad V_h = \prod_{K \in \mathcal{T}_h} \mathbb{P}_p(K), \quad \Sigma_h = \prod_{K \in \mathcal{T}_h} \mathbb{P}_p(K)^2.$$

The products here identify functions that agree almost everywhere on each element; consequently they are subspaces of $L^2(\Omega)$ or its vector analogue. All traces below are Sobolev traces, and all equalities on edges are almost-everywhere equalities. The sets of all, interior, and boundary edges are $\mathcal{E}_h$, $\mathcal{E}_h^\circ$, and $\mathcal{E}_h^\partial$. The length of e is h_e and the diameter of K is h_K. For an interior edge with adjacent outward normals $n^+, n^- = -n^+$, set

$$\{v\} = \tfrac{1}{2}(v^+ + v^-), \quad [\![v]\!] = v^+ n^+ + v^- n^-, \quad \{\tau\} = \tfrac{1}{2}(\tau^+ + \tau^-), \quad [\![\tau]\!] = \tau^+ \cdot n^+ + \tau^- \cdot n^-.$$

On a boundary edge set $[\![v]\!] = vn$ and $\{\tau\} = \tau$. Scalar averages and vector jumps are used only on interior edges. These definitions are independent of interchanging the labels $+$ and $-$. We use $(\cdot,\cdot)$ for the $L^2(\Omega)$ pairing, with a Euclidean dot product when its arguments are vector fields, and $\langle \cdot, \cdot \rangle_e$ for the analogous edge pairing.

4.1 Liftings and their quantitative properties

Lemma 4.1 (Construction, locality, and summation of liftings)**.** *For every $e \in \mathcal{E}_h$ and $\phi \in L^2(e)^2$ there is exactly one $r_e(\phi) \in \Sigma_h$ such that*

$$(r_e(\phi), \tau) = -\langle \phi, \{\tau\} \rangle_e \qquad (\tau \in \Sigma_h). \tag{4.1}$$

For $e \in \mathcal{E}_h^\circ$ and $q \in L^2(e)$ there is exactly one $l_e(q) \in \Sigma_h$ such that

$$(l_e(q), \tau) = -\langle q, [\![\tau]\!] \rangle_e \qquad (\tau \in \Sigma_h). \tag{4.2}$$

These maps are linear. Each of their outputs vanishes on elements not incident to e. For skeleton fields define

$$r(\phi) = \sum_{e \in \mathcal{E}_h} r_e(\phi|_e), \qquad l(q) = \sum_{e \in \mathcal{E}_h^\circ} l_e(q|_e).$$

They are the unique elements of Σ_h satisfying the sums of the right-hand sides of (4.1) and (4.2), respectively. Moreover,

$$\|r(\phi)\|^2 \le 3 \sum_e \|r_e(\phi)\|^2, \qquad \|l(q)\|^2 \le 3 \sum_{e \in \mathcal{E}_h^\circ} \|l_e(q)\|^2. \tag{4.3}$$

Proof. Fix any basis $\tau_1, \dots, \tau_d$ of Σ_h. Its mass matrix $M_{ij} = (\tau_j, \tau_i)$ is invertible: if $a \neq 0$, the basis property gives $\sum_j a_j \tau_j \neq 0$, hence $a^T M a = \| \sum_j a_j \tau_j \|^2 > 0$. The edge functionals in the statement are finite by Cauchy–Schwarz, since polynomials have square-integrable edge restrictions. Solving the system $Ma = b$, where b_i is the corresponding edge functional at τ_i, gives existence. Pairing the difference of two solutions with itself gives uniqueness. Addition and scalar multiplication of the defining identities, followed by uniqueness, give linearity.

If an element K is not incident to e, test with the vector field equal to the lifting on K and zero on all other elements. It belongs to Σ_h, and its relevant edge trace is zero. The identity gives the squared $L^2(K)$ norm of the lifting equal to zero. This proves locality. Summing the identities proves the assertions about r and l and their uniqueness. On each triangle at most three summands can be nonzero. For three vectors, $|a_1 + a_2 + a_3|^2 \le 3 \sum_i |a_i|^2$, by Cauchy–Schwarz in the three-term sum. Apply this inequality pointwise, integrate over the triangle, and sum over the mesh to obtain (4.3). □

Lemma 4.2 (Edge norm equivalence)**.** *There are positive numbers c_-, c_+, depending only on p and κ, such that, for every edge and every $\phi \in \mathbb{P}_p(e)^2$,*

$$c_- h_e^{-1} \|\phi\|_e^2 \le \|r_e(\phi)\|^2 \le c_+ h_e^{-1} \|\phi\|_e^2. \tag{4.4}$$

The constants can be chosen uniformly over all meshes with these bounds.

Proof. The polynomial trace inequality from Lemma 3.5, together with edge–element comparability, supplies $C_{\mathrm{tr}} > 0$ such that

$$\|\tau\|_e \le C_{\mathrm{tr}} h_e^{-1/2} \|\tau\|_K \quad (\tau \in \mathbb{P}_p(K)^2, e \subset \partial K).$$

For an interior edge, $\|\{\tau\}\|_e \le \frac{1}{2}(\|\tau^+\|_e + \|\tau^-\|_e) \le C_{\mathrm{tr}} h_e^{-1/2} \|\tau\|$; the boundary case satisfies the same final bound. Consequently the functional in (4.1) is at most $C_{\mathrm{tr}} h_e^{-1/2} \|\phi\|_e \|\tau\|$ in absolute value. Taking $\tau = r_e(\phi)$ proves the upper bound with $c_+ = C_{\mathrm{tr}}^2$; when that lifting is zero the bound needs no division.

For the lower bound choose one adjacent triangle K. The polynomial extension of Lemma 3.6, applied componentwise, gives $Q \in \mathbb{P}_p(K)^2$ such that $Q|_e = \phi$ and $\|Q\|_K \le$

$C_{\mathrm{ext}} h_e^{1/2} \|\phi\|_e$. Extend Q by zero to the other elements. Its average on e is $a_e \phi$, where $a_e = 1/2$ for an interior edge and $a_e = 1$ for a boundary edge. Thus

$$a_e \|\phi\|_e^2 = |(r_e(\phi), Q)| \le \|r_e(\phi)\| C_{\mathrm{ext}} h_e^{1/2} \|\phi\|_e.$$

If $\phi = 0$ the desired lower bound holds directly; otherwise divide by $C_{\mathrm{ext}} h_e^{1/2} \|\phi\|_e$ and square. One may take $c_- = 1/(4C_{\mathrm{ext}}^2)$. □

For $v \in W_h$ set

$$G_h(v)^2 = \sum_K \|\nabla v\|_K^2, \qquad H_h(v)^2 = \sum_K h_K^2 \|D^2 v\|_K^2,$$
$$J_h(v)^2 = \sum_e h_e^{-1} \|[\![v]\!]\|_e^2, \qquad E_h(v)^2 = G_h(v)^2 + J_h(v)^2,$$
$$N_h(v)^2 = G_h(v)^2 + H_h(v)^2 + J_h(v)^2, \qquad S_h(v)^2 = \sum_e \|r_e([\![v]\!])\|^2.$$

Here $D^2 v$ is the full ordered Cartesian Hessian and its pointwise norm is the Frobenius norm. In particular the mixed derivative occurs twice. If $v = v_h + z$ with $z \in H^2(\Omega) \cap H_0^1(\Omega)$, the trace properties proved in the preceding section give $[\![z]\!] = 0$ on every edge. Hence $[\![v]\!] = [\![v_h]\!]$ is a vector polynomial of degree at most p, independent of the chosen decomposition. It follows that

$$c_- J_h(v)^2 \le S_h(v)^2 \le c_+ J_h(v)^2 \qquad (v \in W_h). \tag{4.5}$$

Thus the lifting-based norm $\big(G_h(v)^2 + H_h(v)^2 + S_h(v)^2\big)^{1/2}$ and N_h are equivalent, with squared equivalence factors $\min\{1, c_-\}$ and $\max\{1, c_+\}$. This argument is deliberately restricted to W_h: arbitrary broken H^2 traces need not be polynomial, and the lower lifting bound does not hold for arbitrary $L^2(e)$ data. Both E_h and N_h are norms by Lemma 3.7. The inverse estimate (3.14) gives a mesh-independent $I \ge 0$ such that

$$H_h(v_h)^2 \le I^2 G_h(v_h)^2, \qquad E_h(v_h)^2 \le N_h(v_h)^2 \le (1 + I^2) E_h(v_h)^2. \tag{4.6}$$

Let $\beta_e \in \mathbb{R}^2$ be constant on each interior edge and suppose $|\beta_e| \le b$. Define

$$R(v) = r([\![v]\!]), \qquad L_\beta(v) = l(\beta \cdot [\![v]\!]), \qquad A_\beta(v) = R(v) + L_\beta(v). \tag{4.7}$$

These linear maps are defined even for $v \in X$.

Lemma 4.3 (Bounds for global liftings)**.** *For $v \in W_h$,*

$$\|R(v)\|^2 \le 3 S_h(v)^2, \qquad \|L_\beta(v)\|^2 \le 12 b^2 S_h(v)^2, \qquad \|A_\beta(v)\| \le \sqrt{3}(1 + 2b) S_h(v). \tag{4.8}$$

Proof. The first inequality is (4.3). Fix an interior edge and a unit normal n from one adjacent triangle. Write $[\![v]\!] = qn$ on this edge. On either adjacent element K, testing against fields supported on K gives

$$l_e(q)|_K = 2 r_e(q n_K)|_K.$$

Indeed the two defining functionals on such fields are respectively $-\langle q, \tau \cdot n_K \rangle_e$ and $-\frac{1}{2}\langle q n_K, \tau \rangle_e$. Their equality after multiplying the second by two and the elementwise uniqueness argument in Lemma 4.1 prove this identity. Since $\beta \cdot [\![v]\!] = (\beta_e \cdot n) q$, with the coefficient constant on e, linearity gives, on K,

$$l_e(\beta \cdot [\![v]\!])|_K = 2(\beta_e \cdot n)(n_K \cdot n) r_e(qn)|_K.$$

The sign $n_K \cdot n$ has absolute value one. Squaring, integrating, and adding the two elements shows $\|l_e(\beta \cdot [\![v]\!])\|^2 \le 4b^2 \|r_e([\![v]\!])\|^2$. Now apply the second inequality in (4.3) and sum over the interior edges. The last estimate follows from the triangle inequality and the square roots of the first two estimates. □

4.2 General linear fluxes: exact domains and implications

Let $\mathcal{T} = \prod_K L^2(\partial K)$ be the space of possibly double-valued scalar traces. Assume given linear maps

$$U : H^1(\mathcal{T}_h) \longrightarrow \mathcal{T}, \qquad Q : X \times H^1(\mathcal{T}_h)^2 \longrightarrow \mathcal{T}^2.$$

Here Q is jointly linear in the pair (w, τ), not bilinear. For $w \in X$ define the extended gradient

$$\mathcal{G}(w) = \nabla_h w - r(\llbracket U(w) - w \rrbracket) - l(\{U(w) - w\}). \tag{4.9}$$

The second lifting uses interior edges only. This is a well-defined linear map $X \to H^1(\mathcal{T}_h)^2$ because the gradient of a broken H^2 function is broken H^1 and each lifting is piecewise polynomial. It need not map X into Σ_h. Write $\widehat{\sigma}(w) = Q(w, \mathcal{G}(w))$. Define the extended primal form on $X \times X$ by

$$\begin{aligned} B(w, v) = (\nabla_h w, \nabla_h v) &+ \sum_e \langle \llbracket U(w) - w \rrbracket, \{\nabla_h v\} \rangle_e - \sum_e \langle \{\widehat{\sigma}(w)\}, \llbracket v \rrbracket \rangle_e \\ &+ \sum_{e \in \mathcal{E}_h^\circ} \langle \{U(w) - w\}, \llbracket \nabla_h v \rrbracket \rangle_e - \sum_{e \in \mathcal{E}_h^\circ} \langle \llbracket \widehat{\sigma}(w) \rrbracket, \{v\} \rangle_e. \end{aligned} \tag{4.10}$$

Every edge pairing is finite by the trace result for broken H^1 functions. Compositions of the displayed linear maps are linear, and each pairing is bilinear; therefore B is a bilinear form. No boundedness assertion for arbitrary U, Q is being inferred from algebraic linearity.

Theorem 4.4 (Equivalence of the flux and primal formulations)**.** *For $f \in L^2(\Omega)$, a pair $(u_h, \sigma_h) \in V_h \times \Sigma_h$ satisfies, on every element,*

$$(\sigma_h, \tau)_K = -(u_h, \nabla \cdot \tau)_K + \langle U(u_h)|_{\partial K}, \tau \cdot n_K \rangle_{\partial K} \qquad (\tau \in \mathbb{P}_p(K)^2), \tag{4.11}$$

$$(\sigma_h, \nabla v)_K = (f, v)_K + \langle Q(u_h, \sigma_h)|_{\partial K} \cdot n_K, v \rangle_{\partial K} \qquad (v \in \mathbb{P}_p(K)), \tag{4.12}$$

if and only if $\sigma_h = \mathcal{G}(u_h)$ and

$$B(u_h, v_h) = (f, v_h) \qquad (v_h \in V_h). \tag{4.13}$$

Proof. For arbitrary elementwise traces a, τ, expansion on an interior edge using $n^- = -n^+$ gives

$$a^+\tau^+ \cdot n^+ + a^-\tau^- \cdot n^- = \llbracket a \rrbracket \cdot \{\tau\} + \{a\} \llbracket \tau \rrbracket.$$

On a boundary edge the corresponding expression is $\llbracket a \rrbracket \cdot \{\tau\}$. Thus summing element boundary pairings gives the sums of these two edge pairings. Integration by parts in the first flux equation, followed by this identity with $a = U(u_h) - u_h$, shows that its assembled form is

$$(\sigma_h, \tau) = (\nabla_h u_h, \tau) + \sum_e \langle \llbracket U(u_h) - u_h \rrbracket, \{\tau\} \rangle_e + \sum_{e \in \mathcal{E}_h^\circ} \langle \{U(u_h) - u_h\}, \llbracket \tau \rrbracket \rangle_e.$$

The lifting identities identify the right-hand side with $(\mathcal{G}(u_h), \tau)$. Both σ_h and $\mathcal{G}(u_h)$ belong to Σ_h, since $\nabla_h V_h \subset \Sigma_h$. Testing with their difference proves equality. Conversely this equality implies the assembled first equation, which implies each local equation by choosing tests supported on one element.

Assembling the second flux equation gives

$$(\mathcal{G}(u_h), \nabla_h v_h) - \sum_e \langle \{\widehat{\sigma}(u_h)\}, \llbracket v_h \rrbracket \rangle_e - \sum_{e \in \mathcal{E}_h^\circ} \langle \llbracket \widehat{\sigma}(u_h) \rrbracket, \{v_h\} \rangle_e = (f, v_h).$$

Since $\nabla_h v_h \in \Sigma_h$, the lifting identities expand the first term into precisely the gradient and the two $U(u_h) - u_h$ terms in (4.10). This is (4.13). Each algebraic equality can be read in reverse; localization of the assembled second equation then proves all local second equations. □

Remark 4.5 (The projection issue)**.** For a nondiscrete $w \in X$, solving only the first flux equation with w in place of u_h and demanding an output in Σ_h gives

$$\sigma_h^{\mathrm{proj}}(w) = \Pi_{\Sigma_h} \nabla_h w - r(\llbracket U(w) - w \rrbracket) - l(\{U(w) - w\}) = \Pi_{\Sigma_h} \mathcal{G}(w).$$

Here Π_{Σ_h} is the L^2 orthogonal projection, constructed by the same invertible mass matrix as in Lemma 4.1. If $U(w)$ is the exact trace, this projected field is $\Pi_{\Sigma_h} \nabla_h w$, generally different from $\nabla_h w$. In contrast, (4.9) is deliberately an extension into broken H^1, for which $\mathcal{G}(w) = \nabla_h w$ in this case. It agrees with the discrete elimination on V_h. Consistency below uses this specified extension and never asserts that the exact gradient belongs to Σ_h.

Theorem 4.6 (Consistent fluxes imply consistency and orthogonality)**.** *Let $u \in H^2(\Omega) \cap H_0^1(\Omega)$ satisfy $-\Delta u = f$ in $L^2(\Omega)$. Suppose $U(u)|_{\partial K} = u|_{\partial K}$ and $Q(u, \nabla u)|_{\partial K} \cdot n_K = \nabla u \cdot n_K$ on each element boundary. Then*

$$B(u, v) = (f, v) \qquad (v \in X).$$

If u_h solves (4.13), then $B(u - u_h, v_h) = 0$ for every $v_h \in V_h$.

Proof. The first assumption makes both lifting inputs in (4.9) zero, so $\mathcal{G}(u) = \nabla u$. The two terms involving $U(u) - u$ in (4.10) vanish. Combining its remaining flux terms by the edge algebra in the preceding proof replaces their sum by $-\sum_K \langle Q(u, \nabla u) \cdot n_K, v \rangle_{\partial K}$. The assumption on the normal component changes this to $-\sum_K \langle \nabla u \cdot n_K, v \rangle_{\partial K}$. Elementwise Green's formula, Lemma 3.3, now gives $B(u, v) = \sum_K (-\Delta u, v)_K = (f, v)$. Subtract (4.13) and use bilinearity to obtain orthogonality. Tangential agreement of the vector flux is unnecessary. □

Theorem 4.7 (Conservation and the boundary condition imply adjoint consistency)**.** *Suppose, for all $w \in X$, the scalar trace $U(w)$ is single-valued on interior edges and zero on boundary edges, and*

$$\llbracket Q(w, \mathcal{G}(w)) \rrbracket = 0 \quad \text{on all interior edges.}$$

For every $\psi \in H^2(\Omega) \cap H_0^1(\Omega)$ satisfying $-\Delta \psi = g \in L^2(\Omega)$,

$$B(w, \psi) = (w, g) \qquad (w \in X).$$

Only conservation of the vector flux's normal component is required.

Proof. The global trace identities give $\llbracket \psi \rrbracket = 0$, $\llbracket \nabla \psi \rrbracket = 0$ on interior edges, and $\{\nabla \psi\} = \nabla \psi$. Green's formula, this time integrating the gradient of ψ, gives

$$(\nabla_h w, \nabla \psi) = (w, g) + \sum_e \langle \llbracket w \rrbracket, \nabla \psi \rangle_e.$$

Substitute these identities into (4.10). The result is

$$B(w, \psi) = (w, g) + \sum_e \langle \llbracket U(w) \rrbracket, \nabla \psi \rangle_e - \sum_{e \in \mathcal{E}_h^\circ} \langle \llbracket Q(w, \mathcal{G}(w)) \rrbracket, \psi \rangle_e.$$

The scalar jump vanishes on interior edges by single-valuedness and on boundary edges because $U(w) = 0$ there. The final sum vanishes by normal conservation. This proves the assertion. □

Remark 4.8 (The statements are implications)**.** Full vector single-valuedness is sufficient for normal conservation but is stronger than needed. Adding an arbitrary element-side tangential field to Q changes neither flux equations nor the primal form, although it can destroy full vector single-valuedness or full-vector consistency. Thus the corresponding full-vector converses cannot be deduced from primal consistency or adjoint consistency. No converse to Theorems 4.6–4.7 is asserted. In particular, interior scalar single-valuedness alone would leave the boundary term $\sum_{e \subset \partial\Omega} \langle U(w), \partial_n \psi \rangle_e$ in the adjoint identity; the boundary condition in Theorem 4.7 is a genuine hypothesis.

Normal conservation also gives element-union conservation for a flux solution: take the test equal to one on an arbitrary collection of elements and zero elsewhere in (4.12). Its elementwise gradient is zero. Each internal edge of the collection cancels, so

$$\int_S f + \int_{\partial S} \widehat{\sigma} \cdot n_S = 0,$$

where the last expression denotes the sum over boundary edges of the collection, with their outward element normals. This interpretation also covers collections meeting only at vertices.

4.3 Four concrete fluxes and extended primal forms

Fix numbers $0 < \eta_{\min} \le \eta_e \le \eta_{\max} < \infty$ for every edge. For a skeleton vector field ϕ define

$$(\alpha_j \phi)|_e = \eta_e h_e^{-1} \phi|_e, \qquad (\alpha_r \phi)|_e = -\eta_e \{r_e(\phi|_e)\}|_e.$$

The average in the second expression is a single vector field on an edge, including the boundary convention. Define symmetric bilinear forms

$$A_j(w, v) = \sum_e \eta_e h_e^{-1} \langle [\![w]\!], [\![v]\!] \rangle_e, \tag{4.14}$$

$$A_r(w, v) = \sum_e \eta_e (r_e([\![w]\!]), r_e([\![v]\!])). \tag{4.15}$$

The lifting identity shows $\sum_e \langle \alpha_r([\![w]\!]), [\![v]\!] \rangle_e = A_r(w, v)$: on edge e, use $\tau = r_e([\![w]\!])$ in the identity for $r_e([\![v]\!])$ and multiply by η_e. Also set, on $X \times X$,

$$C(w, v) = \sum_e \langle \{\nabla_h w\}, [\![v]\!] \rangle_e, \tag{4.16}$$

$$D_\beta(w, v) = \sum_{e \in \mathcal{E}_h^\circ} \langle [\![\nabla_h w]\!], \beta \cdot [\![v]\!] \rangle_e, \tag{4.17}$$

$$B_0(w, v) = (\nabla_h w, \nabla_h v) - C(w, v) - C(v, w). \tag{4.18}$$

In the following flux specifications each displayed vector flux is used on both sides of an interior edge. All scalar fluxes are zero on boundary edges. Terms involving β are omitted on boundary edges:

method	$U(w)$ on interior edges	$Q(w, \tau)$
IP	$\{w\}$	$\{\nabla_h w\} - \alpha_j([\![w]\!])$
Bassi et al. [13]	$\{w\}$	$\{\nabla_h w\} - \alpha_r([\![w]\!])$
Brezzi et al. [22]	$\{w\}$	$\{\tau\} - \alpha_r([\![w]\!])$
LDG	$\{w\} - \beta \cdot [\![w]\!]$	$\{\tau\} + \beta [\![\tau]\!] - \alpha_j([\![w]\!])$

These are linear maps with exactly the domains used above. The first two ignore their second argument. For all four methods the scalar and vector fluxes are single-valued on interior edges. At a global $u \in H^2(\Omega) \cap H_0^1(\Omega)$ the jumps of u and the interior normal jump of ∇u vanish; the trace identities then show $U(u) = u$ and $Q(u, \nabla u) = \nabla u$. Thus all four are consistent and adjoint consistent by the preceding theorems, including their boundary hypotheses.

Proposition 4.9 (Exact extended forms)**.** *The primal forms produced by these fluxes, on all of $X \times X$, are*

$$B_{\mathrm{IP}}(w, v) = B_0(w, v) + A_j(w, v), \tag{4.19}$$

$$B_{13}(w, v) = B_0(w, v) + A_r(w, v), \tag{4.20}$$

$$B_{22}(w, v) = B_0(w, v) + (R(w), R(v)) + A_r(w, v), \tag{4.21}$$

$$B_{\mathrm{LDG}}(w, v) = B_0(w, v) - D_\beta(w, v) - D_\beta(v, w) + (A_\beta(w), A_\beta(v)) + A_j(w, v). \tag{4.22}$$

In particular these forms are bilinear and symmetric on $X \times X$.

Proof. For the first three scalar fluxes, $[\![U(w) - w]\!] = -[\![w]\!]$ on all edges and $\{U(w) - w\} = 0$ on interior edges, so $\mathcal{G}(w) = \nabla_h w + R(w)$. Their vector flux jumps vanish. Substitution in (4.10), followed by (4.14) or (4.15), gives the first two forms. For method [22], the extra vector-flux term is $-\sum_e \langle \{R(w)\}, [\![v]\!] \rangle_e$. Since $R(w) \in \Sigma_h$, the definition of $R(v)$ identifies it with $(R(v), R(w))$ even when w, v are not discrete. This proves the third formula.

For LDG the scalar identities are instead $[\![U(w) - w]\!] = -[\![w]\!]$ and $\{U(w) - w\} = -\beta \cdot [\![w]\!]$, whence $\mathcal{G}(w) = \nabla_h w + A_\beta(w)$. The scalar-flux correction terms in (4.10) are $-C(v, w) - D_\beta(v, w)$. Expanding $Q(w, \mathcal{G}(w))$ gives

$$\{\nabla_h w\} + \beta[\![\nabla_h w]\!] + \{A_\beta(w)\} + \beta[\![A_\beta(w)]\!] - \alpha_j([\![w]\!]).$$

The first two terms, with the minus sign in the primal form, contribute $-C(w, v) - D_\beta(w, v)$, and the last contributes $A_j(w, v)$. The two middle terms contribute

$$-\sum_e \langle \{A_\beta(w)\}, [\![v]\!] \rangle_e - \sum_{e \in \mathcal{E}_h^\circ} \langle [\![A_\beta(w)]\!], \beta \cdot [\![v]\!] \rangle_e = (A_\beta(v), A_\beta(w)),$$

by the two lifting identities, with the admissible polynomial test $A_\beta(w)$. This proves (4.22). Each resulting expression is bilinear and invariant under interchanging w, v, which proves the last assertion. □

Remark 4.10 (Signs and domains in the source formulas)**.** With the scalar flux $\{w\} - \beta \cdot [\![w]\!]$ and the negative duality convention in (4.2), the two D_β terms in (4.22) have minus signs. The plus signs printed in equation (3.27) of the source article are inconsistent with its flux definitions (3.25)–(3.26); the calculation above fixes the signs directly and agrees with its discrete square-completion formula in Table 4.1. Also a squared bound for L_β requires b^2, as in (4.8), rather than the unsquared factor printed in its (4.9). Finally, the compact identities in the next subsection use $\nabla_h V_h \subset \Sigma_h$ and are stated only for discrete arguments. Consistency and duality always use (4.19)–(4.22), whose domains include the exact solutions.

4.4 Boundedness, coercivity, and discrete solvability

Choose the trace constant $T > 0$ so that the side-trace bound of Lemma 3.8 reads

$$\sum_K \sum_{e \subset \partial K} h_e \|\nabla v|_K\|_e^2 \le T^2 \big(G_h(v)^2 + H_h(v)^2\big) \qquad (v \in X). \tag{4.23}$$

The bound depends only on κ, not on the number or sizes of elements. Put $a = \sqrt{3}(1 + 2b)$.

Theorem 4.11 (Uniform boundedness)**.** *Each of the four extended forms satisfies*

$$|B(w, v)| \le C_b N_h(w) N_h(v) \qquad (w, v \in W_h).$$

Valid choices for C_b are

$$\begin{aligned} C_{b,\mathrm{IP}} &= 1 + 2T + \eta_{\max}, \\ C_{b,13} &= 1 + 2T + \eta_{\max} c_+, \\ C_{b,22} &= 1 + 2T + (3 + \eta_{\max}) c_+, \\ C_{b,\mathrm{LDG}} &= 1 + 2T + 2\sqrt{2} b T + a^2 c_+ + \eta_{\max}. \end{aligned}$$

Proof. Cauchy–Schwarz gives $|(\nabla_h w, \nabla_h v)| \le G_h(w) G_h(v)$. On interior edges $|\{\nabla_h w\}|^2 \le (|\nabla_h w^+|^2 + |\nabla_h w^-|^2)/2$; on boundary edges the average is the one trace. Apply weighted Cauchy–Schwarz to (4.16), then (4.23), to obtain $|C(w, v)| \le T(G_h(w)^2 + H_h(w)^2)^{1/2} J_h(v)$.

Similarly $|[\![\nabla_h w]\!]|^2 \le 2(|\nabla_h w^+|^2+|\nabla_h w^-|^2)$, so $|D_\beta(w,v)| \le \sqrt{2}bT(G_h(w)^2+H_h(w)^2)^{1/2}J_h(v)$. The penalty terms satisfy

$$|A_j(w,v)| \le \eta_{\max} J_h(w)J_h(v), \qquad |A_r(w,v)| \le \eta_{\max} S_h(w)S_h(v) \le \eta_{\max} c_+ J_h(w)J_h(v).$$

The first penalty bound is weighted Cauchy–Schwarz over the edges; the second is Cauchy–Schwarz over the finite list of lifting fields, followed by (4.5). The same inequality and Lemma 4.3 give

$$|(R(w),R(v))| \le 3c_+ J_h(w)J_h(v), \qquad |(A_\beta(w),A_\beta(v))| \le a^2 c_+ J_h(w)J_h(v).$$

Each factor G_h, J_h, or $(G_h^2+H_h^2)^{1/2}$ is at most N_h. Sum the absolute-value bounds for the terms in each of (4.19)–(4.22) to get the stated constants. □

Lemma 4.12 (Identities restricted to discrete arguments)**.** *For $w_h, v_h \in V_h$ one has*

$$\begin{aligned}
B_{\mathrm{IP}}(w_h,v_h) &= (\nabla_h w_h, \nabla_h v_h) + (R(w_h), \nabla_h v_h) + (\nabla_h w_h, R(v_h)) + A_j(w_h,v_h),\\
B_{13}(w_h,v_h) &= (\nabla_h w_h, \nabla_h v_h) + (R(w_h), \nabla_h v_h) + (\nabla_h w_h, R(v_h)) + A_r(w_h,v_h),\\
B_{22}(w_h,v_h) &= (\nabla_h w_h + R(w_h), \nabla_h v_h + R(v_h)) + A_r(w_h,v_h),\\
B_{\mathrm{LDG}}(w_h,v_h) &= (\nabla_h w_h + A_\beta(w_h), \nabla_h v_h + A_\beta(v_h)) + A_j(w_h,v_h).
\end{aligned}$$

Proof. In the definition of $R(v_h)$ choose the test $\tau = \nabla_h w_h \in \Sigma_h$. This gives $(R(v_h), \nabla_h w_h) = -C(w_h,v_h)$. In the definition of $L_\beta(v_h)$ the same test gives $(L_\beta(v_h), \nabla_h w_h) = -D_\beta(w_h,v_h)$. Insert these identities and their versions with the arguments interchanged into (4.19)–(4.22) and expand the displayed inner products. □

Theorem 4.13 (Uniform coercivity and explicit sufficient thresholds)**.** *Assume the method's lower penalty bound satisfies*

$$\eta_{\min} > 6c_+ \quad (\mathrm{IP}), \qquad \eta_{\min} > 3 \quad ([13]), \qquad \eta_{\min} > 0 \quad ([22] \text{ and } LDG).$$

Then, for all $v_h \in V_h$,

$$B(v_h,v_h) \ge kE_h(v_h)^2 \ge C_s N_h(v_h)^2, \qquad C_s = \frac{k}{1+I^2} > 0,$$

where the following explicit choices are valid:

$$\begin{aligned}
k_{\mathrm{IP}} &= \min\{1/2, \eta_{\min} - 6c_+\},\\
k_{13} &= \min\{1/2, (\eta_{\min}-3)/(6+c_-^{-1})\},\\
k_{22} &= \min\{1/2, \eta_{\min}/(6+c_-^{-1})\},\\
k_{\mathrm{LDG}} &= \min\{1/2, \eta_{\min}/(2a^2c_+ + 1)\}.
\end{aligned}$$

The sufficient IP threshold is expressed in terms of the explicit polynomial-trace constant $c_+ = C_{\mathrm{tr}}^2$ of Lemma 4.2; no unquantified "sufficiently large" condition remains.

Proof. Abbreviate $G = G_h(v_h)$, $J = J_h(v_h)$ and $S = S_h(v_h)$. For IP, Lemma 4.12, the penalty lower bound, Cauchy–Schwarz, and the elementary inequality $2xy \le \frac{1}{2}x^2 + 2y^2$ give

$$B_{\mathrm{IP}}(v_h,v_h) \ge G^2 - 2G\|R(v_h)\| + \eta_{\min}J^2 \ge \tfrac{1}{2}G^2 + (\eta_{\min} - 6c_+)J^2.$$

Here $\|R(v_h)\|^2 \le 3S^2 \le 3c_+J^2$ was used in the last step. This is the asserted bound for IP.

Set $q = \|\nabla_h v_h + R(v_h)\|$. The triangle inequality and $|x+y|^2 \le 2|x|^2 + 2|y|^2$ give

$$E_h(v_h)^2 = G^2 + J^2 \le 2q^2 + 6S^2 + c_-^{-1}S^2.$$

For method [22], the exact square identity and $A_r(v_h, v_h) \geq \eta_{\min} S^2$ give $B_{22}(v_h, v_h) \geq q^2 + \eta_{\min} S^2$. By the definition of k_{22}, the right-hand side is at least $k_{22}(2q^2 + (6 + c_-^{-1})S^2)$, hence at least $k_{22} E_h(v_h)^2$. For method [13], the discrete identity can instead be written

$$B_{13}(v_h, v_h) = q^2 - \|R(v_h)\|^2 + A_r(v_h, v_h) \geq q^2 + (\eta_{\min} - 3)S^2.$$

The identical comparison, with $\eta_{\min}$ replaced by $\eta_{\min} - 3$, gives k_{13}.

For LDG put $q_\beta = \|\nabla_h v_h + A_\beta(v_h)\|$. The global lifting bound and (4.5) imply

$$E_h(v_h)^2 \leq 2q_\beta^2 + (2a^2 c_+ + 1)J^2, \qquad B_{\mathrm{LDG}}(v_h, v_h) \geq q_\beta^2 + \eta_{\min} J^2.$$

The definition of k_{LDG} proves the comparison exactly as above. Finally (4.6) turns each kE_h^2 lower bound into $kN_h^2/(1 + I^2)$. All displayed constants are positive under the respective strict penalty hypotheses. □

Theorem 4.14 (Existence and uniqueness of the discrete solution)**.** *Under the assumptions of Theorem 4.13, every $f \in L^2(\Omega)$ has a unique $u_h \in V_h$ satisfying (4.13). There is a unique corresponding flux pair, with $\sigma_h = \mathcal{G}(u_h)$, for the chosen method.*

Proof. Define the linear map $\mathscr{B} : V_h \to V_h^*$ by $(\mathscr{B} w_h)(v_h) = B(w_h, v_h)$. If $\mathscr{B} w_h = 0$, then $0 = B(w_h, w_h) \geq C_s N_h(w_h)^2$, so $w_h = 0$ since N_h is a norm. Thus $\mathscr{B}$ is injective. The finite-dimensional real spaces V_h and V_h^* have the same dimension: if a basis of V_h has d elements, the coordinate evaluations form a basis of its dual with d elements. Rank–nullity therefore makes $\mathscr{B}$ surjective. The functional $v_h \mapsto (f, v_h)$ belongs to V_h^* by Cauchy–Schwarz and linearity, so it has exactly one preimage. Theorem 4.4 supplies the flux pair and also its uniqueness. This argument does not require completeness of W_h or an application of an infinite-dimensional existence theorem. □

4.5 Quasioptimality and the energy and L^2 estimates

Theorem 4.15 (Quasioptimality)**.** *Suppose $u \in H^2(\Omega) \cap H_0^1(\Omega)$ solves $-\Delta u = f$ and u_h is the discrete solution of one of the four methods under its stability hypotheses. Then*

$$N_h(u - u_h) \leq (1 + C_b/C_s) \inf_{v_h \in V_h} N_h(u - v_h). \tag{4.24}$$

Proof. Fix $v_h \in V_h$ and put $d_h = v_h - u_h$. Consistency gives $B(u - u_h, d_h) = 0$, so bilinearity yields $B(d_h, d_h) = B(v_h - u, d_h)$. Coercivity and boundedness therefore give

$$C_s N_h(d_h)^2 \leq C_b N_h(v_h - u) N_h(d_h).$$

If $N_h(d_h) = 0$, then $d_h = 0$ and $N_h(u - u_h) = N_h(u - v_h)$ already proves the desired comparison. Otherwise divide by $C_s N_h(d_h) > 0$ to get $N_h(d_h) \leq (C_b/C_s) N_h(u - v_h)$. The triangle inequality applied to $u - u_h = (u - v_h) + d_h$ proves the comparison for every v_h. Taking the infimum proves (4.24); no best approximant is assumed to exist. □

Theorem 4.16 (Energy approximation and optimal order)**.** *Let $1 \leq m \leq p$ be an integer and suppose in addition $u \in H^{m+1}(\Omega)$. Write $C_{a,m} = C_{\mathrm{app}}(m, \kappa)$ for the approximation constant from Lemma 3.10. Then*

$$N_h(u - u_h) \leq (1 + C_b/C_s) C_{a,m} \left(\sum_K h_K^{2m} |u|_{H^{m+1}(K)}^2 \right)^{1/2}. \tag{4.25}$$

In particular, writing $h = \max_K h_K$, for $u \in H^{p+1}(\Omega)$,

$$N_h(u - u_h) \leq C_E h^p |u|_{H^{p+1}(\Omega)}, \qquad C_E = (1 + C_b/C_s) C_{a,p}. \tag{4.26}$$

The constant is independent of the mesh sizes and of u, f.

Proof. The approximation lemma constructs $\Pi_h^m u \in V_h$ with $N_h(u - \Pi_h^m u)$ bounded by the right-hand side of (4.25) without the factor $1 + C_b/C_s$. Use this particular element in (4.24). For $m = p$, the inequality $h_K^{2p} \le h^{2p}$ and the additivity of the integral of every derivative square over the mesh yield (4.26). □

Theorem 4.17 (L^2 estimate by adjoint consistency)**.** *Assume Ω is a bounded convex polygon and let C_{reg} be the constant in Theorem 2.10: for $g \in L^2(\Omega)$ its weak solution ψ belongs to $H^2(\Omega) \cap H_0^1(\Omega)$ and $\|\psi\|_{H^2(\Omega)} \le C_{\mathrm{reg}}\|g\|$. For the same discrete solutions,*

$$\|u - u_h\| \le C_b C_{a,1} C_{\mathrm{reg}}\, h\, N_h(u - u_h). \tag{4.27}$$

Consequently, if $u \in H^{p+1}(\Omega)$,

$$\|u - u_h\| \le C_L h^{p+1} |u|_{H^{p+1}(\Omega)}, \qquad C_L = C_b C_{a,1} C_{\mathrm{reg}} C_E. \tag{4.28}$$

All constants are fixed before choosing the mesh, f, or u: their allowed dependence is on the domain, p, κ, $\eta_{\min}$, $\eta_{\max}$, and, for LDG, b.

Proof. Let $e = u - u_h \in W_h \subset L^2(\Omega)$. If $\|e\| = 0$, both claims hold. Otherwise solve $-\Delta\psi = e$ with homogeneous Dirichlet data by the regularity theorem. Adjoint consistency applies with $w = e \in W_h \subset X$ and gives $\|e\|^2 = B(e, \psi)$. The degree-one approximant $\psi_I = \Pi_h^1 \psi$ belongs to V_h because $p \ge 1$. Galerkin orthogonality gives $B(e, \psi_I) = 0$. Consequently boundedness and the degree-one approximation estimate give

$$\|e\|^2 = B(e, \psi - \psi_I) \le C_b N_h(e) N_h(\psi - \psi_I) \le C_b C_{a,1} h N_h(e) |\psi|_{H^2(\Omega)} \le C_b C_{a,1} C_{\mathrm{reg}} h N_h(e) \|e\|.$$

Both arguments in the boundedness step belong to W_h: $e = u - u_h$ and $\psi - \psi_I$ are explicitly sums of a discrete function and a global $H^2 \cap H_0^1$ function. Divide by the positive $\|e\|$ to obtain (4.27), then insert (4.26). □

Proof of Theorems 1.1, 1.2, and 1.3. Fix the data quantified before the meshes in Theorem 1.1. Choose c_-, c_+ by Lemma 4.2, the trace and inverse constants T, I by Lemma 3.8, and set $\eta_*(p, \kappa) = 6c_+$. In the constants of Theorems 4.11 and 4.13, use $\eta_{\min} = \eta_-$ and $\eta_{\max} = \eta_+$, with the fixed bound b for LDG. The hypotheses of the public theorem make its selected C_s positive. Set $C_E = (1 + C_b/C_s)C_{a,p}$ and $C_0 = C_b C_{a,1} C_{\mathrm{reg}} C_E$, where $C_{a,m} = C_{\mathrm{app}}(m, \kappa)$ and C_{reg} is supplied by Theorem 2.10. Every constant is consequently chosen before the mesh, edge parameters, right-hand side, and solution.

Proposition 4.9 identifies the forms of Section 1 with those analyzed here: $B_{13} = B_{\mathrm{Bassi}}$ and $B_{22} = B_{\mathrm{Brezzi}}$. It proves symmetry and bilinearity. Theorems 4.11, 4.13, and 4.14 prove parts (i) and (ii) of Theorem 1.1. Theorem 2.10 identifies the weak solution in (1.9) with an element of $H^2(\Omega) \cap H_0^1(\Omega)$ satisfying $-\Delta u = f$. Equality in distributions here is equality in L^2, since both sides belong to L^2 and compactly supported smooth tests are dense there. If this solution has the additional H^{p+1} regularity assumed in part (iii), Theorems 4.16 and 4.17 give exactly (1.12) with the constants just chosen.

For the flux assertion, the scalar-flux computations in the proof of Proposition 4.9 give $\mathcal{G}(u_h) = \nabla_h u_h + R(u_h)$ for the first three methods and $\mathcal{G}(u_h) = \nabla_h u_h + A_\beta(u_h)$ for LDG. Theorem 4.4 proves both directions of Theorem 1.2, and the normal-conservation cancellation following Remark 4.8 proves its element-union balance assertion. Finally the four fluxes were checked to have the exact-trace and conservation properties immediately before Proposition 4.9. Theorems 4.6 and 4.7, restricted from X to W_h, prove all of Theorem 1.3. □

A shorter argument. For the four concrete forms, a shorter proof of adjoint consistency is available: symmetry and primal consistency at the dual solution give $B(w,\psi) = B(\psi,w) = (-\Delta\psi, w) = (w, g)$. The error estimates can use this identity directly. The general normal-conservative flux theorem above is retained to prove the broader flux correspondence, where symmetry is not assumed.

The mesh supplies finite sets of triangles and edges, incidence maps, side normals, and polynomial spaces. The liftings can be constructed from finite positive-definite mass matrices, so no infinite-dimensional Riesz theorem is needed. The algebraic flux equivalence and the implications concerning consistency and adjoint consistency are separate from the quantitative trace and approximation estimates. All discrete coercivity statements quantify only over V_h; boundedness, consistency, and duality quantify over the explicitly specified extension W_h (or X where stated). The enhanced norm N_h eliminates liftings from norm definitions while (4.5) proves equivalence to the lifting-based norm. The convergence assertion is an h^p estimate at fixed p, with constants allowed to depend on p; it does not claim a bound uniform as $p \to \infty$. Superpenalty and the two weakly stable methods are outside the assertions of this section.

5 Conforming averaging, NIPG, and superpenalties

This section proves the assertions for ordinary NIPG in Table 6.1 of [1], and the superpenalty assertions of its Section 5.2. The pure jump-penalty method is the method of Babuška–Zlámal [7]; the pure lifting-penalty method is that of Brezzi et al. [23]. The reference numbers here are those in the source article. The latter method is different from the stabilized Brezzi et al. [22] method already treated. We also prove the optimal estimates for the superpenalized NIPG form. The source's superpenalty proof requires a continuous interpolant. We construct one using values of *polynomials*, so no pointwise evaluation of the exact Sobolev solution, Sobolev embedding theorem, or unproved nodal interpolation estimate is needed.

5.1 Polynomial nodes and uniformly bounded element stars

Fix an integer $m \geq 1$. If a_1, a_2, a_3 are the vertices of K, let $\lambda_1, \lambda_2, \lambda_3$ be its affine barycentric coordinates, characterized by $\lambda_i(a_j) = \delta_{ij}$ and $\sum_i \lambda_i = 1$. Define its degree-m node set by

$$\mathcal{N}_m(K) = \left\{ \frac{\alpha_1 a_1 + \alpha_2 a_2 + \alpha_3 a_3}{m} : \alpha_i \in \mathbb{N}_0, \quad \alpha_1 + \alpha_2 + \alpha_3 = m \right\}.$$

Nodes on a shared edge agree geometrically, regardless of the ordering of the vertices. The global node set is the union of these finite sets, with equal points identified. A boundary node is one lying on $\partial\Omega$. A node's element star $\mathcal{T}(a)$ is the set of triangles containing that node. An element-interior node has one triangle in its star, an edge-interior node has one or two, and a vertex node has the triangles incident to that vertex.

Lemma 5.1 (Unisolvence and polynomial evaluation bounds)**.** *Values at $\mathcal{N}_m(K)$ determine exactly one polynomial in $\mathbb{P}_m(K)$. There is a constant $C_{\mathrm{node}}(m,\kappa)$, uniform over the mesh family, such that for $q \in \mathbb{P}_m(K)$ and $s = 0, 1, 2$,*

$$|q|^2_{H^s(K)} \leq C^2_{\mathrm{node}} h_K^{2-2s} \sum_{a \in \mathcal{N}_m(K)} |q(a)|^2. \tag{5.1}$$

For $b \in e$ and $q_e \in \mathbb{P}_m(e)$ there is $C_{\mathrm{ev}}(m)$ such that

$$|q_e(b)|^2 \leq C_{\mathrm{ev}}(m)^2 h_e^{-1} \|q_e\|^2_{L^2(e)}. \tag{5.2}$$

The latter bound includes both endpoints of e.

Proof. The number of nodes is $(m+1)(m+2)/2$, the dimension of $\mathbb{P}_m(K)$. We show that a polynomial vanishing at all nodes is zero. Its restriction to the line $\lambda_3 = 0$ has $m+1$ distinct zeros and degree at most m, and so is identically zero. It is therefore divisible, as a polynomial in two affine coordinates, by λ_3: expanding in powers of λ_3 shows that the constant coefficient polynomial is zero. Suppose inductively that

$$q = \prod_{j=0}^{k-1} (\lambda_3 - j/m)\, q_k, \qquad \deg q_k \le m - k.$$

For $k < m$, the $m - k + 1$ nodes on $\lambda_3 = k/m$ make the restriction of q_k to that line zero; the preceding product is nonzero at those nodes. Thus q_k has the additional factor $\lambda_3 - k/m$. At $k = m$ the remaining q_m is constant and vanishes at the last node, so is zero. This proves injectivity. Equality of dimensions proves that every list of nodal values occurs.

On the fixed reference triangle let ℓ_a be the unique polynomial equal to one at node a and zero at all other nodes. Then $\widehat{q} = \sum_a \widehat{q}(a)\ell_a$. Cauchy–Schwarz in this finite sum bounds each of its derivative norms by the Euclidean norm of its nodal values times $(\sum_a |\ell_a|^2_{H^s(\widehat{K})})^{1/2}$. These numbers are finite and depend only on m. The affine derivative and volume scalings in (3.2), and the bounds on B_K^{-1}, give (5.1). For $m = 1, s = 2$ its left side is zero.

For the edge claim pull back to $[0,1]$ and express the polynomial in monomials. The coefficient vector is bounded in Euclidean norm by its $L^2(0,1)$ norm: this follows from the positive minimum of the polynomial's squared integral on the coefficient unit sphere, as in (3.8). For $0 \le t \le 1$, the Euclidean norm of $(1, t, \dots, t^m)$ is at most $\sqrt{m+1}$. Cauchy–Schwarz therefore bounds every point value uniformly by the $L^2(0,1)$ norm. Edge scaling gives (5.2). □

Lemma 5.2 (Star size, connectivity, and local size comparison)**.** *There are an integer $M = M(\kappa)$ and a number $\Lambda = \Lambda(\kappa) \ge 1$ with the following properties. Every vertex belongs to at most M triangles. Its incident triangles can be ordered cyclically at an interior vertex, or linearly at a boundary vertex, so that consecutive triangles share an edge containing the vertex. At a boundary vertex the first and last triangles have a boundary edge containing it. Any two triangles in one vertex star have diameters with ratio at most Λ.*

For $K \in \mathcal{T}_h$, let

$$\omega_K = \{L \in \mathcal{T}_h : L \text{ shares a vertex with } K\}.$$

Then $\#\omega_K \le 3M$, $h_L/h_K \in [\Lambda^{-1}, \Lambda]$ for $L \in \omega_K$, and each triangle belongs to at most $3M$ of these patches.

Proof. For any angle θ of a triangle, with adjacent side lengths $b, c \le h_K$, (3.1) gives

$$\sin\theta = \frac{2|K|}{bc} \ge \frac{2}{\kappa}.$$

Put $\kappa_* = \max\{4, \kappa\}$. Thus every angle is at least $\theta_0 = \arcsin(2/\kappa_*) > 0$; this formula is defined even for a parameter value whose admissible mesh class is empty. Choose a circle about a given mesh vertex so small that it meets no other vertex, no nonincident triangle, and no edge not containing the vertex, including the opposite edges of incident triangles. The finitely many excluded closed sets have positive distance from the vertex, so such a radius exists. The incident triangles cut out disjoint successive circular arcs. Their union is the full circle at an interior vertex and the polygon's angular interval at a boundary vertex. A missing arc would contradict that the triangles cover $\overline{\Omega}$; an overlapping arc would contradict disjoint interiors. Conformity identifies consecutive arc endpoints with full shared edges through the vertex. The two endpoints of a boundary interval belong to boundary edges. This proves the ordering claims, and the sum of the angles, at most 2π, gives the bound, for example $M = \lceil 2\pi/\theta_0 \rceil + 1$. Triangles sharing an edge have diameter ratio at most $\kappa/2$ by (3.1). A chain in a star has

length less than M, so $\Lambda = (\kappa_*/2)^M$ suffices. The patch is contained in the union of the three vertex stars of K. Finally $L \in \omega_K$ if and only if $K \in \omega_L$, so its cardinality bound also proves the overlap bound. None of these assertions uses quasiuniformity. □

5.2 Conforming averaging without values of a Sobolev function

For a discontinuous $P \in \prod_K \mathbb{P}_m(K)$, define one real number at every global node by the following formula. An elementwise polynomial L^2 class has a unique polynomial representative, since a polynomial zero almost everywhere is identically zero; only this representative is evaluated.

$$a_m(P;a) = \begin{cases} \dfrac{1}{\#\mathcal{T}(a)} \displaystyle\sum_{L\in\mathcal{T}(a)} P|_L(a), & a \notin \partial\Omega, \\ 0, & a \in \partial\Omega. \end{cases}$$

Let $\mathcal{A}_h^m P$ be the elementwise polynomial with these nodal values. Lemma 5.1 proves its existence and uniqueness. The nodal averages are linear in P, so uniqueness makes $\mathcal{A}_h^m$ a linear operator. Only values of finitely many polynomials occur in this definition.

Lemma 5.3 (Conformity and membership in H_0^1)**.** *$\mathcal{A}_h^m P$ has equal polynomial traces on every interior edge and zero trace on every boundary edge. It belongs to $H_0^1(\Omega) \cap \prod_K \mathbb{P}_m(K)$. More generally, an elementwise H^1 function with matching interior traces and zero boundary traces belongs to $H_0^1(\Omega)$.*

Proof. Two traces of $\mathcal{A}_h^m P$ on an interior edge have the same $m+1$ edge nodal values. Their difference is a one-variable polynomial of degree at most m, hence is zero. On a boundary edge all its $m+1$ nodal values vanish, so the trace vanishes.

Here is the asserted Sobolev conclusion, including its boundary condition. Extend the piecewise H^1 function v by zero to $\mathbb{R}^2$, obtaining $\widetilde{v} \in L^2(\mathbb{R}^2)$. For $\varphi \in C_c^\infty(\mathbb{R}^2)$, apply (3.6) with the vector field equal to φ times the ith coordinate vector, and sum over K. Matching traces and opposite interior normals cancel the interior terms, and zero boundary traces cancel the boundary terms. Thus

$$\int_{\mathbb{R}^2} \widetilde{v}\, \partial_i \varphi = -\sum_K \int_K \partial_i (v|_K)\varphi.$$

The right side is integration against the zero extension of a piecewise L^2 function. Hence $\widetilde{v} \in H^1(\mathbb{R}^2)$.

To obtain the defining $C_c^\infty(\Omega)$ approximation, fix an interior point x_0 and a ball $B_r(x_0) \subset \Omega$. For $0 < \lambda < 1$ set

$$v_\lambda(x) = \widetilde{v}\big(x_0 + (x - x_0)/\lambda\big).$$

Its support is contained in $x_0 + \lambda(\overline{\Omega} - x_0)$, which is a compact subset of Ω by (2.3). The affine weak chain rule and the continuity of translation and dilation in L^2, applied to the function and its weak first derivatives, give $v_\lambda \to \widetilde{v}$ in $H^1(\mathbb{R}^2)$ as $\lambda \uparrow 1$. For fixed λ, convolve with a smooth mollifier whose radius is smaller than the distance of this compact support from $\partial\Omega$. The convolution belongs to $C_c^\infty(\Omega)$ and tends to v_λ in H^1 by (2.2), also applied to its derivatives. A diagonal sequence proves $v \in H_0^1(\Omega)$. This establishes the reverse trace-to-closure implication needed here, rather than assuming it as a trace theorem. □

Lemma 5.4 (The correction is controlled by polynomial jumps)**.** *For $\delta = P - \mathcal{A}_h^m P$, $K \in \mathcal{T}_h$, and $s = 0, 1, 2$,*

$$|\delta|^2_{H^s(K)} \le C(m,\kappa) h_K^{2-2s} \sum_{e\in\mathcal{F}_K} h_e^{-1} \|[\![P]\!]\|^2_{L^2(e)}, \tag{5.3}$$

where $\mathcal{F}_K$ consists of all edges containing a vertex of K. Every triangle incident to such an edge belongs to ω_K. Boundary jumps in this formula are Pn, as before.

Proof. At an element-interior node, $\delta(a) = 0$. At an interior edge-interior node, the average uses the two incident polynomials. Their difference at that node is bounded by (5.2) applied to their edge difference; the magnitude of that difference equals the magnitude of the vector jump. At a boundary edge-interior node the same argument uses the single polynomial and its boundary jump.

Consider an interior vertex a of K. For every incident triangle L, choose a path from K to L in the ordering of Lemma 5.2. Along it,

$$P|_K(a) - P|_L(a) = \sum_{j=0}^{q-1} \big(P|_{K_j}(a) - P|_{K_{j+1}}(a)\big), \qquad q < M.$$

Cauchy–Schwarz in this sum and (5.2) on the crossed edges bound the square by $C(m,\kappa)\sum_{e\ni a} h_e^{-1}\|[\![P]\!]\|_e^2$. Since $\delta|_K(a)$ is the average of these differences, Jensen's inequality gives the same bound for it. At a boundary vertex, choose instead a path from K to an incident triangle L having a boundary edge through a. Write $P|_K(a)$ as the sum of the successive differences and the final value $P|_L(a)$. Estimate the differences as above and the final value by the boundary-edge version of (5.2). Since the averaged boundary value is zero, this is exactly a bound on $\delta|_K(a)$.

There are $(m+1)(m+2)/2$ nodes of K. All edges just used belong to $\mathcal{F}_K$. Summing the nodal bounds and applying (5.1) proves (5.3). The incidence assertion follows because every such edge contains a vertex of K and each triangle incident to it contains that vertex as well. □

Lemma 5.5 (Constructive conforming approximation)**.** *For every fixed $m \geq 1$ and κ, there is $C_{\mathrm{conf}}(m,\kappa) > 0$ with the following uniform property. For every mesh in this shape-regular class and every $u \in H^{m+1}(\Omega) \cap H_0^1(\Omega)$, put*

$$I_h^m u = \mathcal{A}_h^m \Pi_h^m u,$$

where Π_h^m is the derivative-average polynomial of (3.15). This is a linear operator. Then $I_h^m u \in H_0^1(\Omega) \cap \prod_K \mathbb{P}_m(K)$ and

$$\big(G_h(u - I_h^m u)^2 + H_h(u - I_h^m u)^2\big)^{1/2} \leq C_{\mathrm{conf}}(m,\kappa) \left(\sum_K h_K^{2m} |u|^2_{H^{m+1}(K)}\right)^{1/2}. \tag{5.4}$$

Moreover

$$\|u - I_h^m u\|_{L^2(\Omega)} \leq C_{\mathrm{conf}}(m,\kappa) \left(\sum_K h_K^{2m+2} |u|^2_{H^{m+1}(K)}\right)^{1/2}. \tag{5.5}$$

Increasing the same constant if necessary makes both assertions hold. For $m \leq p$, $I_h^m u \in V_h$ and its error has zero jumps on all edges. In particular $m = p$ gives order h^p in the enhanced energy norm, and $m = 1$ gives order h for every H^2 dual solution, including when $p = 1$.

Proof. Let $P = \Pi_h^m u$, $\eta = u - P$, and $\delta = P - \mathcal{A}_h^m P$. Lemma 5.3 gives the membership assertion. Global trace matching and the zero boundary trace of u, proved in (3.5), imply $[\![P]\!] = -[\![\eta]\!]$ on every edge. This identity is an identity of Sobolev traces; no value $u(a)$ is used.

For an edge incident to L, its contribution from $\eta|_L$ satisfies, by (3.1), (3.4), and (3.16),

$$\begin{aligned} h_e^{-1}\|\eta|_L\|^2_{L^2(e)} &\leq C(\kappa)\big(h_L^{-2}\|\eta\|^2_{L^2(L)} + \|\nabla\eta\|^2_{L^2(L)}\big) \\ &\leq C(m,\kappa) h_L^{2m} |u|^2_{H^{m+1}(L)}. \end{aligned}$$

The squared interior jump is at most twice the sum of its two one-sided squares, and the boundary jump has just one square. Therefore (5.3) implies

$$|\delta|^2_{H^s(K)} \le C(m,\kappa) h_K^{2-2s} \sum_{L\in\omega_K} h_L^{2m} |u|^2_{H^{m+1}(L)}.$$

Combining this with the local bound for η and Lemma 5.2 yields, for $s = 0, 1, 2$,

$$|u - I_h^m u|^2_{H^s(K)} \le C(m,\kappa) h_K^{2m+2-2s} \sum_{L\in\omega_K} |u|^2_{H^{m+1}(L)}. \tag{5.6}$$

In particular, for $m = 1, s = 2$, the factor is h_K^0; the moment-approximation endpoint is valid, and the averaging correction is affine, with zero second derivative.

For $s = 1$, sum (5.6). For $s = 2$, first multiply it by h_K^2 and then sum. In either case the size factor is h_K^{2m}. For $L \in \omega_K$ replace it by at most $\Lambda^{2m} h_L^{2m}$ and use the overlap bound $3M$. This proves (5.4). For $s = 0$ the same argument uses the factor h_K^{2m+2} and proves (5.5). All constants arose from the fixed degree, reference polynomials, and κ; they were chosen before the mesh or function. Finally both u and $I_h^m u$ have matching interior traces and zero boundary traces, proving the zero-jump assertion. □

5.3 The superpenalties and their precise norms

Fix once and for all Ω, $p \ge 1$, a shape bound κ, and numbers $0 < \eta_{\min} \le \eta_{\max} < \infty$. All constants in the remainder of this section depend only on these fixed data. They will work for every admissible mesh, every choice $\eta_{\min} \le \eta_e \le \eta_{\max}$, and every right-hand side. Set $D_* = \max\{1, \operatorname{diam}\Omega\}$, so that $h \le D_*$. For $w, v \in W_h$ define

$$A_j^{\sup}(w,v) = \sum_e \eta_e h_e^{-(2p+1)} \langle [\![w]\!], [\![v]\!] \rangle_e, \tag{5.7}$$

$$A_r^{\sup}(w,v) = \sum_e \eta_e h_e^{-2p} (r_e([\![w]\!]), r_e([\![v]\!])), \tag{5.8}$$

$$S_{a,h}(v)^2 = A_a^{\sup}(v,v), \qquad \mathcal{N}_{a,h}(v)^2 = G_h(v)^2 + H_h(v)^2 + S_{a,h}(v)^2 \quad (a \in \{j,r\}). \tag{5.9}$$

The liftings are exactly those of (4.1); the factor h_e^{-2p} is outside their defining mass-matrix equations.

Let $c_-, c_+ > 0$ be the uniform constants of (4.4) and put

$$\lambda_j = 1, \qquad \lambda_r = c_-, \qquad d_a = (\eta_{\min}\lambda_a)^{-1/2}, \qquad \Gamma_a = T d_a,$$

where T is the trace constant in (3.12). Because a member of W_h has polynomial jumps, the lifting equivalence applies and proves

$$c_- S_{j,h}(v)^2 \le S_{r,h}(v)^2 \le c_+ S_{j,h}(v)^2, \qquad J_h(v) \le d_a h^p S_{a,h}(v) \quad (v \in W_h). \tag{5.10}$$

The second assertion follows by multiplying the edge lower bounds by $\eta_e h_e^{-2p}$ and using $h_e \le h$. In particular these are genuine norms, and

$$\min\{1, c_-\} \mathcal{N}_{j,h}(v)^2 \le \mathcal{N}_{r,h}(v)^2 \le \max\{1, c_+\} \mathcal{N}_{j,h}(v)^2. \tag{5.11}$$

For definiteness, if $\mathcal{N}_{a,h}(v) = 0$, (5.10) gives $G_h(v) = J_h(v) = 0$, so Lemma 3.7 gives $v = 0$. Homogeneity and the triangle inequality follow from their definitions as direct-sum L^2 norms of linear maps. The equivalences have only been asserted on W_h; a lower lifting bound on arbitrary nonpolynomial edge traces would be false.

With $g(w,v) = (\nabla_h w, \nabla_h v)$ and C as in (4.16), the three superpenalized forms are conveniently written

$$\mathcal{B}_{a,\varepsilon}(w,v) = g(w,v) + \varepsilon\big(-C(w,v) + C(v,w)\big) + A_a^{\sup}(w,v), \tag{5.12}$$

for $(a,\varepsilon) = (j,0), (r,0), (j,1)$. They are respectively Babuška–Zlámal, Brezzi et al. [23], and superpenalized NIPG. In every case the discrete problem is

$$\text{find } u_h \in V_h : \qquad \mathcal{B}_{a,\varepsilon}(u_h, v_h) = (f, v_h) \quad (v_h \in V_h). \tag{5.13}$$

Lemma 5.6 (Residual bound, boundedness, and coercivity)**.** *Uniformly over the meshes and coefficients fixed above,*

$$|C(w,v)| \le \Gamma_a h^p \big(G_h(w)^2 + H_h(w)^2\big)^{1/2} \mathcal{N}_{a,h}(v) \quad (w, v \in W_h). \tag{5.14}$$

If $w \in H^2(\Omega) \cap H_0^1(\Omega)$, its right side is at most $\Gamma_a D_* h^p \|w\|_{H^2(\Omega)} \mathcal{N}_{a,h}(v)$. The three forms obey

$$|\mathcal{B}_{a,\varepsilon}(w,v)| \le C_b^{a,\varepsilon} \mathcal{N}_{a,h}(w) \mathcal{N}_{a,h}(v), \qquad \mathcal{B}_{a,\varepsilon}(v_h, v_h) \ge C_s \mathcal{N}_{a,h}(v_h)^2, \tag{5.15}$$

where the first assertion has domain $W_h \times W_h$, the second has domain V_h, and one may choose

$$C_b^{a,\varepsilon} = 1 + 2\varepsilon\Gamma_a D_*^p, \qquad C_s = (1 + I^2)^{-1}. \tag{5.16}$$

Here I is the inverse constant in (3.10). Each problem (5.13) has exactly one solution.

Proof. Cauchy–Schwarz with edge weights h_e, h_e^{-1}, followed by (3.13) and (5.10), gives

$$|C(w,v)| \le \left(\sum_e h_e \|\{\nabla_h w\}\|_e^2\right)^{1/2} J_h(v) \le T\big(G_h(w)^2 + H_h(w)^2\big)^{1/2} d_a h^p S_{a,h}(v).$$

For global H^2 functions, $G_h(w)^2 + H_h(w)^2 \le D_*^2 \|w\|_{H^2(\Omega)}^2$. The sum of the gradient and penalty pairings is bounded by the product of $\mathcal{N}_{a,h}$ norms, by Cauchy–Schwarz in their direct sum. Apply (5.14) to each of the two remaining terms and use $h^p \le D_*^p$. This proves boundedness.

On the diagonal the two C terms cancel exactly, leaving

$$\mathcal{B}_{a,\varepsilon}(v_h, v_h) = G_h(v_h)^2 + S_{a,h}(v_h)^2.$$

Since $H_h(v_h)^2 \le I^2 G_h(v_h)^2$, this is at least $(1 + I^2)^{-1} \mathcal{N}_{a,h}(v_h)^2$. Thus no large-penalty threshold is required here: positivity of $\eta_{\min}$ suffices, with the displayed constants. No small-mesh condition is needed because the fixed upper bound D_* has been retained. Finally, the matrix of the finite-dimensional discrete system has zero kernel: test a homogeneous solution against itself, use coercivity, and then definiteness. A square matrix with zero kernel is invertible, proving existence and uniqueness for every load. □

Lemma 5.7 (Exact consistency and adjoint-defect identities)**.** *Let $u \in H^2(\Omega) \cap H_0^1(\Omega)$ solve $-\Delta u = f$. For every $v \in W_h$,*

$$\mathcal{B}_{a,\varepsilon}(u,v) - (f,v) = (1-\varepsilon) C(u,v). \tag{5.17}$$

If $\psi \in H^2(\Omega) \cap H_0^1(\Omega)$ solves $-\Delta\psi = q$, then

$$\mathcal{B}_{a,\varepsilon}(v,\psi) = (q,v) + (1+\varepsilon) C(\psi,v). \tag{5.18}$$

Thus the pure penalties have primal and adjoint defect coefficient one, and NIPG has primal defect zero and adjoint defect coefficient two. For the discrete error $e = u - u_h$,

$$\mathcal{B}_{a,\varepsilon}(e, v_h) = 0 \quad (v_h \in V_h \cap H_0^1(\Omega)). \tag{5.19}$$

Proof. The mesh Green identity and the global gradient traces give $g(u,v) = (f,v) + C(u,v)$. The scalar jump of u is zero, so every penalty term containing u and the term $C(v,u)$ vanish. Substitution in (5.12) proves (5.17). Similarly $g(v,\psi) = (q,v) + C(\psi,v)$, while $C(v,\psi) = 0$ and the penalty vanishes. This proves (5.18). Subtract the discrete equation from the primal identity. The remaining defect is zero for a conforming, zero-boundary test because its scalar jumps vanish. This proves (5.19), including the inconsistent pure-penalty cases. □

Theorem 5.8 (Optimal superpenalty energy and full L^2 errors)**.** *Fix $\Omega, p, \kappa, \eta_{\min}, \eta_{\max}$ as above. For each of the three choices (a,ε) in (5.12), there exist positive constants $C_E^{a,\varepsilon}, C_L^{a,\varepsilon}$, depending only on these fixed data, such that the following holds for every admissible mesh and edge coefficient choice and every $f \in L^2(\Omega)$. Let u be the unique zero-Dirichlet Poisson solution and u_h the unique solution of (5.13). If additionally $u \in H^{p+1}(\Omega)$, then*

$$\mathcal{N}_{a,h}(u-u_h) \le C_E^{a,\varepsilon} h^p \|u\|_{H^{p+1}(\Omega)}, \qquad \|u-u_h\|_{L^2(\Omega)} \le C_L^{a,\varepsilon} h^{p+1} \|u\|_{H^{p+1}(\Omega)}. \tag{5.20}$$

In the superpenalized NIPG case, the full norm on the right can be replaced by $|u|_{H^{p+1}(\Omega)}$. One possible explicit choice is

$$C_E^{a,\varepsilon} = \left(1 + \frac{C_b^{a,\varepsilon}}{C_s}\right) C_{\mathrm{conf}}(p,\kappa) + \frac{(1-\varepsilon)\Gamma_a D_*}{C_s}, \tag{5.21}$$

$$C_L^{a,\varepsilon} = C_E^{a,\varepsilon} C_{\mathrm{reg}} \left(C_b^{a,\varepsilon} C_{\mathrm{conf}}(1,\kappa) + (1+\varepsilon)\Gamma_a D_*^p \right), \tag{5.22}$$

where C_{reg} is the fixed polygon regularity constant from Theorem 2.10.

Proof. All constants appearing in the statement were obtained uniformly over the class before choosing the particular mesh, coefficients, or solution. Existence of u follows from Theorem 2.10; existence of u_h follows from Lemma 5.6. Set $u_I = I_h^p u$ and $d_h = u_I - u_h$. The interpolation error has zero jumps, so Lemma 5.5 gives

$$\mathcal{N}_{a,h}(u-u_I) \le C_{\mathrm{conf}}(p,\kappa) h^p |u|_{H^{p+1}(\Omega)}.$$

Using the primal defect identity and the discrete equation, we have

$$\begin{aligned} C_s \mathcal{N}_{a,h}(d_h)^2 &\le \mathcal{B}_{a,\varepsilon}(d_h, d_h) \\ &= \mathcal{B}_{a,\varepsilon}(u_I - u, d_h) + (1-\varepsilon) C(u, d_h) \\ &\le \left(C_b^{a,\varepsilon} \mathcal{N}_{a,h}(u_I - u) + (1-\varepsilon)\Gamma_a D_* h^p \|u\|_{H^2(\Omega)} \right) \mathcal{N}_{a,h}(d_h). \end{aligned}$$

If the last norm is zero there is nothing to divide; otherwise divide by $C_s \mathcal{N}_{a,h}(d_h)$. The triangle inequality for $u - u_h = (u - u_I) + d_h$ proves the energy estimate with (5.21), since $\|u\|_{H^2} \le \|u\|_{H^{p+1}}$. If $\varepsilon = 1$, the full-norm residual term is absent and the bound uses only the highest seminorm.

For the L^2 estimate put $e = u - u_h$ and let $-\Delta\psi = e$, $\psi \in H^2(\Omega) \cap H_0^1(\Omega)$. The regularity theorem gives $\|\psi\|_{H^2(\Omega)} \le C_{\mathrm{reg}} \|e\|_{L^2(\Omega)}$. Choose the conforming degree-one approximation $\psi_I = I_h^1 \psi \in V_h \cap H_0^1(\Omega)$. It has zero jumps, and

$$\mathcal{N}_{a,h}(\psi - \psi_I) \le C_{\mathrm{conf}}(1,\kappa) h \, |\psi|_{H^2(\Omega)}.$$

Equations (5.18) and (5.19) give the exact identity

$$\|e\|_{L^2}^2 = \mathcal{B}_{a,\varepsilon}(e, \psi - \psi_I) - (1+\varepsilon) C(\psi, e).$$

Bound the first term by boundedness and the preceding interpolation estimate. Bound the second by (5.14). Since $p \ge 1$ and $h \le D_*$, $D_* h^p \le D_*^p h$, whence

$$\|e\|_{L^2}^2 \le C_{\mathrm{reg}} \left(C_b^{a,\varepsilon} C_{\mathrm{conf}}(1,\kappa) + (1+\varepsilon)\Gamma_a D_*^p \right) h \, \mathcal{N}_{a,h}(e) \|e\|_{L^2}.$$

For zero L^2 norm the conclusion is immediate; otherwise divide and then use the energy estimate. This proves (5.22) and the full L^2 estimate, rather than a bound on a projection of the error. □

5.4 Ordinary NIPG and its guaranteed L^2 rate

The ordinary NIPG form, with bounded rather than superpenalized edge weights, is

$$B_N(w,v) = g(w,v) - C(w,v) + C(v,w) + A_j(w,v), \tag{5.23}$$

where A_j is exactly (4.14). Thus its penalty weight is $\eta_e h_e^{-1}$, not $\eta_e h_e^{-(2p+1)}$. We first prove the discrete Poincare estimate needed for a guaranteed L^2 error bound without adjoint consistency.

Lemma 5.9 (A uniform broken Poincare inequality)**.** *For every admissible mesh and every $v \in H^2(\mathcal{T}_h)$,*

$$\|v\|_{L^2(\Omega)} \le C_P E_h(v) \le C_P N_h(v), \qquad C_P = C_{\mathrm{reg}}\sqrt{1+T^2D_*^2}. \tag{5.24}$$

Proof. Solve $-\Delta\psi = v$ with $\psi \in H^2(\Omega)\cap H_0^1(\Omega)$. Mesh Green's identity and single-valued gradient traces imply

$$\|v\|_{L^2}^2 = (\nabla_h v, \nabla\psi) - C(\psi, v).$$

Cauchy–Schwarz, the averaged gradient trace estimate, and regularity give

$$\begin{aligned}\|v\|_{L^2}^2 &\le G_h(v)\|\nabla\psi\|_{L^2} + TJ_h(v)\big(G_h(\psi)^2 + H_h(\psi)^2\big)^{1/2}\\ &\le C_{\mathrm{reg}}\big(G_h(v) + TD_*J_h(v)\big)\|v\|_{L^2}\\ &\le C_{\mathrm{reg}}\sqrt{1+T^2D_*^2}\,E_h(v)\|v\|_{L^2}.\end{aligned}$$

Division when the norm is nonzero proves the claim; the zero case requires no division. The presence of the boundary edges in J_h is what makes this estimate possible for arbitrary broken functions. □

Theorem 5.10 (Ordinary NIPG stability and error estimates)**.** *Fix the same domain, degree, shape bound, and positive lower and upper coefficient bounds before choosing a mesh, coefficients, or load. There are constants depending only on those data such that the ordinary NIPG discrete problem $B_N(u_h, v_h) = (f, v_h)$ has a unique solution for every $f \in L^2(\Omega)$. Its bilinear form is bounded on $W_h \times W_h$ and coercive on V_h in N_h. If the exact solution additionally belongs to $H^{p+1}(\Omega)$, then*

$$N_h(u-u_h) \le C_{E,N}h^p|u|_{H^{p+1}(\Omega)}, \qquad \|u-u_h\|_{L^2(\Omega)} \le C_P C_{E,N}h^p|u|_{H^{p+1}(\Omega)}. \tag{5.25}$$

One may take

$$C_{b,N} = 1 + 2T + \eta_{\max}, \qquad C_{s,N} = \frac{\min\{1,\eta_{\min}\}}{1+I^2}, \qquad C_{E,N} = \left(1 + \frac{C_{b,N}}{C_{s,N}}\right)C_{\mathrm{app}}(p,\kappa). \tag{5.26}$$

Its primal consistency is exact, while for a dual solution $-\Delta\psi = q$ it satisfies the exact adjoint-defect identity

$$B_N(v,\psi) = (q,v) + 2C(\psi,v). \tag{5.27}$$

Proof. The bound $|C(w,v)| \le TN_h(w)J_h(v)$ follows from (3.13). The gradient and penalty pairings have bounds $N_h(w)N_h(v)$ and $\eta_{\max}N_h(w)N_h(v)$, respectively, proving the displayed boundedness constant. On the diagonal the skew terms cancel:

$$B_N(v_h,v_h) = G_h(v_h)^2 + A_j(v_h,v_h) \ge \min\{1,\eta_{\min}\}E_h(v_h)^2 \ge C_{s,N}N_h(v_h)^2.$$

The same zero-kernel finite-dimensional argument as in Lemma 5.6 proves unique solvability. The derivation of Lemma 5.7 uses only that the penalty vanishes when one argument has

zero jump; it therefore gives $B_N(u,v) = (f,v)$ and (5.27) verbatim. Subtracting the discrete equation gives exact Galerkin orthogonality against all V_h.

For any $v_h \in V_h$, set $d_h = v_h - u_h$. Coercivity, orthogonality, and boundedness yield

$$C_{s,N} N_h(d_h)^2 \le B_N(d_h,d_h) = B_N(v_h - u, d_h) \le C_{b,N} N_h(v_h - u) N_h(d_h).$$

Divide in the nonzero case and use the triangle inequality; the zero case gives the same result directly. Taking $v_h = \Pi_h^p u$ and applying (3.18) proves the first estimate in (5.25). Apply Lemma 5.9 to $u - u_h \in W_h$ to prove the second. This is the guaranteed h^p bound asserted here; (5.27) records exactly why the preceding adjoint-consistent proof of an extra power of h does not apply. No claim excluding higher rates in particular meshes or examples is needed for this theorem. □

The estimates (5.20) correspond to the source's equations (5.9)–(5.12), with the lifting analogue made explicit. Equations (5.25) provide the ordinary NIPG energy and L^2 entries of Table 6.1. The conformity construction and Lemma 5.9 supply two dependencies that are needed for a complete proof but are implicit in the source's short presentation.

5.5 Ordinary pure penalties and the common flux framework

Proposition 5.11 (The remaining boundedness and stability entries)**.** *With ordinary bounded coefficients $0 < \eta_{\min} \le \eta_e \le \eta_{\max}$, the forms $B_{P,j} = g + A_j$ and $B_{P,r} = g + A_r$ are bounded on $W_h \times W_h$, coercive on V_h in N_h, and uniquely solvable for every L^2 load. One may take*

	C_b	C_s
$B_{P,j}$	$1 + \eta_{\max}$	$\min\{1, \eta_{\min}\}/(1+I^2)$
$B_{P,r}$	$1 + \eta_{\max} c_+$	$\min\{1, \eta_{\min} c_-\}/(1+I^2)$.

They have primal and adjoint defects $C(u,v)$ and $C(\psi,v)$, respectively, for exact and dual solutions.

Proof. The estimates

$$|A_j(w,v)| \le \eta_{\max} J_h(w) J_h(v), \qquad |A_r(w,v)| \le \eta_{\max} c_+ J_h(w) J_h(v)$$

were proved in Theorem 4.11. Add the gradient bound. On the diagonal use $A_j(v_h,v_h) \ge \eta_{\min} J_h(v_h)^2$, or $A_r(v_h,v_h) \ge \eta_{\min} c_- J_h(v_h)^2$, and then $N_h(v_h)^2 \le (1+I^2) E_h(v_h)^2$. This proves the displayed constants. Testing a homogeneous solution with itself gives a zero kernel; equality of finite dimensions gives unique solvability. The mesh Green identity gives $g(u,v) = (f,v) + C(u,v)$, while the penalty vanishes on an exact zero-boundary solution. Interchanging the arguments gives the adjoint defect. Stability therefore does not imply the consistency needed for the standard error estimate; the superpenalty theorem proves the advertised convergence conclusion. □

For the pure penalties, the general flux maps can be chosen as $U(w)|_{\partial K} = w|_{\partial K}$ on every element side and $Q(w,\tau) = -\alpha_a(\llbracket w \rrbracket)$, where $a = j$ or r and α_a was defined before (4.14). Then $\mathcal{G}(w) = \nabla_h w$; substitution in (4.10) gives $g + A_a$. For ordinary NIPG, use the scalar flux of Baumann–Oden:

$$U(w)|_{\partial K} = \{w\} + n_K \cdot \llbracket w \rrbracket \text{ on interior edges}, \qquad U(w) = 2w \text{ on boundary edges},$$

and $Q(w,\tau) = \{\nabla_h w\} - \alpha_j(\llbracket w \rrbracket)$. The identities $\llbracket U(w) - w \rrbracket = \llbracket w \rrbracket$ on all edges and $\{U(w) - w\} = 0$ on interior edges yield $\mathcal{G}(w) = \nabla_h w - R(w)$. Substitution gives $g - C + C^{\mathrm{op}} + A_j$.

Replacing the penalty coefficients by their superpenalty values gives the three corresponding superpenalty flux formulations.

These vector fluxes are single-valued edge fields, hence have zero normal jumps. Together with the four earlier fluxes and the BO/BR choices in Section 6, they verify the general flux–primal equivalence and element-union balance for all nine original methods. The scalar fluxes of the nonsymmetric and pure-penalty methods do not satisfy the conservativity and boundary-zero hypotheses of Theorem 4.7. Their explicit defect identities, rather than a misapplied adjoint-consistency theorem, are used in their error proofs.

6 The two weakly stable methods

We retain the polygon, conforming triangular mesh, fixed degree p, and shape bound $h_K/\rho_K \le \kappa$ from the preceding sections. All constants in this section are uniform over meshes with that bound; they may depend on p, κ and the fixed polygon. Write $X_h = H^2(\mathcal{T}_h)$, and retain the definitions of G_h, H_h, J_h, N_h, C, and R. The two extended forms, defined on $X_h \times X_h$, are

$$B_{\mathrm{BO}}(w,v) = (\nabla_h w, \nabla_h v) + C(v,w) - C(w,v), \tag{6.1}$$

$$B_{\mathrm{BR}}(w,v) = (\nabla_h w, \nabla_h v) - C(v,w) - C(w,v) + (R(w), R(v)). \tag{6.2}$$

The first is the Baumann–Oden method; the second is the original unstabilized Bassi–Rebay method. Both are consistent:

$$B(u,v) = (-\Delta u, v) \quad (u \in H^2(\Omega) \cap H_0^1(\Omega),\ v \in X_h). \tag{6.3}$$

Indeed $[\![u]\!] = 0$, hence $R(u) = 0$ and $C(v,u) = 0$; the remaining expression is the broken Green identity. The BR form is symmetric, so it is also adjoint consistent. The trace and lifting estimates already proved give

$$\begin{aligned} |B_{\mathrm{BO}}(w,v)| &\le (1+2T) N_h(w) N_h(v), \\ |B_{\mathrm{BR}}(w,v)| &\le (1+2T+3c_+) N_h(w) N_h(v) \qquad (w,v \in W_h). \end{aligned} \tag{6.4}$$

Only the BO bound will also be used for arbitrary broken H^2 arguments; it holds on $X_h \times X_h$ by the same trace proof, which does not use polynomial jumps. The BR lifting bound as stated is needed only on W_h.

For completeness these extended forms have the numerical-flux interpretation of the original article. BR uses $U(w) = \{w\}$ on interior edges, $U(w) = 0$ on boundary edges, and $Q(w,\tau) = \{\tau\}$. BO uses, on the side belonging to K,

$$\begin{aligned} U(w)|_{\partial K} &= \{w\} + n_K \cdot [\![w]\!] && \text{on interior edges,} \\ U(w)|_{\partial K} &= 2w && \text{on boundary edges,} \\ Q(w,\tau) &= \{\nabla_h w\} && \text{on all edges.} \end{aligned}$$

For BO these choices give $[\![U(w) - w]\!] = [\![w]\!]$ on all edges and $\{U(w) - w\} = 0$ on interior edges. For BR the first of these signs is reversed. Substitution in (4.10) therefore gives (6.1) and (6.2); in the BR case the additional lifting term is obtained by testing the definition of $R(v)$ with $R(w)$. The general flux–primal equivalence theorem applies. BO's boundary scalar flux is deliberately $2w$, not zero; its adjoint inconsistency must not be erased by assigning it the boundary flux of the symmetric methods.

6.1 A quadratic normal-moment construction

For a triangle K and one of its edges e, let a_e be the opposite vertex and define

$$\chi_{K,e}(x) = \frac{|x - a_e|^2}{4|K|}. \tag{6.5}$$

For every edge e' of K,

$$\int_{e'} \partial_{n_K} \chi_{K,e} = \delta_{ee'}. \tag{6.6}$$

To verify this, note that $\nabla \chi_{K,e} = (x - a_e)/(2|K|)$. On either edge through a_e this vector has zero normal component. On the opposite edge its normal component is $a_{K,e}/(2|K|) = 1/h_e$, using $2|K| = h_e a_{K,e}$. Integration proves (6.6). The geometry estimates and $|x - a_e| \le h_K$ give, for $s = 0, 1, 2$,

$$|\chi_{K,e}|_{H^s(K)} \le C(\kappa) h_K^{1-s}, \qquad \|\chi_{K,e}\|_{L^2(e')} \le C(\kappa) h_{e'}^{1/2} \quad (e' \subset \partial K). \tag{6.7}$$

For $s = 0$, this follows from the uniform pointwise bound $|\chi_{K,e}| \le \kappa/4$ and $|K|^{1/2} \le h_K$. For $s = 1, 2$, differentiate the displayed polynomial, use $|K| \ge h_K^2/\kappa$, and integrate the resulting bounds over K. The edge estimate follows from the same pointwise bound.

Lemma 6.1 (Interpolation with zero normal-derivative moments)**.** *Let $p \ge 2$ and $1 \le m \le p$. There is an elementwise linear operator $\mathcal{I}_h^m : H^{m+1}(\Omega) \to V_h$ such that, on every element and every one of its edges,*

$$\int_e \partial_{n_K} (u - \mathcal{I}_h^m u)|_K = 0. \tag{6.8}$$

For $s = 0, 1, 2$,

$$|u - \mathcal{I}_h^m u|_{H^s(K)} \le C(m, \kappa) h_K^{m+1-s} |u|_{H^{m+1}(K)}. \tag{6.9}$$

If $u \in H^{m+1}(\Omega) \cap H_0^1(\Omega)$, then

$$N_h(u - \mathcal{I}_h^m u) \le C_{\mathrm{BO,a}}(m, \kappa) \left(\sum_K h_K^{2m} |u|_{H^{m+1}(K)}^2 \right)^{1/2}. \tag{6.10}$$

Proof. Let $P_K = \Pi_K^m u$ be the derivative-moment polynomial from (3.15) and put

$$d_{K,e} = \int_e \partial_{n_K} (u - P_K), \qquad (\mathcal{I}_h^m u)|_K = P_K + \sum_{e \subset \partial K} d_{K,e} \chi_{K,e}.$$

This has degree at most $\max\{m, 2\} \le p$ and is linear in u. Equation (6.6) proves the required normal moments. By Cauchy–Schwarz on e and the triangle trace estimate applied to each component of $\nabla(u - P_K)$,

$$|d_{K,e}| \le C(\kappa) \big(\|\nabla(u - P_K)\|_K + h_K \|D^2(u - P_K)\|_K \big) \le C(m, \kappa) h_K^m |u|_{H^{m+1}(K)}.$$

For $m = 1$, the second derivative of P_K is zero; the same estimate uses the endpoint $s = m + 1 = 2$ of the polynomial approximation lemma. Multiplying this bound by (6.7), summing the three corrections, and adding the bound for $u - P_K$ proves (6.9).

To control jumps, write $\eta = u - \mathcal{I}_h^m u$. On interior edges $|[\![\eta]\!]|^2 \le 2(|\eta^+|^2 + |\eta^-|^2)$, and the corresponding boundary expression is $|\eta|^2$. The same trace and local size comparisons as in the proof of Lemma 3.10 give

$$J_h(\eta)^2 \le C(\kappa) \sum_K \big(h_K^{-2} \|\eta\|_K^2 + \|\nabla \eta\|_K^2 \big).$$

Use (6.9) with $s = 0, 1$ on this right-hand side and with $s = 1, 2$ on $G_h^2 + H_h^2$. This proves (6.10). □

Let $P_0 v$ denote the elementwise L^2 projection onto constants, namely $(P_0 v)|_K = |K|^{-1} \int_K v$. For the interpolation error $\eta = u - \mathcal{I}_h^m u$ and every piecewise constant s,

$$B_{\mathrm{BO}}(\eta, s) = B_{\mathrm{BO}}(s, \eta) = 0. \tag{6.11}$$

In fact the gradient term and $C(s, \eta)$ vanish, and each contribution to $C(\eta, s)$ is a constant times a sum of the normal-derivative moments in (6.8). Boundary contributions use the same identity with just one adjacent triangle.

Lemma 6.2 (Removing element means)**.** *For $v_h \in V_h$,*

$$N_h(v_h - P_0 v_h) \le C_{\mathrm{mean}}(p, \kappa) G_h(v_h). \tag{6.12}$$

Proof. Put $q_h = v_h - P_0 v_h$. Elementwise Poincare gives $\|q_h\|_K \le \sqrt{2} h_K \|\nabla v_h\|_K$. The trace inequality, local size comparison, and $|[\![q_h]\!]|^2 \le 2(|q_h^+|^2 + |q_h^-|^2)$ give

$$J_h(q_h)^2 \le C(\kappa) \sum_K (h_K^{-2} \|q_h\|_K^2 + \|\nabla q_h\|_K^2) \le C(\kappa) G_h(v_h)^2.$$

Also $G_h(q_h) = G_h(v_h)$ and the polynomial inverse inequality gives $H_h(q_h) \le I G_h(v_h)$. Adding these three squared bounds proves the assertion. □

Lemma 6.3 (Quadratic testing controls all constant jumps)**.** *For $p \ge 2$ and each piecewise constant function s, there is $v_s \in V_h$ with*

$$C(v_s, s) = J_h(s)^2, \qquad N_h(v_s) \le C_{\mathrm{lift}}(\kappa) J_h(s). \tag{6.13}$$

Proof. Choose one incident element K_e for every interior edge e, and write N_e for the other element. Set $a_e = 2(s|_{K_e} - s|_{N_e})$. On a boundary edge take its unique incident element as K_e and put $a_e = s|_{K_e}$. Define

$$v_s|_K = \sum_{e : K_e = K} a_e \chi_{K,e}.$$

There are at most three summands on an element. On an interior edge, only its own correction has a nonzero integral of the outward normal derivative; all corrections associated with other edges have zero normal-derivative moments there. The contribution to $C(v_s, s)$ is therefore $\frac{1}{2}(s|_{K_e} - s|_{N_e}) a_e = (s|_{K_e} - s|_{N_e})^2$. On a boundary edge it is $s|_{K_e} a_e = s|_{K_e}^2$. Since h_e is the edge length, these are exactly the contributions to $h_e^{-1} \|[\![s]\!]\|_e^2$. Their sum proves the first identity.

For the bound, (6.7) gives on every element

$$\|\nabla v_s\|_K^2 + h_K^2 \|D^2 v_s\|_K^2 \le C(\kappa) \sum_{e : K_e = K} a_e^2.$$

It also gives $\sum_{e' \subset \partial K} h_{e'}^{-1} \|v_s|_K\|_{e'}^2 \le C(\kappa) \sum_{e : K_e = K} a_e^2$. Summing, and bounding jumps by the two side traces, proves $N_h(v_s)^2 \le C(\kappa) \sum_e a_e^2$. Finally $\sum_e a_e^2 \le 4 J_h(s)^2$ by their definitions. □

Theorem 6.4 (Baumann–Oden: solvability and full error estimates)**.** *Fix $p \ge 2$. For every $f \in L^2(\Omega)$ there is a unique $u_h \in V_h$ satisfying*

$$B_{\mathrm{BO}}(u_h, v_h) = (f, v_h) \qquad (v_h \in V_h).$$

If its exact weak solution u belongs to $H^{p+1}(\Omega)$, then

$$N_h(u - u_h) \le C_{\mathrm{BO,E}} h^p |u|_{H^{p+1}(\Omega)}, \qquad \|u - u_h\| \le C_{\mathrm{BO},0} h^p |u|_{H^{p+1}(\Omega)}. \tag{6.14}$$

The constants depend only on the fixed polygon, p, and κ.

Proof. The skew edge terms in (6.1) cancel when the two arguments coincide, giving $B_{\mathrm{BO}}(v,v) = G_h(v)^2$. Suppose w_h is in the kernel of the discrete map $w_h \mapsto B_{\mathrm{BO}}(w_h,\cdot)$. Test with w_h to get $G_h(w_h) = 0$, so elementwise Poincare makes $w_h = s$ piecewise constant. Test with v_s from Lemma 6.3. Since $B_{\mathrm{BO}}(s,v_s) = C(v_s,s) = J_h(s)^2$, it follows that $J_h(s) = 0$. Definiteness of N_h now gives $s = 0$. The discrete map is injective between finite-dimensional spaces of the same dimension, hence bijective; this proves existence and uniqueness for the stated right-hand functional.

For the error, set $u_I = \mathcal{I}_h^p u$, $\eta = u - u_I$, $w_h = u_I - u_h$, $s = P_0 w_h$, and $q_h = w_h - s$. Consistency gives

$$B_{\mathrm{BO}}(w_h, v_h) = -B_{\mathrm{BO}}(\eta, v_h) \qquad (v_h \in V_h).$$

Using $v_h = w_h$, (6.11), boundedness, and Lemma 6.2, we obtain

$$G_h(w_h)^2 = -B_{\mathrm{BO}}(\eta, q_h) \le C N_h(\eta) N_h(q_h) \le C N_h(\eta) G_h(w_h).$$

Division if $G_h(w_h) > 0$ gives $G_h(w_h) \le C N_h(\eta)$; if it is zero this conclusion already holds. Thus $N_h(q_h) \le C N_h(\eta)$ as well.

The constant part is controlled by a different test:

$$J_h(s)^2 = B_{\mathrm{BO}}(s, v_s) = -B_{\mathrm{BO}}(\eta, v_s) - B_{\mathrm{BO}}(q_h, v_s) \le C\big(N_h(\eta) + N_h(q_h)\big) J_h(s).$$

Divide if $J_h(s) > 0$; otherwise no division is needed. This proves $J_h(s) \le C N_h(\eta)$. Since $N_h(s) = J_h(s)$,

$$N_h(u - u_h) \le N_h(\eta) + N_h(q_h) + N_h(s) \le C N_h(\eta) \le C h^p |u|_{H^{p+1}(\Omega)}.$$

This proves the full enhanced-norm estimate, including its jump term. In particular it is stronger than merely controlling the broken gradient seminorm.

For the L^2 estimate let $e = u - u_h$ and solve $-\Delta\psi = e$ with zero Dirichlet data. Theorem 2.10 gives $\|\psi\|_{H^2} \le C_{\mathrm{reg}} \|e\|$. Choose $\psi_I = \mathcal{I}_h^1 \psi \in V_h$ from Lemma 6.1. Primal consistency at ψ, the algebraic identity $B_{\mathrm{BO}}(w,v) + B_{\mathrm{BO}}(v,w) = 2(\nabla_h w, \nabla_h v)$, and Galerkin orthogonality give

$$\|e\|^2 = B_{\mathrm{BO}}(\psi, e) = 2(\nabla\psi, \nabla_h e) - B_{\mathrm{BO}}(e, \psi - \psi_I).$$

Hence

$$\|e\|^2 \le C\|e\| G_h(e) + C N_h(e) h \|\psi\|_{H^2} \le C(1+h) N_h(e) \|e\|.$$

If $\|e\| = 0$ the desired bound holds directly; otherwise divide. Finally $h \le \operatorname{diam}\Omega$ and the already proved energy estimate give the second assertion. No adjoint consistency was assumed for BO. □

Proposition 6.5 (A concrete failure at degree one)**.** *For $p = 1$, the Baumann–Oden discrete problem need not be solvable.*

Proof. Take Ω to be one nondegenerate triangle and take the mesh to consist of that triangle. For affine w, v, Green's formula and $\Delta w = \Delta v = 0$ give

$$C(w,v) = \int_{\partial\Omega} (\partial_n w) v = \int_\Omega \nabla w \cdot \nabla v = C(v,w).$$

Therefore $B_{\mathrm{BO}}(w,v) = (\nabla w, \nabla v)$. For $f = 1$, the test $v = 1$ would require $0 = |\Omega|$, which is false. Constants also show failure of uniqueness for $f = 0$. Thus a general degree-one solvability or convergence assertion would be false; the quadratic-test hypothesis $p \ge 2$ is substantive. □

6.2 A polynomial vector construction for the original Bassi–Rebay method

Put $k = p-1 \geq 0$ and interpret $\mathbb{P}_{-1} = \{0\}$. For later use, $P_k^K v$ denotes the unique polynomial in $\mathbb{P}_k(K)$ satisfying $\int_K (P_k^K v - v)q = 0$ for every $q \in \mathbb{P}_k(K)$. Existence and uniqueness follow by solving the polynomial mass matrix: its quadratic form is the strictly positive squared $L^2(K)$ norm of a nonzero polynomial. These defining equations give linearity, reproduction of $\mathbb{P}_k(K)$, and orthogonality of the error. Define $(P_k v)|_K = P_k^K(v|_K)$; in particular $(P_k v, v) = \|P_k v\|^2$ after summing the local orthogonality identities. This is the projection used throughout the BR argument.

Let $\widetilde{\mathbb{P}}_k$ denote homogeneous polynomials of total degree exactly k in two coordinates, and define on a triangle

$$\mathcal{R}_k(K) = \mathbb{P}_k(K)^2 + x\widetilde{\mathbb{P}}_k(K). \tag{6.15}$$

Translations of the coordinate origin give the same space: translating a homogeneous degree-k polynomial changes only lower-degree terms, which are absorbed into $\mathbb{P}_k^2$ after multiplication by x. This space is often called the Raviart–Thomas space; every property needed here is proved below. In particular no interpolation or surjectivity theorem about that space is being imported.

Lemma 6.6 (Degrees of freedom and unisolvence)**.** *A field in $\mathcal{R}_k(K)$ is uniquely determined by*

$$\int_e (\tau \cdot n_K)q \quad (q \in \mathbb{P}_k(e),\ e \subset \partial K), \qquad \int_K \tau \cdot r \quad (r \in \mathbb{P}_{k-1}(K)^2). \tag{6.16}$$

Conversely, arbitrary values for these functionals on fixed bases of their test spaces are realized by one field in $\mathcal{R}_k(K)$. Also $\tau \cdot n_K \in \mathbb{P}_k(e)$ on every edge and $\operatorname{div}\tau \in \mathbb{P}_k(K)$.

Proof. Write $\tau = a + xq$, with $a \in \mathbb{P}_k^2$ and q homogeneous of degree k. On an edge $x \cdot n_K$ is constant, so the normal component has degree at most k. Euler's identity for a homogeneous polynomial, proved term by term on monomials, gives $\operatorname{div}(xq) = (k+2)q$. Thus the divergence has degree at most k. The representation is a direct sum because xq is homogeneous of degree $k+1$, whereas a has degree at most k. Consequently

$$\dim \mathcal{R}_k = 2\frac{(k+1)(k+2)}{2} + (k+1) = (k+1)(k+3),$$

which equals the number $3(k+1) + k(k+1)$ of functionals in (6.16).

Suppose all functionals vanish. The normal component is a degree-k polynomial orthogonal to all such polynomials on its edge, so it is identically zero there. For every $q_0 \in \mathbb{P}_k(K)$, integration by parts gives

$$\int_K (\operatorname{div}\tau)q_0 = -\int_K \tau \cdot \nabla q_0 + \sum_e \int_e (\tau \cdot n_K)q_0 = 0.$$

Here $\nabla q_0 \in \mathbb{P}_{k-1}^2$, including the zero case $k = 0$. Take $q_0 = \operatorname{div}\tau$ to obtain zero divergence. Its degree-k homogeneous part is $(k+2)q$, so $q = 0$ and $\tau \in \mathbb{P}_k^2$.

We may reduce the remaining argument to $\widehat{K} = \{(x,y) : x > 0, y > 0, x + y < 1\}$. Indeed an affine map $F(\widehat{x}) = B\widehat{x} + b$, with positive determinant d, sends the reference space to the physical space by

$$\tau \circ F = d^{-1}B\widehat{\tau}. \tag{6.17}$$

Substitution into (6.15) proves preservation of the space; lower-degree terms due to b are absorbed into $\mathbb{P}_k^2$. The elementary normal-change identity $n_K\, ds = dB^{-T}\widehat{n}\, d\widehat{s}$ proves preservation of zero normal moments. To verify it, parametrize a reference edge with unit tangent $\widehat{t}$, oriented consistently with its outward normal. The physical tangent length vector is $B\widehat{t}\, d\widehat{s}$.

Rotation through a right angle converts tangent length vectors into normal length vectors; the matrix identity $JB = (\det B)B^{-T}J$, checked entry by entry for a two-by-two matrix, gives the displayed formula. Positive determinant preserves the chosen orientation. Change of variables in the interior moments gives test fields $B^T r \circ F$, which range over $\mathbb{P}_{k-1}(\widehat{K})^2$, so those zero moments are preserved as well.

On the reference triangle, zero normal component on $x = 0$ and $y = 0$ gives $\widehat{\tau} = (xA, yB_0)$ for polynomials $A, B_0 \in \mathbb{P}_{k-1}$ when $k \geq 1$. Test the interior moments with (A, B_0) to get $\int_{\widehat{K}}(xA^2 + yB_0^2) = 0$. Both terms are nonnegative and $x, y > 0$ in the triangle, so both polynomials vanish almost everywhere and hence identically. When $k = 0$, the field is constant and its two normal components on $x = 0, y = 0$ already force it to vanish. This proves injectivity of the degree-of-freedom map. Equality of dimensions proves its surjectivity and all unisolvence assertions. □

For $\tau \in H^1(K)^2$, define $\mathcal{R}_K\tau \in \mathcal{R}_k(K)$ by matching all moments (6.16). Traces make the edge moments finite, and Lemma 6.6 defines a unique linear interpolant.

Lemma 6.7 (Boundedness, approximation, and divergence identity)**.** *The vector interpolant reproduces $\mathcal{R}_k(K)$ and satisfies*

$$\|\mathcal{R}_K\tau\|_K \leq C(k,\kappa)(\|\tau\|_K + h_K\|D\tau\|_K), \tag{6.18}$$

$$\operatorname{div}\mathcal{R}_K\tau = P_k^K(\operatorname{div}\tau), \tag{6.19}$$

where P_k^K is the $L^2(K)$ projection onto $\mathbb{P}_k(K)$. For $1 \leq m \leq k+1$ and $w \in H^{m+1}(K)$,

$$\|\nabla w - \mathcal{R}_K\nabla w\|_K \leq C(m,k,\kappa)h_K^m|w|_{H^{m+1}(K)}. \tag{6.20}$$

Proof. Reproduction follows from uniqueness. On the reference triangle, choose bases for the spaces in (6.16). Each edge functional is bounded by the H^1 norm of its input, by the trace inequality and Cauchy–Schwarz; each interior functional is bounded by its L^2 norm. The inverse of the fixed finite degree-of-freedom matrix exists by Lemma 6.6. Its finite matrix norm, followed by the coefficient-to-L^2 norm bound for a fixed polynomial basis, gives $\|\widehat{\mathcal{R}}\widehat{\tau}\|_{\widehat{K}} \leq C_k\|\widehat{\tau}\|_{H^1(\widehat{K})}$.

The moment identities show that interpolation commutes with the change (6.17): after change of variables both candidate fields match all reference moments, so unisolvence makes them equal. For $\widehat{\tau} = dB^{-1}\tau \circ F$, change of variables gives

$$\|\widehat{\tau}\| \leq d^{1/2}\|B^{-1}\|\|\tau\|_K, \qquad \|D\widehat{\tau}\| \leq d^{1/2}\|B^{-1}\|\|B\|\|D\tau\|_K.$$

Conversely the physical L^2 norm of a field obtained by (6.17) is at most $d^{-1/2}\|B\|$ times its reference norm. The geometry bounds give $\|B\| \leq \sqrt{2}h_K$ and $\|B^{-1}\| \leq \kappa/(\sqrt{2}h_K)$. Combining these four inequalities proves (6.18).

For $q \in \mathbb{P}_k(K)$, integration by parts and the defining moments give

$$\int_K \operatorname{div}(\mathcal{R}_K\tau - \tau)q = -\int_K (\mathcal{R}_K\tau - \tau)\cdot\nabla q + \sum_e \int_e (\mathcal{R}_K\tau - \tau)\cdot n_K\, q = 0.$$

The interpolated divergence belongs to $\mathbb{P}_k$, so this is exactly the defining orthogonality of P_k^K, proving (6.19).

To prove the error bound, approximate each component of ∇w by its derivative-moment polynomial a of degree $m - 1 \leq k$. The proof of Lemma 3.9, applied componentwise, gives

$$\|\nabla w - a\|_K + h_K\|D(\nabla w - a)\|_K \leq C_m h_K^m|w|_{H^{m+1}(K)}.$$

For $m = 1$, take a to be the componentwise mean; the same assertion is precisely mean-zero Poincare together with $D(\nabla w - a) = D^2 w$. Since $a \in \mathbb{P}_k^2$ is reproduced,

$$\|\nabla w - \mathcal{R}_K \nabla w\|_K \le \|\nabla w - a\|_K + \|\mathcal{R}_K(\nabla w - a)\|_K.$$

Use (6.18) and the preceding approximation bound to obtain (6.20). □

Lemma 6.8 (A globally conservative vector interpolant)**.** *For $w \in H^2(\Omega) \cap H_0^1(\Omega)$ set $\sigma_I(w)|_K = \mathcal{R}_K(\nabla w|_K)$. Then $\sigma_I(w) \in \Sigma_h$ and its interior normal jumps vanish. Its distributional divergence equals its elementwise polynomial divergence. If $-\Delta w = g$ and $g|_K \in \mathbb{P}_{p-1}(K)$, then*

$$-\operatorname{div} \sigma_I(w) = g, \qquad (\sigma_I(w), \nabla_h v_h + R(v_h)) = (g, v_h) \quad (v_h \in V_h). \tag{6.21}$$

Moreover, for $1 \le m \le p$ and $w \in H^{m+1}(\Omega)$,

$$\|\nabla w - \sigma_I(w)\| \le C_{\mathrm{RT},m} \left(\sum_K h_K^{2m} |w|^2_{H^{m+1}(K)} \right)^{1/2}. \tag{6.22}$$

Proof. Every interpolated field has degree at most $k + 1 = p$, hence belongs to Σ_h. The two traces of ∇w on an interior edge coincide, with opposite outward normals. The interpolant's two normal components therefore have opposite moments against every degree-k edge polynomial. Their sum is itself such a polynomial; testing with that sum shows its squared norm is zero. For any $\varphi \in C_c^\infty(\Omega)$, elementwise integration by parts cancels all interior boundary contributions and has no external boundary contribution. This proves that the distributional divergence is the piecewise divergence, an L^2 function. This is exactly the assertion that $\sigma_I(w) \in H(\mathrm{div}; \Omega)$, if that notation is used.

Equation (6.19) and the polynomial assumption on g give the first identity in (6.21). The lifting definition, with the admissible test $\sigma_I(w) \in \Sigma_h$, and the broken Green identity give

$$\begin{aligned}(\sigma_I(w), \nabla_h v_h + R(v_h)) &= (\sigma_I(w), \nabla_h v_h) - \sum_e \langle \{\sigma_I(w)\}, [\![v_h]\!] \rangle_e \\ &= (-\operatorname{div}_h \sigma_I(w), v_h) + \sum_{e \in \mathcal{E}_h^\circ} \langle [\![\sigma_I(w)]\!], \{v_h\} \rangle_e = (g, v_h).\end{aligned}$$

Finally square (6.20), sum over elements, and take the square root to obtain (6.22). □

6.3 The BR kernel and precise solvability statement

Define the discrete linear map

$$\mathcal{D}_h v_h = \nabla_h v_h + R(v_h), \qquad Z_h = \ker \mathcal{D}_h \subset V_h. \tag{6.23}$$

On $V_h \times V_h$, the lifting identities give

$$B_{\mathrm{BR}}(w_h, v_h) = (\mathcal{D}_h w_h, \mathcal{D}_h v_h). \tag{6.24}$$

For $v_h \in V_h$ and $\tau_h \in \Sigma_h$, another useful identity is

$$(\mathcal{D}_h v_h, \tau_h) = -(v_h, \operatorname{div}_h \tau_h) + \sum_{e \in \mathcal{E}_h^\circ} \langle \{v_h\}, [\![\tau_h]\!] \rangle_e. \tag{6.25}$$

Both follow by expanding R against its polynomial test and then using the broken Green identity. In particular there is no boundary average term on the right of (6.25).

Lemma 6.9 (Divergences of zero-normal vector polynomials)**.** *Let* $\mathcal{R}_k^0(K) = \{\tau \in \mathcal{R}_k(K) : \tau \cdot n_K = 0 \text{ on } \partial K\}$*. Its divergences are exactly the zero-mean polynomials in* $\mathbb{P}_k(K)$*.*

Proof. Every such divergence belongs to $\mathbb{P}_k$ and has zero integral, by the divergence formula. Suppose $q \in \mathbb{P}_k$ is orthogonal to all these divergences. Integration by parts gives $\int_K \nabla q \cdot \tau = 0$ for all $\tau \in \mathcal{R}_k^0(K)$. By Lemma 6.6 there is a field in this space whose interior moments agree with those of ∇q: prescribe zero edge moments and prescribe $\int_K \tau \cdot r = \int_K \nabla q \cdot r$ for every $r \in \mathbb{P}_{k-1}^2$. Taking $r = \nabla q$ gives $\|\nabla q\|_K^2 = 0$. Thus q is constant. For $k = 0$ this last conclusion holds without any interior tests. The orthogonal complement of the divergence range inside the finite-dimensional space $\mathbb{P}_k$ is therefore exactly the constants. Taking orthogonal complements again proves the assertion. This last step is finite-dimensional linear algebra: extend an orthonormal basis of the range to one of $\mathbb{P}_k$ to identify both complements. □

Lemma 6.10 (Independent normal traces in $\mathbb{P}_p^2$)**.** *For* $p \geq 1$*, arbitrary polynomials* $q_e \in \mathbb{P}_p(e)$ *on the three edges of* K *are the outward normal traces of a field* $\tau \in \mathbb{P}_p(K)^2$*.*

Proof. Let e join vertices i, j, with remaining vertex k, and let $\lambda_i, \lambda_j, \lambda_k$ be barycentric coordinates. The $p+1$ functions $\lambda_i^a \lambda_j^{p-a}$, $0 \leq a \leq p$, restricted to e, form a basis of $\mathbb{P}_p(e)$. For example, independence follows on $0 < t < 1$ by dividing a linear combination of $t^a(1-t)^{p-a}$ by $(1-t)^p$; a polynomial in $t/(1-t)$ which vanishes on an interval has all coefficients zero.

If $1 \leq a \leq p-1$, multiply the barycentric monomial by a constant vector c with $c \cdot n_e = 1$. The field has the prescribed normal monomial on e and is zero on the other two edges. For $a = p$, choose c tangent to the edge joining i and k and normalize it so that $c \cdot n_e = 1$; this is possible because the triangle is nondegenerate. Then $c\lambda_i^p$ has the required normal trace on e, zero normal trace on edge ik by tangency, and vanishes on the edge opposite i. For $a = 0$ use the analogous vector tangent to edge jk. Thus every basis polynomial on one edge has an extension with zero normal trace on the other two. Extend each q_e by linear combination and add the three extensions. □

Theorem 6.11 (Exact characterization of the BR nullspace)**.** *For* $v_h \in V_h$*, the following are equivalent:*

$$v_h \in Z_h \quad \Longleftrightarrow \quad P_{p-1}^K(v_h|_K) = 0 \text{ for every } K, \qquad \{v_h\}|_e = 0 \text{ for every interior edge } e. \tag{6.26}$$

There is no boundary-average constraint in this characterization.

Proof. Suppose first $\mathcal{D}_h v_h = 0$. In (6.25), use fields from $\mathcal{R}_k^0(K)$ extended by zero to the other elements. Lemma 6.9 implies that $v_h|_K$ is orthogonal to every zero-mean polynomial in $\mathbb{P}_k(K)$. Thus $P_k^K(v_h|_K) = c_K$ is a constant.

Next fix an edge of K and prescribe any degree-k polynomial q as the outward normal trace there, with zero normal traces on the other two edges. Lemma 6.6 constructs such a field in $\mathcal{R}_k(K)$ by prescribing its edge moments and, for example, zero interior moments. The resulting normal trace equals the prescribed one because their difference is a degree-k polynomial orthogonal to every degree-k polynomial. Extend it by zero to the other elements. Its divergence lies in $\mathbb{P}_k(K)$, so the volume term in (6.25) equals $-c_K \int_K \operatorname{div} \tau = -c_K \int_e q$. For an interior edge the equation consequently becomes

$$\int_e (\{v_h\} - c_K) q = 0 \qquad (q \in \mathbb{P}_k(e)).$$

Applying this from its other incident triangle gives the same identity with c_N. Taking $q = 1$ in both gives $c_K = c_N$. For a boundary edge there is no edge term in (6.25); taking $q = 1$ gives $c_K = 0$. The triangle adjacency graph is connected, as proved in Lemma 3.7, and has a boundary triangle. Hence every $c_K = 0$.

It remains to remove a possible degree-p edge component of $\{v_h\}$. On any interior edge choose one incident triangle and use Lemma 6.10 to construct a field in $\mathbb{P}_p(K)^2$ whose normal trace equals $\{v_h\}$ there and is zero on the other two edges. Extend it by zero elsewhere. The divergence has degree at most $p-1=k$, and we have just proved $P_k^K v_h = 0$; thus its volume pairing with v_h vanishes. Identity (6.25) now gives $\|\{v_h\}\|_e^2 = 0$, as required.

Conversely suppose all projections and interior averages on the right of (6.26) vanish. For every $\tau_h \in \Sigma_h$, its divergence on each element has degree at most $p-1$. Both terms on the right of (6.25) therefore vanish. Take $\tau_h = \mathcal{D}_h v_h \in \Sigma_h$ to obtain $\|\mathcal{D}_h v_h\|^2 = 0$, proving membership in Z_h. □

Theorem 6.12 (BR compatibility and uniqueness modulo the kernel)**.** *The problem*

$$B_{\mathrm{BR}}(u_h, v_h) = (f, v_h) \qquad (v_h \in V_h) \tag{6.27}$$

has a solution if and only if $(f, z_h) = 0$ for all $z_h \in Z_h$. When this condition holds its solution set is $u_h + Z_h$; its flux $\mathcal{D}_h u_h$ and its elementwise projection $P_{p-1}u_h$ are unique. In particular, compatibility holds whenever $f|_K \in \mathbb{P}_{p-1}(K)$ for every K.

Proof. Testing (6.27) with a kernel element and using (6.24) proves necessity. For sufficiency take the $L^2(\Omega)$ orthogonal complement $Y_h = Z_h^\perp \cap V_h$. The form restricted to Y_h has an injective map into Y_h^*: if its pairing with all of Y_h is zero, testing with the field itself shows $\mathcal{D}_h v_h = 0$, hence $v_h \in Z_h \cap Y_h = \{0\}$. Equal finite dimensions imply surjectivity. Solve the problem on Y_h, and decompose an arbitrary test as a sum of a Y_h and a Z_h element. The Y_h equation and the assumed compatibility give the full equation. If two solutions exist, testing their difference equation with that difference gives membership of the difference in Z_h. Conversely adding any member of Z_h preserves the equation. The flux is unique by definition of Z_h; the projected solution is unique by Theorem 6.11. Finally piecewise degree-$(p-1)$ data are orthogonal to Z_h by the same characterization. □

The kernel can be nontrivial, so uniqueness of u_h itself would be incorrect. On a one-triangle mesh with $p=1$, the characterization gives $Z_h = \{v \in \mathbb{P}_1(K) : \int_K v = 0\}$, a two-dimensional space. Taking any nonzero such z and setting $f = z$ gives an incompatible right-hand side because $(f, z) = \|z\|^2 > 0$. For $f = 0$ the exact Dirichlet solution is $u = 0$, whereas every $u_h = tz$, $t \in \mathbb{R}$, is a discrete solution. Its flux and degree-zero projection are both zero, but $\|u - u_h\| = |t|\|z\|$ can be arbitrarily large. This explicitly rules out an unprojected L^2 error bound for all BR solutions, even with the source's admissible piecewise constant forcing.

6.4 BR flux and projected scalar error estimates

We use the constructive conforming approximation from Lemma 5.5: for $1 \le m \le p$ and $w \in H^{m+1}(\Omega) \cap H_0^1(\Omega)$ there is $I_h^m w \in V_h \cap H_0^1(\Omega)$ with vanishing jumps and

$$G_h(w - I_h^m w) + H_h(w - I_h^m w) \le C_{\mathrm{conf},m} \left(\sum_K h_K^{2m} |w|^2_{H^{m+1}(K)} \right)^{1/2}. \tag{6.28}$$

One may take $C_{\mathrm{conf},m} = \sqrt{2} C_{\mathrm{conf}}(m, \kappa)$ to pass from the square-root-of-squares norm in that lemma to the sum on the left of (6.28). Only this estimate, including its degree-one instance, is used below. Its construction by averages of polynomial data does not require unproved point-evaluation or named-element interpolation theorems.

Theorem 6.13 (Original Bassi–Rebay error estimates)**.** *Fix $p \ge 1$. Suppose $f|_K \in \mathbb{P}_{p-1}(K)$ for every element and that the exact weak solution belongs to $H^{p+1}(\Omega)$. Let u_h be any solution of (6.27), whose existence is guaranteed by Theorem 6.12, and put $\sigma_h = \mathcal{D}_h u_h$. Then*

$$\|\nabla u - \sigma_h\| \le C_{\mathrm{BR},\sigma} h^p |u|_{H^{p+1}(\Omega)}, \qquad \|P_{p-1}(u - u_h)\| \le C_{\mathrm{BR},0} h^{p+1} |u|_{H^{p+1}(\Omega)}. \tag{6.29}$$

Here P_{p-1} is the elementwise L^2 projection, and both left-hand sides are independent of the representative u_h modulo Z_h. The constants depend only on the fixed polygon, p, and κ. No bound on the unprojected $\|u - u_h\|$ or $N_h(u - u_h)$ is asserted.

Proof. Choose $u_I = I_h^p u$, so $[\![u_I]\!] = 0$ and $R(u_I) = 0$. Put $d_h = u_I - u_h$ and $\sigma_I = \sigma_I(u)$ from Lemma 6.8. By the discrete equation, the conservative identity, and (6.24),

$$\begin{aligned}\|\mathcal{D}_h d_h\|^2 = B_{\mathrm{BR}}(d_h, d_h) &= B_{\mathrm{BR}}(u_I, d_h) - (f, d_h)\\ &= (\nabla_h u_I - \sigma_I, \mathcal{D}_h d_h).\end{aligned}$$

Thus, dividing only when $\|\mathcal{D}_h d_h\| > 0$,

$$\|\mathcal{D}_h d_h\| \le \|\nabla_h u_I - \nabla u\| + \|\nabla u - \sigma_I\| \le C h^p |u|_{H^{p+1}(\Omega)}. \tag{6.30}$$

The same conclusion holds directly if the norm is zero. Since $\nabla_h u_I - \sigma_h = \mathcal{D}_h d_h$, the triangle inequality and (6.28) prove the first estimate in (6.29).

For the second estimate put $e = u - u_h$ and $q = P_{p-1} e$. Projection orthogonality gives $(q, e) = \|q\|^2$. If $q = 0$ there is nothing to prove; otherwise solve $-\Delta\psi = q$ with zero Dirichlet data and use $\|\psi\|_{H^2} \le C_{\mathrm{reg}} \|q\|$. Choose the conforming degree-one approximant $\psi_I = I_h^1 \psi$, which lies in V_h because $p \ge 1$. Symmetry and primal consistency at ψ, followed by Galerkin orthogonality at ψ_I, give

$$\|q\|^2 = B_{\mathrm{BR}}(e, \psi) = B_{\mathrm{BR}}(e, \psi - \psi_I).$$

Let $\eta = u - u_I$ and $\zeta = \psi - \psi_I$. Both have zero jumps, so their extended form is exactly $B_{\mathrm{BR}}(\eta, \zeta) = (\nabla_h \eta, \nabla_h \zeta)$. For the remaining discrete argument, the conservative identity for $\sigma_I(\psi)$ is applicable because its forcing q is piecewise degree $p - 1$. It gives

$$B_{\mathrm{BR}}(d_h, \psi) = B_{\mathrm{BR}}(\psi, d_h) = (\sigma_I(\psi), \mathcal{D}_h d_h), \qquad B_{\mathrm{BR}}(d_h, \psi_I) = (\nabla_h \psi_I, \mathcal{D}_h d_h).$$

The latter equality uses (6.24) only for the two discrete arguments and the fact $R(\psi_I) = 0$. Consequently

$$\|q\|^2 = (\sigma_I(\psi) - \nabla_h \psi_I, \mathcal{D}_h d_h) + (\nabla_h \eta, \nabla_h \zeta).$$

The degree-one case of (6.22), the degree-one conforming estimate, (6.30), and the degree-p conforming estimate bound the right-hand side by

$$C h^{p+1} |u|_{H^{p+1}(\Omega)} \, |\psi|_{H^2(\Omega)} \le C C_{\mathrm{reg}} h^{p+1} |u|_{H^{p+1}(\Omega)} \|q\|.$$

Divide by $\|q\| > 0$. Independence of the representative follows from the uniqueness statements in Theorem 6.12. □

Scope. The BO theorem controls the full enhanced norm by an explicit test for elementwise constants; it does not infer jump control from a broken-gradient estimate. The degree-one counterexample prevents silently extending that result to $p = 1$. The BR theorem is a theorem on a quotient by Z_h: its meaningful unique outputs are the flux and the projected scalar field. The vector polynomial construction, its unisolvence, normal continuity, commuting divergence, and the kernel characterization have all been reduced to finite-dimensional linear algebra, elementary polynomial identities, and the trace and approximation estimates already proved. No Morley, BDM, or RT interpolation result is an external assumption.

7 Analytic tools

This section proves the analytic tools used in the polygon argument: cutoffs, L^2 approximation and the weak-derivative graph (Section 7.1), the normalization of the Fourier transform and its weak-derivative rule (Section 7.2), the interval sine and cosine bases (Section 7.3), and one-dimensional representatives and polar coordinates (Section 7.4). Their inputs are standard facts of measure theory and functional analysis: completeness of L^2 and of finite products of Hilbert spaces, the Riesz representation theorem, Fubini's theorem, dominated convergence, density of smooth compactly supported functions in L^2, smooth bump functions and convolution, uniqueness of weak derivatives (the fundamental lemma of the calculus of variations), the change-of-variables formula, the Plancherel theorem on $L^2(\mathbb{R}^d)$, and the Fourier basis of L^2 on the circle. For the Fourier transform we write $\mathcal{F}_c$ for the unitary transform on $L^2(\mathbb{R}^d;\mathbb{C})$ in the cyclic-frequency convention $\hat{f}(\zeta) = \int f(x)e^{-2\pi i x\cdot\zeta}\,dx$.

7.1 Support control, L^2 approximation, and weak graphs

Lemma 7.1 (Cutoffs inside an open set)**.** *If $K \subset U \subset \mathbb{R}^d$, K is compact and U is open, there exists $\chi \in C_c^\infty(U)$ with $0 \le \chi \le 1$ and $\chi = 1$ on a neighborhood of K.*

Proof. At each $x \in K$ choose a smooth bump b_x supported in U and equal to one at x. Its set $\{b_x > 1/2\}$ is a neighborhood of x. A finite subcover gives $b_1, \dots, b_N$; their sum s exceeds $1/2$ on a neighborhood of K and has compact support in U. Choose a smooth scalar function q with $q = 0$ on $(-\infty, 1/4]$, $q = 1$ on $[1/2, \infty)$, and $0 \le q \le 1$, and put $\chi = q \circ s$. For an explicit such q, normalize the integral of a nonnegative nonzero smooth bump supported in $(1/4, 1/2)$ and take its cumulative integral. The fundamental theorem of calculus makes that cumulative integral smooth and constant outside this interval. The support of χ is a closed subset of the compact support of s, hence is compact and lies in U. □

Lemma 7.2 (Approximation)**.** *Translations and invertible affine maps tending to the identity act strongly continuously on $L^2(\mathbb{R}^d)$. A smooth nonnegative kernel ρ supported in the unit ball, normalized by $\int \rho = 1$, satisfies*

$$\rho_\varepsilon * f \longrightarrow f \quad \text{in } L^2(\mathbb{R}^d), \qquad \rho_\varepsilon(x) = \varepsilon^{-d}\rho(x/\varepsilon).$$

For every open U, $C_c^\infty(U)$ is dense in $L^2(U)$.

Proof. The required kernel exists by taking a nonnegative smooth bump positive at zero and dividing it by its integral. That integral is finite by compact support and positive because continuity makes the bump positive on a ball of positive Lebesgue measure.

Let $A_j x + b_j \to x$, where $A_j \to I$ and $b_j \to 0$. For a smooth compactly supported g, its compositions with these maps converge pointwise to g. For large j their supports lie in one fixed compact ball: A_j^{-1} and b_j are bounded. Their absolute values are bounded by $\|g\|_\infty$. Dominated convergence, with the integrable majorant $4\|g\|_\infty^2$ times that ball's indicator, proves convergence in L^2. For general f, choose such a g with $\|f - g\|_2 < \delta$ using the density of smooth compactly supported functions. The Jacobian formula gives

$$\|(f - g) \circ (A_j\ \cdot + b_j)\|_2 = |\det A_j|^{-1/2}\|f - g\|_2.$$

The factors are bounded, so the triangle inequality followed by $\delta \downarrow 0$ proves the first assertion. The same proof with the first power in place of the square gives strong continuity on L^1 when that local version is needed.

For almost every x, Cauchy–Schwarz with the probability measure $\rho_\varepsilon(y)\,dy$ gives

$$|(\rho_\varepsilon * f)(x) - f(x)|^2 \le \int \rho_\varepsilon(y)|f(x - y) - f(x)|^2\,dy.$$

Tonelli integrates the nonnegative right side; translation invariance then bounds it by $4\|f\|_2^2$. Its integral tends to zero because translation continuity makes $\sup_{|y|\le\varepsilon}\|f(\cdot-y)-f\|_2$ tend to zero. These facts also establish existence of the convolution almost everywhere. Convolution with a smooth compactly supported kernel is smooth; local integrability follows from Cauchy–Schwarz on compact sets.

For the last assertion choose compact sets $K_j \subset U$ increasing to U. For example take points with $|x| \le j$ and distance at least $1/j$ from the closed complement; if the complement is empty take closed balls. Every point of U eventually belongs to these sets. Dominated convergence gives $\|f - 1_{K_j} f\|_{L^2(U)} \to 0$. Extend $1_{K_j} f$ by zero to $\mathbb{R}^d$. For fixed j and sufficiently small ε, the support of $\rho_\varepsilon * (1_{K_j} f)$ lies in $K_j + \overline{B}_\varepsilon \Subset U$. This smooth compactly supported function approximates $1_{K_j} f$ in L^2. Choosing first j and then ε proves density. Empty K_j cause no problem: their truncated function is zero. □

Lemma 7.3 (A concrete Hilbert model for the document's Sobolev spaces)**.** *For a finite integer m and an open set U, the functions with all weak derivatives through order m can be represented as a closed linear subspace of a finite product of $L^2(U)$ spaces, with the document's ordered-derivative norm. The first component uniquely determines all components. The closure defining $H_0^1(U)$ is a closed linear subspace of this Hilbert space.*

Proof. Use the ℓ^2 finite product (`PiLp 2`) and index it by all ordered lists $\alpha = (i_1, \dots, i_j)$ of coordinate indices of length $0 \le j \le m$. A tuple (v_α) belongs to the subspace when, for every such list and every $\varphi \in C_c^\infty(U)$,

$$\int_U v_\alpha \varphi = (-1)^j \int_U v_\varnothing D^\alpha \varphi.$$

Each constraint is a continuous linear equality on the product by Cauchy–Schwarz, so their intersection is closed. A closed subspace of the complete finite product of L^2 Hilbert spaces is complete. Uniqueness of the components follows from the uniqueness of weak derivatives; repeated-index permutations agree since smooth test derivatives commute. The constraints also imply $D_i v_\alpha = v_{(\alpha,i)}$: insert $D_i\varphi$ in the constraint for v_α and compare it with the constraint for $v_{(\alpha,i)}$. Thus the distributional and iterated definitions agree, not merely their formal symbols. Thus the tuple model and the usual weak definition are isometric, including the multiplicity of mixed derivatives in the ordered norm. Smooth compactly supported functions map to this graph by integration by parts; their linear image has a closed linear closure, which defines $H_0^1(U)$.

Restrictions, smooth multipliers with bounded derivatives, and invertible affine changes of variables preserve the constraints by testing, the product rule, and the Jacobian formula. The proof of Lemma 2.1 gives these identities component by component. The independently proved inequality (2.4) makes the gradient norm equivalent to the graph norm on $H_0^1(U)$ when U is bounded. Only its elementary inequality proof, not the subsequent Riesz existence conclusion of Lemma 2.4, is used here. A gradient-norm Cauchy sequence is consequently a graph-norm Cauchy sequence, converges in the graph, stays in the closed zero-boundary subspace, and converges in gradient norm. This verifies precisely the completeness hypothesis needed when applying the Riesz representation theorem in the weak Poisson existence proof. □

Lemma 7.2 supplies the previously implicit whole-space density and approximate-identity inputs in the homothety proof of Lemma 2.3. That lemma proves smooth-up-to-boundary H^m density on a bounded convex set; it does *not* assert $C_c^\infty(U)$ density in $H^m(U)$. The latter assertion would be false, for example for the constant function one when $m = 1$ on a bounded domain.

7.2 Fourier normalization and weak derivatives

Lemma 7.4 (Cyclic and angular unitary transforms)**.** *Let $d \in \{1, 2\}$ and let $\mathcal{F}_c$ be the cyclic-frequency $L^2(\mathbb{R}^d; \mathbb{C})$ transform. Define*

$$(\mathcal{F}_a f)(\xi) = (2\pi)^{-d/2} (\mathcal{F}_c f)(\xi/(2\pi)). \tag{7.1}$$

This is a complex-linear unitary equivalence. On Schwartz functions,

$$(\mathcal{F}_a f)(\xi) = (2\pi)^{-d/2} \int_{\mathbb{R}^d} e^{-ix\cdot\xi} f(x)\, dx, \qquad \mathcal{F}_a(D_j f) = i\xi_j \mathcal{F}_a f.$$

It preserves the complex inner product, maps the Schwartz space onto itself, and obeys

$$\mathcal{F}_a(\overline{f})(\xi) = \overline{(\mathcal{F}_a f)(-\xi)} \quad \text{as } L^2 \text{ classes.} \tag{7.2}$$

Proof. For any $h \in L^2$, the substitution $\xi = 2\pi\zeta$ gives

$$\int |(2\pi)^{-d/2} h(\xi/(2\pi))|^2 \, d\xi = (2\pi)^{-d} (2\pi)^d \int |h(\zeta)|^2 \, d\zeta.$$

Thus the rescaling operator is an isometry. Its inverse sends k to $\zeta \mapsto (2\pi)^{d/2} k(2\pi\zeta)$, so it is onto. Composing with the Plancherel isometry $\mathcal{F}_c$ proves the first assertion and preservation of inner products. On Schwartz functions $\mathcal{F}_c$ is given by the Fourier integral. Substituting $\zeta = \xi/(2\pi)$ in the derivative identity $2\pi i\zeta_j \mathcal{F}_c f$ gives exactly $i\xi_j \mathcal{F}_a f$. Linear rescaling preserves Schwartz decay and differentiability, and the Fourier transform is a bijection of the Schwartz space; hence the angular transform is also a Schwartz-space automorphism.

For a Schwartz function, conjugating its absolutely convergent integral proves (7.2). Smooth compactly supported functions are Schwartz and dense in L^2. Conjugation, reflection, and both Fourier transforms are isometries on L^2, so passage to the limit proves (7.2) for all L^2 functions. Consequently the transform of a real function has Hermitian symmetry. Conversely Hermitian symmetry implies that f is real almost everywhere, by (7.2) and injectivity. □

Lemma 7.5 (The L^2 weak-derivative multiplier theorem)**.** *For $u \in L^2(\mathbb{R}^d; \mathbb{C})$ and a coordinate j, the following are equivalent:*

1. *u has a weak derivative $D_j u = g \in L^2$;*
2. *$\xi_j \mathcal{F}_a u \in L^2$.*

In that case $\mathcal{F}_a g = i\xi_j \mathcal{F}_a u$ almost everywhere. In cyclic frequency the corresponding identity is $\mathcal{F}_c g = 2\pi i \xi_j \mathcal{F}_c u$. For each integer $m \geq 1$,

$$u \in H^m(\mathbb{R}^d) \quad \Longleftrightarrow \quad (1 + |\xi|^m)\mathcal{F}_a u \in L^2, \qquad |u|_{H^m}^2 = \int |\xi|^{2m} |\mathcal{F}_a u(\xi)|^2 \, d\xi. \tag{7.3}$$

Proof. All weak identities with real tests extend to complex tests by separating real and imaginary parts. Suppose first $D_j u = g$. We may also test against any Schwartz function φ. Indeed take a smooth χ equal to one on B_1, supported in B_2, with $0 \leq \chi \leq 1$, and set $\chi_R(x) = \chi(x/R)$. Then $\chi_R \varphi \to \varphi$ in L^2, and

$$D_j(\chi_R \varphi) - D_j \varphi = (\chi_R - 1) D_j \varphi + (D_j \chi_R)\varphi \longrightarrow 0 \quad \text{in } L^2.$$

The first convergence follows by dominated convergence; the second extra term has norm at most $R^{-1} \|D_j \chi\|_\infty \|\varphi\|_2$. Cauchy–Schwarz therefore passes the weak identity to φ. Using the complex inner product, or equivalently testing against $\overline{\varphi}$, and then using Plancherel gives

$$\int (\mathcal{F}_a g)\overline{\mathcal{F}_a \varphi} = -\int (\mathcal{F}_a u)\overline{i\xi_j \mathcal{F}_a \varphi} = \int i\xi_j (\mathcal{F}_a u)\overline{\mathcal{F}_a \varphi}.$$

For every $\zeta \in C_c^\infty(\mathbb{R}^d)$ choose $\varphi = \mathcal{F}_a^{-1}\zeta$, which is Schwartz by Lemma 7.4. We conclude that the locally integrable function $\mathcal{F}_a g - i\xi_j \mathcal{F}_a u$ integrates to zero against every compactly supported smooth test. It is locally integrable because multiplication by ξ_j is bounded on each compact set and L^2 embeds into L^1 on sets of finite measure. Uniqueness of weak derivatives implies that it is zero almost everywhere. In particular $\xi_j \mathcal{F}_a u \in L^2$; no such weighted integrability was assumed in this direction.

Conversely let $i\xi_j \mathcal{F}_a u \in L^2$ and let g be its inverse angular Fourier transform. For $\varphi \in C_c^\infty$, the same three integrals in reverse order give $\int g\overline{\varphi} = -\int u\overline{D_j\varphi}$. Thus g is the weak derivative. The cyclic statement follows from (7.1).

Iterate this first-order result. The squared sum of the multipliers for all ordered derivatives of length m is

$$\sum_{i_1,\dots,i_m} \xi_{i_1}^2 \cdots \xi_{i_m}^2 = (\xi_1^2 + \cdots + \xi_d^2)^m = |\xi|^{2m}.$$

If the right side of (7.3) is integrable, all lower-order products are square integrable as well, since $|\xi|^j \le 1 + |\xi|^m$ for $0 \le j \le m$. The first-order converse can therefore be applied successively. In the opposite direction, all ordered derivatives exist in L^2, and summing their Plancherel identities gives precisely the displayed formula. □

In particular Lemma 2.5 is a consequence of the Plancherel theorem and the lemma above, rather than of an assumed Sobolev Fourier theorem. More directly, a distributional equation $-\Delta V = F \in L^2$, with $V \in L^2$, may be tested against Schwartz functions by the same cutoff argument with second derivatives. It gives $|\xi|^2 \mathcal{F}_a V = \mathcal{F}_a F$ locally as distributions and therefore almost everywhere. Equation (7.3) then gives $V \in H^2$ and $\|D^2 V\|_2 = \|F\|_2$.

7.3 Interval sine and cosine bases from the circle basis

Lemma 7.6 (Normalized real interval bases)**.** *Let $\omega > 0$, $I = (0, \omega)$, and $\nu_n = n\pi/\omega$. The functions*

$$e_n(\theta) = \sqrt{2/\omega}\,\sin(\nu_n \theta) \quad (n \ge 1)$$

form an orthonormal basis of real or complex $L^2(I)$. The functions

$$c_0(\theta) = \omega^{-1/2}, \qquad c_n(\theta) = \sqrt{2/\omega}\,\cos(\nu_n \theta) \quad (n \ge 1)$$

form another orthonormal basis. For $a_n = \int_I f e_n$ and $b_n = \int_I f c_n$,

$$\|f\|_{L^2(I)}^2 = \sum_{n \ge 1} |a_n|^2, \qquad f = \lim_{N\to\infty} \sum_{n=1}^{N} a_n e_n,$$

$$\|f\|_{L^2(I)}^2 = |b_0|^2 + \sum_{n \ge 1} |b_n|^2, \qquad f = \lim_{N\to\infty} \sum_{n=0}^{N} b_n c_n,$$

where both limits are in L^2.

Proof. Use the circle $\mathbb{R}/T\mathbb{Z}$ of period $T = 2\omega$, with normalized Haar measure. For $T > 0$, the quotient map identifies it with $[a, a+T)$ as a measurable space. Here take $a = -\omega$. Choose a Borel representative of f on I, extended by zero at the endpoints. The Borel function

$$r(t) = t - 2\omega \left\lfloor \frac{t+\omega}{2\omega} \right\rfloor \in [-\omega, \omega)$$

is unchanged when t is replaced by $t + 2\omega k$, $k \in \mathbb{Z}$, by the integer-translation identity for the floor function. Two numbers in this half-open interval differing by $2\omega k$ are equal, so r supplies

the unique representative of each circle class. Periodizing the following interval formulas by composition with r therefore gives well-defined measurable circle functions. The Haar interval formula shows that changing f on a null set changes them only on a Haar-null set: reflection preserves null sets and the selected interval has only two endpoint exceptions. Define the odd and even extensions, respectively, by

$$Of(\theta) = \sqrt{\omega}\,\operatorname{sgn}(\theta) f(|\theta|), \qquad Ef(\theta) = \sqrt{\omega}\, f(|\theta|).$$

Their values at zero and at the identified endpoint may be set to zero. These choices do not change the L^2 classes. Reflection and the interval description of Haar measure give

$$\|Of\|^2_{L^2(\mathrm{Haar})} = \|Ef\|^2_{L^2(\mathrm{Haar})} = \frac{\omega}{2\omega}\, 2 \int_0^\omega |f|^2 = \|f\|^2_{L^2(I)}.$$

They are thus complex-linear isometries onto the odd and even subspaces, respectively; restriction to $(0,\omega)$ divided by $\sqrt{\omega}$ is their inverse on those subspaces.

The circle coefficients are computed as interval integrals. For $n \geq 1$ splitting the integral at zero and substituting $\theta \mapsto -\theta$ in its negative half gives

$$\widehat{Of}(0) = 0, \qquad \widehat{Of}(n) = -\frac{i}{\sqrt{2}} a_n, \qquad \widehat{Of}(-n) = \frac{i}{\sqrt{2}} a_n,$$

and

$$\widehat{Ef}(0) = b_0, \qquad \widehat{Ef}(n) = \widehat{Ef}(-n) = \frac{1}{\sqrt{2}} b_n.$$

These equalities hold for complex f also; no conjugate of a_n or b_n is inserted when the sign of the integer index changes. All these integrals exist because I has finite measure and Cauchy–Schwarz gives $L^2(I) \subset L^1(I)$.

Circle Parseval now gives both claimed norm identities. The two coefficients with indices n and $-n$ combine into $\sqrt{2} a_n \sin(\nu_n \theta)$ in the odd case and $\sqrt{2} b_n \cos(\nu_n \theta)$ in the even case. Circle Fourier series converge in L^2, so their symmetric partial sums with indices $-N, \ldots, N$ converge too. This also follows directly from the squared tail norm in Parseval. Restriction and division by $\sqrt{\omega}$ give the displayed interval expansions. Orthogonality follows either from the circle orthonormality under these same two-mode combinations, or by integrating their inner products. Every real f has real interval coefficients, so the same expansions prove the assertion for the real Hilbert space. □

Lemma 7.7 (Angular weak derivatives and the strip coefficient map)**.** *For $h \in H^1_0(I)$ with sine coefficients a_n,*

$$\|h'\|^2_{L^2(I)} = \sum_{n \geq 1} \nu_n^2 |a_n|^2, \qquad \|h\|_2 \leq \nu_1^{-1} \|h'\|_2. \tag{7.4}$$

Conversely a sequence satisfying $\sum_{n\geq 1}(1+\nu_n^2)|a_n|^2 < \infty$ is the sine coefficient sequence of a unique member of $H^1_0(I)$. For $Q = \mathbb{R} \times I$, the map

$$f \longmapsto \left(t \longmapsto \int_I f(t,\theta) e_n(\theta)\, d\theta \right)_{n \geq 1}$$

is a unitary map from complex $L^2(Q)$ onto the Hilbert direct sum $\ell^2(\mathbb{N}_{\geq 1}; L^2(\mathbb{R}))$, and the same holds over $\mathbb{R}$.

Proof. For $h \in C^\infty_c(I)$, integration by parts gives $\int_I h' c_0 = 0$ and $\int_I h' c_n = \nu_n a_n$ for $n \geq 1$. Cosine Parseval proves the first formula and sine Parseval proves the second. Approximation in

the definition of $H_0^1(I)$ passes them to all its members; the coefficient maps are continuous in L^2. For the converse, finite sine sums have the asserted H^1 norms. Each e_n belongs to $H_0^1(I)$: multiplying by smooth cutoffs which remove endpoint intervals of length ε gives convergence in H^1, since $|e_n(\theta)| \le C_n\theta$ at zero and $|e_n(\theta)| \le C_n(\omega - \theta)$ at the other endpoint. Each extra cutoff-derivative term has squared integral bounded by $C_n\varepsilon$. Completeness of the closed weak graph and the closed zero-boundary subspace now give the converse.

For $f \in L^2(Q)$, Tonelli gives $f(t,\cdot) \in L^2(I)$ for almost every t. The coefficients are measurable: on bounded t intervals the integrands are integrable by Cauchy–Schwarz and Fubini applies. Applying interval Parseval for almost every t and then Tonelli to the nonnegative series gives

$$\sum_n \int_{\mathbb{R}} |f_n(t)|^2 \, dt = \int_Q |f(t,\theta)|^2 \, dt \, d\theta.$$

Conversely, from any square-summable list $f_n \in L^2(\mathbb{R})$ form the partial sums $\sum_{n=1}^N f_n(t)e_n(\theta)$. Their squared difference norm is exactly the sum of the omitted $\|f_n\|_2^2$, by finite orthonormality and Fubini. They are Cauchy in the complete $L^2(Q)$ and converge to a function with the prescribed coefficients. This proves surjectivity and the unitary claim. The same reasoning, applied to the weak t derivatives and the angular identities above, proves the strip energy-space description used in Lemma 2.7. □

7.4 One-dimensional representatives and polar coordinates

The one-dimensional primitive construction rests on the Lebesgue differentiation theorem, which identifies the almost-everywhere derivative of the indefinite integral of a locally integrable function. Absolute continuity of that primitive can also be checked directly: for disjoint intervals with total length less than δ, the sum of the norms of the primitive increments is bounded by the integral of $|f|$ over their union, which tends uniformly to zero with δ by absolute continuity of an L^1 integral. The latter follows by truncating $|f|$ at height M: the bounded part contributes at most $M\delta$, and the tail integral is made small first. This justifies the absolutely continuous representatives in Lemma 2.2 and therefore the constant-coefficient ODE and point evaluation arguments in the corner proof.

For completeness, the polar map used there is

$$\Phi(t,\theta) = e^t(\cos\theta, \sin\theta), \qquad (t,\theta) \in \mathbb{R} \times (0,\omega), \quad 0 < \omega < \pi.$$

Its derivative matrix and determinant are

$$D\Phi = e^t \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}, \qquad \det D\Phi = e^{2t}.$$

It is injective: equal images have equal norms and hence equal t; their unit vectors have equal angles modulo 2π, and two angles in an interval of length less than π cannot differ by a nonzero multiple of 2π. Its image is the sector by definition. After rotation by $-\omega/2$, that sector lies in $x > 0$, and a smooth inverse is

$$t = \tfrac{1}{2}\log(x^2 + y^2), \qquad \theta = \omega/2 + \arctan(y/x).$$

Thus the measurable-source, injectivity, and differentiability hypotheses of the change-of-variables formula are satisfied. It gives $dx = e^{2t}\,dt\,d\theta$. The chain rule with the displayed matrix gives the gradient energy identity. For the Hessian, write $e_r = (\cos\theta, \sin\theta)$ and $e_\theta = (-\sin\theta, \cos\theta)$, so $\partial_\theta e_r = e_\theta$ and $\partial_\theta e_\theta = -e_r$. Differentiating $\nabla v = v_r e_r + r^{-1}v_\theta e_\theta$ in the two unit directions gives

$$(D^2v)_{rr} = v_{rr}, \qquad (D^2v)_{r\theta} = r^{-1}v_{r\theta} - r^{-2}v_\theta, \qquad (D^2v)_{\theta\theta} = r^{-2}v_{\theta\theta} + r^{-1}v_r.$$

Putting $r = e^t$ gives exactly the three entries in (2.23). Local mollification and the bounded Jacobians on compact annular subsectors pass these identities from smooth functions to the local H^2 functions in the proof. All support and exhaustion arguments at the vertex remain those in Lemma 2.9; the vertex is outside the open sector, so no interior distribution supported at it is lost.

8 Correspondence with the source

The statements of Section 1 correspond to the results of [1] as follows.

Result in this document	Source correspondence
Theorems 1.2, 4.4	Section 3.2, (3.8)–(3.11)
Theorem 4.6	Section 3.3, (3.13)–(3.14)
Theorem 4.7	Section 3.3, (3.16), with boundary hypotheses explicit
Theorem 4.11	Section 4.1, (4.4)
Theorem 4.13	Section 4.2, (4.10)
Theorems 4.15, 4.16	First argument of Section 5.1
Theorem 4.17	Duality argument of Section 5.1
Theorem 1.1	First four rows of Table 6.1
Theorem 5.10	NIPG row of Table 6.1
Theorem 5.8	Section 5.2
Theorem 6.4	Section 5.3.1
Theorems 6.11, 6.12	Section 5.3.2
Theorem 6.13	(5.27)–(5.28)

References

[1] Douglas N. Arnold, Franco Brezzi, Bernardo Cockburn, and L. Donatella Marini, *Unified analysis of discontinuous Galerkin methods for elliptic problems*, SIAM Journal on Numerical Analysis **39** (2002), no. 5, 1749–1779, doi:10.1137/S0036142901384162.